

Industrial cameras · Made in China



2023

PRODUCT MANUAL OF HUATENGVISION

New CoaXPress camera series

Company Profile

HuaTeng Vision is a professional industrial camera manufacturer. Since its establishment in 2013, the company has been focusing on the R&D, production and manufacturing of industrial cameras. After nearly 10 years of development, it has become the head enterprise of domestic industrial cameras. The company was founded in Shenzhen and has production center and R&D base in Changsha, Hunan and Wuxi, Jiangsu with more than 200 employees.

HuaTeng Vision focuses on providing development services for various industrial camera products. We are a pure camera hardware supplier. Our company has a complete range of products, and its core products include area scan cameras, line scan cameras, thermal imaging cameras, smart cameras, and 3D cameras, binocular camera module, customized camera machine, camera module and other product lines, with a total of more than 100 product patents and software copyrights. Huatengvision's products have been widely used in scientific research, intelligent transportation, biology, medical treatment, non-standard automation and other fields. At the same time, it has maintained long-term and stable cooperation with many universities, scientific research institutions, and leading companies in major domestic industries.

Because of focus, so professional! HuaTeng never forgets its original aspiration, firmly takes the camera as the core development route, and cooperates with customers in various industries to win-win and create brilliance together.



CONTENTS

Catalogue

◆ 01 Gigabit Network Industrial Camera	P01 – P14
02 GigE Board Level Camera /Module	P15 – P16
◆ 03 Gigabit Network Cable Scanning Industrial Camera	P17 – P19
04 10 Gigabit Network Array Industrial Camera-General Series	P20 – P24
◆ 05 10 Gigabit Network Array Industrial Camera-256 TDI Series	P25 – P34
06 10GigE Area Scan Industrial Camera	P35 – P44
◆07 USB3.0 Industrial Camera	P45 – P48
08 USB3.0 Board Level Camera / Module	P49 – P52
09 USB2.0 Industrial Camera	P53 – P52
◆ 10 CoaXPress High Speed Camera	P53 – P55
11 CoaXPress Linear Array Camera	P56 – P57
◆ 12 X86 Smart Industrial Camera	P58 – P63
13 ARM Smart Industrial Camera	P64 – P66
◆ 14 Short Wave Infrared Industrial Camera	P67 – P68
15 Thermal Imaging Industrial Camera	P69 – P70
◆ 16 Short Wave Infrared Industrial Camera	P71 – P72
17 Pen Mini Camera	P73 – P74
◆ 18 Miniature Industrial Camera	P75 – P76
19 Binocular Camera / Module	P77 – P79
◆ 20 90 Degree Right Angle Industrial Camera	P80 – P82
19 3D Industrial Camera	P83
◆ 22 10 Gigabit Ethernet Card	P84
23 PCIE Expansion Card	P85– P86
◆ 24 Customized Camera	P87

GigE Industrial Camera Series

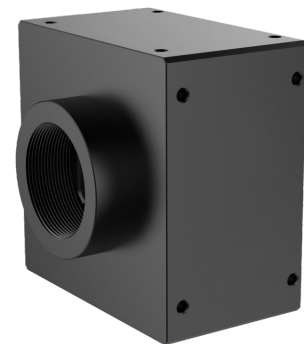
This series of cameras are compatible with GigE vision and genicam standards. It includes various of color and monochrome CMOS and CCD, global shutter and rolling shutter photosensitive chip ,Front-end chips for area and line arrays are also included. The pixel support ranges from 0.3MP to 65MP. It is fully compatible with LabVIEW, Halcon and visionpro And other mainstream visual software. It widely used in electronic semiconductors, factory automation, food and beverage, pharmaceutical packaging, image measurement.



GE Series.CMOS Sensor



GEF Series.Large target surface



GEC Series



GED Series.CCD Sensor

Product Features

- Unique packet retransmission technology to ensure the data reliable transmission
 - The built-in hardware accelerates image processing and reduces CPU utilization on the host side
 - Support simultaneous operation of multiple cameras, unlimited number, and arbitrary networking
 - Pixels from 0.3MP to 65MP, CCD and CMOS are optional
- Excellent SDK design, as simple as using a USB camera, plug and play
 - Support external trigger and flash synchronization, up to seven channels of GPIO, all photoelectric isolation
 - Compatible with the vision standard, drive free and directly support Halcon, visionpro and other software
 - Gigabit network interface, 100m long-distance stable transmission, supporting Poe power supply (optional)

Product Selection Table

Model Number	Effective Pixels	Sensor type	Shutter method	Maximum resolution	Pixel size	Frame rate (FPS)	Target size	Minimum exposure	Sensor Model	Colour
HT-GE33GC/M	0.3MP	CMOS	Global	640×480	4.0μm	159.75	1/5.6"	0.06ms	SmartSens	Color/Mono
HT-GE34GC/M	0.3MP	CMOS	Global	640×480	4.8μm	392.25	1/4"	0.004ms	PYTHON	Color/Mono
HT-GE40GC/M	0.4MP	CMOS	Global	720×540	6.9μm	299	1/2.9"	0.005ms	IMX287	Color/Mono
HT-GE50GC/M	0.48MP	CMOS	Global	800×600	4.8μm	119	1/3.6"	0.009ms	PYTHON480	Color/Mono
HT-GE130M	1.3MP	CMOS	Rolling	1280×1024	5.2μm	30	1/2"	0.032ms	MT9M001	Mono
HT-GE131GC	1.2MP	CMOS	Global	1280×960	3.75μm	72	1/3"	0.014ms	AR0134	Color
HT-GE130GM	1.2MP	CMOS	Global	1280×960	3.75μm	60.75	1/3"	0.016ms	Ar0135	Mono
HT-GE130RC	1.2MP	CMOS	Rolling	1280×960	3.75μm	60.5	1/3"	0.016ms	AR0130	Color
HT-GE133GC/M	1.3MP	CMOS	Global	1280×1024	4μm	93.5	1/2.7"	0.01ms	SC130GS	Color/Mono
HT-GE134GC/M	1.3MP	CMOS	Global	1280×1024	4.8μm	91.5	1/2"	0.008ms	PYTHON	Color/Mono
HT-GE200C	2MP	CMOS	Rolling	1920×1080	2.9μm	30	1/2.8"	0.02ms	SONY	Color
HT-GE200GC/M	2MP	CMOS	Global	1600×1200	4.5μm	60	1/1.8"	0.015ms	EV76C570	Color/Mono
HT-GE202GC/M	2MP	CMOS	Global	1920×1200	4.8μm	51	2/3"	0.005ms	PYTHON2000	Color/Mono
HT-GE231GC/M	2.3MP	CMOS	Global	1920×1200	5.86μm	40.25	1/1.2"	0.02ms	IMX249	Color/Mono
HT-GE232GC/M	2.3MP	CMOS	Global	1920×1200	3.0μm	52	1/2.6"	0.014ms	AR0234	Color/Mono
HT-GE300GC/M	3MP	CMOS	Global	2048×1536	3.45μm	38	1/1.8"	0.016ms	IMX265	Color/Mono
HT-GE501GC/M	5MP	CMOS	Global	2448×2048	3.45μm	24	2/3"	0.019ms	IMX264	Color/Mono
HT-GE502C/M	5MP	CMOS	Rolling	2592×1944	2.2μm	24	1/2.5"	0.02ms	Ar0521	Color/Mono
HT-GE502GC/M	5MP	CMOS	Global	2592×2048	4.8μm	23	1"	0.005ms	PYTHON5000	Color/Mono
HT-GE505GC/M	5MP	CMOS	Global	2592×2048	3.2μm	22	2/3"	0.044ms	XGS5000	Color/Mono
HT-GE630C/M	6.3MP	CMOS	Rolling	3088×2064	2.4μm	18.75	1/1.8"	0.025ms	IMX178	Color/Mono
HT-GE800C/M	8MP	CMOS	Rolling	3840×2160	2.0μm	14.25	1/1.8"	0.062ms	OEM	Color/Mono
HT-GE1000C/M	10MP	CMOS	Rolling	3664×2748	1.67μm	8	1/2.3"	0.043ms	MT9J003	Color/Mono
HT-GE1201C/M	12MP	CMOS	Rolling	4000×3000	1.85μm	9.86	1/1.7"	0.01ms	IMX226	Color/Mono
HT-GE1202C/M	12MP	CMOS	Rolling	4000×3000	1.65μm	9.5	1/2"	0.034ms	AR1202	Color/Mono
HT-GE1600C/M	16MP	CMOS	Rolling	4608×3456	1.34μm	7	1/2.3"	0.039ms	IMX206	Color/Mono
HT-GE2000C/M	20MP	CMOS	Rolling	5488×3672	2.4μm	6	1"	0.044ms	IMX183	Color/Mono
HT-GED32C/M	0.3MP	CCD	Global	640×480	7.4μm	165	1/3"	0.0001ms	SHARP	Color/Mono
HT-GED130C/M	1.3MP	CCD	Global	1280×960	3.75μm	43	1/3"	0.0001ms	SHARP	Color/Mono
HT-GED200C/M	2MP	CCD	Global	1600×1200	4.4μm	27	1/1.8"	0.0001ms	SHARP	Color/Mono
HT-GED500C/M	5MP	CCD	Global	2448×2048	3.45μm	9	2/3"	0.0001ms	SHARP	Color/Mono
HT-GEC501M	5MP	CMOS	Rolling	2640×1968	6.6μm	21.4	4/3"	0.011ms	OEM	Mono
HT-GEF401GC/M	4MP	CMOS	Global	2040×2048	5.5μm	28.529	1"	0.016ms	OEM	Color/Mono
HT-GEF890GC/M	8.9MP	CMOS	Global	4096×2160	3.45μm	13	1.1"	0.034ms	IMX267	Color/Mono
HT-GEF1200GC/M	12MP	CMOS	Global	4096×3000	3.45μm	9.6	1.1"	0.034ms	IMX304	Color/Mono
HT-GEF1205GC/M	12MP	CMOS	Global	4096×3072	3.2μm	9	1"	0.031ms	XGS12000	Color/Mono

Technical Parameters

Parameter \ Model	HT-GE33GC/M	HT-GE34GC/M	HT-GE40GC/M	HT-GE50GC/M
Resolution@ frame rate	640×480@159.75FPS	640×480@392.25FPS	720×540@299FPS	800×600@119FPS
Pixel size	4.0μm×4.0μm	4.8μm×4.8μm	6.9μm×6.9μm	4.8μm×4.8μm
Pixel bit depth	8bit	10bit	8bit	10bit
Sensitivity	8V/lux-s 540nm	7.3V/lux-s 540nm	3660mV 1/30s	7.7V/lux-s 550nm
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	16	16.5	256	8
Exposure time range (ms)	0.006~49.1	0.0040~524.3	0.005~2999.7	0.009~589.8
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color: Bayer8 Mono: Mono8	Color: Bayer8 Mono: Mono8	Color: Bayer8/Bayer12 Mono: Mono8/Mono12	Color: Bayer8/Bayer10 Mono: Mono8/Mono10
GPIO	1 optical isolation input, 1 optical isolation output; optional 3 inputs and 4 outputs			
Lens Mount	C-Mount(CS port Optional)			
Power supply	12~24V(POE is optional)			
Power	<2.5W			
Dimensions	29×29×40mm (Without lens mount and rear shell interface)			
Weight	<75g			

Parameter \ Model	HT-GE130M	HT-GE131GC	HT-GE130GM	HT-GE130RC
Resolution@ frame rate	1280×1024@30FPS	1280×960@72FPS	1280×960@60.75FPS	1280×960@60.5FPS
Pixel size	5.2μm×5.2μm	3.75μm×3.75μm	3.75μm×3.75μm	3.75μm×3.75μm
Pixel bit depth	10bit	12bit	12bit	12bit
Sensitivity	2.1V/lux-s 550nm	5.3V/lux-s 550nm	6.1V/lux-s 550nm	5.5V/lux-s 550nm
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	15	4	4	8
Exposure time range (ms)	0.032~524.3	0.014~53.8	0.016~30.7	0.016~1048.6
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Mono: Mono8/Mono12	Color: Bayer8/Bayer12	Mono: Mono8/Mono12	Color: Bayer8/Bayer12
GPIO	1 optical isolation input, 1 optical isolation output; optional 3 inputs and 4 outputs;			
Lens Mount	C-Mount(CS port Optional)			
Power supply	12~24V(POE is optional)			
Power	<2.5W			
Dimensions	29×29×40mm (Without lens mount and rear shell interface)			
Weight	<75g			

Technical Parameters

Parameter \ Model	HT-GE133GC/M	HT-GE134GC/M	HT-GE200C	HT-GE200GC/M
Resolution@ frame rate	1280×1024@93.5FPS	1280×1024@91.5FPS	1920×1080@30FPS	1600×1200@60FPS
Pixel size	4μm×4μm	4.8μm×4.8μm	2.9μm×2.9μm	4.5μm×4.5μm
Pixel bit depth	8bit	10bit	8bit	10bit
Sensitivity	8 V/Lux.s	7.3V/lux-s 540nm	1300mV 1/30s	7.4V/lux-s 3200K
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	32	16.5	8	8
Exposure time range (ms)	0.01~81.9	0.008~1048.6	0.02~121.9	0.015~95.7
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color: Bayer8 Mono: Mono8	Color: Bayer8/Bayer12 Mono: Mono8/Mono12	Color: Bayer8 Mono: Mono8	Color: Bayer BG 8/Bayer BG 12 Mono: Mono8/Mono12
GPIO	1 optical isolation input, 1 optical isolation output; optional 3 inputs and 4 outputs			
Lens Mount	C-Mount(CS port Optional)			
Power supply	12~24V(POE is optional)			
Power	<2.5W			
Dimensions	29×29×40mm （Without lens mount and rear shell interface）			
Weight	<75g			

Model Parameter	HT-GE202GC/M	HT-GE231GC/M	HT-GE232GC/M	HT-GE300GC/M
Resolution@ frame rate	1920×1200@51FPS	1920×1200@40FPS	1920×1200@52FPS	2048×1536@38FPS
Pixel size	4.8μm×4.8μm	5.86μm×5.86μm	3.0μm×3.0μm	3.45μm×3.45μm
Pixel bit depth	10bit	12bit	8bit	12bit
Sensitivity	7.5V/lux.s 550nm	Color: 1016mV 1/30s Mono: 825mV 1/30s	Color: 3.1V/lux*s Mono: 3.6V/lux*s	1146mV 1/30s
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	8	64	16	64
Exposure time range (ms)	0.005~327.7	0.02~41943	0.014~917.5	0.016~16777
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color: Bayer8/Bayer1 Mono: Mono8/Mono12		Color: Bayer8 Mono: Mono8	Color: Bayer8/Bayer12 Mono: Mono8/Mono12
GPIO	1 optical isolation input, 1 optical isolation output; optional 3 inputs and 4 outputs			
Lens Mount	C-Mount(CS port Optional)			
Power supply	12~24V(POE is optional)			
Power	<3.036W	<2.5W		<3.1W
Dimensions	29×29×40mm (Without lens mount and rear shell interface)			
Weight	<75g			

Technical Parameters

Model Parameter	HT-GE501GC/M	HT-GE502C/M	HT-GE502GC/M	HT-GE505GC/M
Resolution@ frame rate	2448×2048@24FPS	2592×1944@24FPS	2592×2048@22FPS	2592×2048@22FPS
Pixel size	3.45μm×3.45μm	2.2μm×2.2μm	4.8μm×4.8μm	3.2μm×3.2μm
Pixel bit depth	12bit	10bit	10bit	12bit
Sensitivity	Color:1146mV 1/30s Mono:915mV 1/30s	Color:18.8 ke-/lux*sec Mono:36 ke-/lux*sec	7.5 V/lux.s 550nm	Color:1146mV 1/30s Mono:915mV 1/30s
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	64	15	8	22
Exposure time range (ms)	0.019~19922.7	0.02~155.5	0.005~327	0.044~11534
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color:Bayer8/Bayer12 Mono:Mono8/Mono12			
GPIO	1 optical isolation input, 1 optical isolation output; optional 3 inputs and 4 outputs			
Lens Mount	C-Mount(CS port Optional)			
Power supply	12~24V(POE is optional)			
Power	<2.5W			
Dimensions	29×29×40mm (Without lens mount and rear shell interface)			
Weight	<75g			

Model Parameter	HT-GE630C/M	HT-GE800C/M	HT-GE1000C/M	HT-GE1201C/M
Resolution@ frame rate	3088×2064@18.7FPS	3840X2160@14.25FPS	3664×2748@8FPS	4000×3000@9.86FPS
Pixel size	2.4μm×2.4μm	2μm×2μm	1.67μm×1.67μm	1.85μm×1.85μm
Pixel bit depth	12bit	12bit	12bit	8bit
Sensitivity	425mV 1/30s	2200mV/lux.s	0.31V/lux-s 550nm	250mV 1/30s
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	32	15	8	22
Exposure time range (ms)	0.025~14448	0.062~2031.6	0.043~950	0.012~1310.7
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color:Bayer8/Bayer12 Mono:Mono8/Mono12			
GPIO	1 optical isolation input, 1optical isolation output; optional 3 inputs and 4 outputs			
Lens Mount	C-Mount(CS port Optional)			
Power supply	12~24V(POE is optional)			
Power	<2.5W			
Dimensions	29×29×40mm (Without lens mount and rear shell interface)			
Weight	<75g			

Technical Parameters

Model Parameter	HT-GE1202C/M	HT-GE1600C/M	HT-GE2000C/M	HT-GED32C/M
Resolution@ frame rate	4000×3000@9.5FPS	4608×3456@7FPS	5488×3672@5.9FPS	640×480@165FPS
Pixel size	1.65μm×1.65μm	1.34μm×1.34μm	2.4μm×2.4μm	7.4μm×7.4μm
Pixel bit depth	8bit	12bit	12bit	12bit
Sensitivity	Color:1146mV Mono:915mV	142mV 1/30s	Color:462mV 1/30s Mono:388mV 1/30s	Color:3000mV 1/30s Mono:4500mV 1/30s
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	64	16	22	Color:100 Mono:125
Exposure time range (ms)	0.034~9792	0.039~129393	0.044~11534.3	0.0001~6500
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color:Bayer8 Mono:Mono8	Color:Bayer8/Bayer12 Mono:Mono8/Mono12		
GPIO	1 optical isolation input, 1 optical isolation output; optional 3 inputs and 4 outputs			
Lens Mount	C-Mount(CS port Optional)			
Power supply	12~24V(POE is optional)			
Power	<2.5W			
Dimensions	29×29×40mm (Without lens mount and rear shell interface)			
Weight	<75g			

Model Parameter	HT-GED130C/M	HT-GED200C/M	HT-GED500C/M	HT-GEC501M
Resolution@ frame rate	1280×960@43FPS	Color:1600×1200@27FPS Mono:1600×1200@15.25FPS	2448×2048@9FPS	2640×1968@21.4FPS
Pixel size	3.75μm× 3.75μm	4.4μm×4.4μm	3.45μm×3.45μm	6.6μm×6.6μm
Pixel bit depth	12bit	12bit	12bit	12bit
Sensitivity	Color:950mV 1/30s Mono:1430mV 1/30s	Color:1100mV 1/30s Mono:1650mV 1/30s	Color:530mV 1/30s Mono:800mV 1/30s	389mV 1/30s
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	32	32	32	160
Exposure time range (ms)	0.0001~24000	Color:0.0001~48000 Mono:0.0001~24000	0.0001~24000	0.011~5767.2
Frame buffer	32M Bytes	32M Bytes	32M Bytes	128M Bytes
User-defined data area	2K Bytes			
Video output format	Color:Bayer8/Bayer12 Mono:Mono8/Mono12			
GPIO	1 optical isolation input, 1 optical isolation output; optional 3 inputs and 4 outputs			1-channel trigger input,1-channel flash output,1-channel GPI input, 1-way GPI output, 1-way non-isolated bidirectional IO
Lens Mount	C-Mount(CS port Optional)			
Power supply	12~24V(POE is optional)			
Power	<2.5W			<4.2W
Dimensions	29×29×40mm (Without lens mount and rear shell interface)			59.5×59.5×35.5mm (Without lens mount and rear shell interface)
Weight	<75g			<250g

Technical Parameters

Model Parameter	HT-GEF401GC/M	HT-GEF890GC/M	HT-GEF1200GC/M	HT-GEF1205GC/M
Resolution@ frame rate	2040×2048@29FPS	4096×2160@13FPS	4096×3000@9.6FPS	4096×3072@9 FPS
Pixel size	5.5μm×5.5μm	3.45μm×3.45μm	3.45μm×3.45μm	3.2μm×3.2μm
Pixel bit depth	12bit	12bit	12bit	8bit
Sensitivity	7.5V/lux-s 540nm	Color:1146mV Mono:915mV	Color:1146mV Mono:915mV	Color:1146mV Mono:915mV
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	16.5	250	250	8
Exposure time range (ms)	0.016~11534.3	0.034~71303.1	0.034~71303.1	0.031~2031.6
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color:Bayer8/Bayer12 Mono:Mono8/Mono12			
GPIO	1 optical isolation input, 1 optical isolation output; optional 3 inputs and 4 outputs			
Lens Mount	C-Mount(CS port Optional)			
Power supply	12~24V(POE is optional)			
Power	<2.5W	<3.828W	<3.744W	<3.2W
Dimensions	29×39.5×40mm (Without lens mount and rear shell interface)	29×29×40mm (Without lens mount and rear shell interface)		
Weight	<100g			

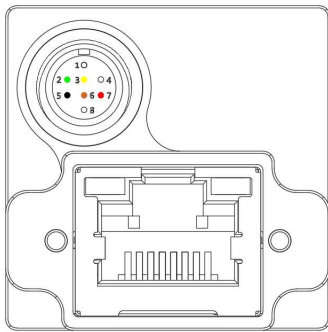
General parameters

Vision Standard Agreement	GigE Vision V1 2、GenIcam
Data interface	Rj45 Gigabit Ethernet interface, backward compatible with 100M network standard
Operating system	WIINXP, WIN7/8/10 32@64 bit system, Linux and ARM Linux driver, Android platform driver, MAC OS system
Driver	Directshow component Halcon special component Labview special driver OCX component TWAIN component
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Working temperature @humidity	0~50℃ @ 20%~80%(No condensation)
Storage temperature @humidity	-30~60℃ @ 20%~95%(No condensation)
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

GE/GEF/GED/GEC Camera Suffix Selection Table

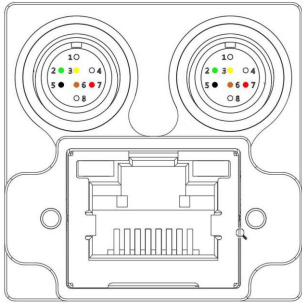
Function	Aircraft head line sequence Definition diagram	POE Power supply	Dual air bearextend IO	Shrapnel typeAircraft head interface	Threaded typeAircraft head interface	Lens interface C-mount	Lens interface CS-mount	State
Suffix								
-T-CL	1				●	●		Recommend
-T-L	1				●		●	Book
-TPO-CL	2	●	●		●	●		Book
-T1-C	3			●		●		Recommend
-T1P-C	3	●		●		●		Book
-T	4			●			●	Planned shutdown
-TPO	5	●	●	●			●	Planned shutdown
-TPO-C	5	●	●	●		●		Planned shutdown

-T-CL、-T-L Line Sequence Definition1



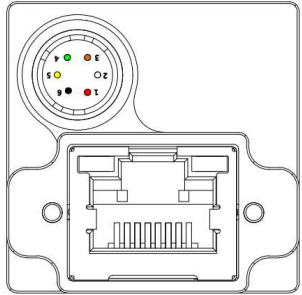
Port	Pin number	Line color	Signal name	Signal description	Remarks
Port A	1	White	GPI1+/TRIG_IN+	GPI1 Positive end/Trigger input positive end	The default is trigger
	2	Green	GPO1+/STRB_OUT +	GPO1 Positive end / Flash output positive end	The default is flash output
	3	Yellow	GPO1-/STRB_OUT-	GPO1 Negative terminal / flash output negative terminal	The default is flash output
	4	Empty foot			
	5	Black	PWRGND	Camera power input negative terminal	
	6	Brown(high soft blue)	GPI1-/TRIG_IN-	GPI1 Negative end / trigger input negative end	The default is trigger input
	7	Red	PWR+	Camera power input positive terminal	12~24V
	8	Empty foot			

-TPO-CL Line Sequence Definition2



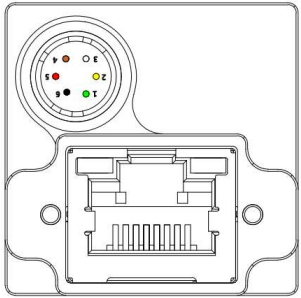
Port	Pin Number	Line Color	Signal Name	Signal Description	Remarks
Port A	1	White	GPI1+/TRIG_IN+	GPI1 positive end/Trigger input positive end	Default to trigger input
	2	Green	GPO1+/STRB_OUT +	GPO1 positive end/Positive end of flash output	Default to flash output
	3	Yellow	GPO1-/STRB_OUT-	GPO1 negative end/Flash output negative terminal	Default to flash output
	4	Empty foot			
	5	Black	PWRGND	Camera power input negative terminal	
	6	Brown (High soft blue)	GPI1-/TRIG_IN-	GPI1 negative end/Trigger input negative terminal	Default to trigger input
	7	Red	PWR+	Camera power input positive terminal	12~24V
	8	Empty foot			
Port B	1	White	GPO4+	GPO4 positive end output	
	2	Green	GPO2+	GPO2 positive end output	
	3	Yellow	GPO3+	GPO3 positive end output	
	4	Empty foot			
	5	Black	GPIO_COM	GPIO common negative terminal	
	6	Brown	GPI2+	GPI2 positive end output	
	7	Red	GPI3+	GPI3 positive end output	
	8	Empty foot			

-T1-C、-T1P-C Line Sequence Definition3



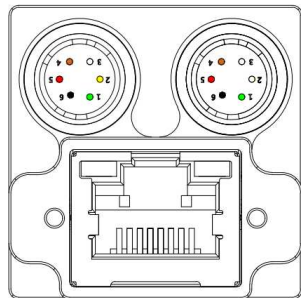
Port	Pin Number	Line Color	Signal Name	Signal Description	Remarks
Port A	1	Red	PWR+	Camera power input positive terminal	12~24V
	2	White	GPI1+/TRIG_IN+	GPI1 positive end/Trigger input positive end	Default to trigger input
	3	Brown	GPO2+	GPO2 output positive end	
	4	Green	GPO1+/STRB_OUT+	GPO1 positive end/Positive end of flash output	Default to flash output
	5	Yellow	GPO1-/STRB_OUT-/TRIG_IN-	GPO1 negative end/Flash output negative terminal/Trigger input negative terminal	GPIO common negative terminal
	6	Black	PWRGND	Camera power input negative terminal	

-T Line Sequence Definition4



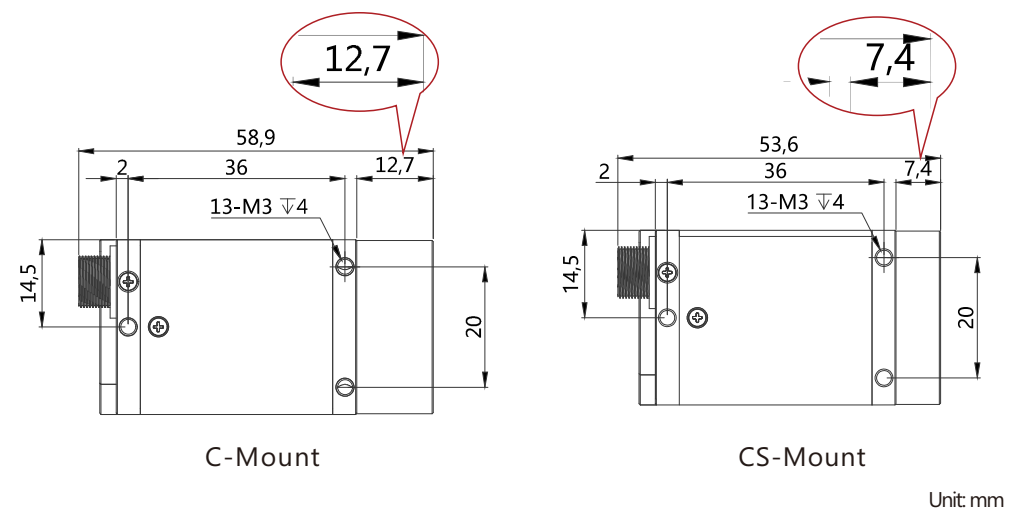
Port	Pin number	Line color	Signal name	Signal name	Remarks
Port A	1	Green	GPO1+/STRB_OUT+	GPO1positive end/Positive end of flash	Default to flash output
	2	Yellow	GPO1-/STRB_OUT-	GPO1negative end/Flash output negative	Default to flash output
	3	White	GPI1+/TRIG_IN+	GPI1positive end/Trigger input positive end	Default to trigger input
	4	Brown	GPI1-/TRIG_IN-	GPI1Negative end/Trigger input negative terminal	Default to trigger input
	5	Red	PWR+	Camera power input positive terminal	12~24V
	6	Black	PWRGND	Camera power input negative terminal	

-TPO、-TPO-C Line Sequence Definition5

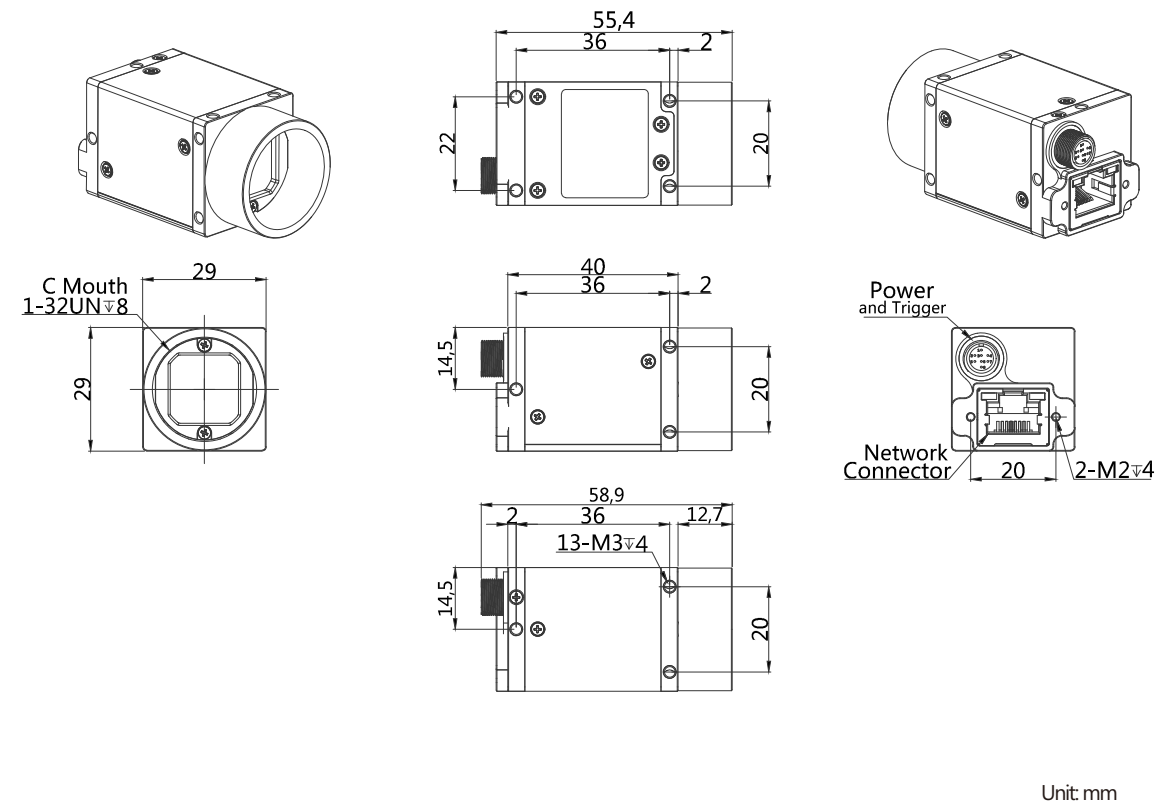


Port	Pin number	Line color	Signal name	Signal description	Remarks
Port A	1	Green	GPO1+/STRB_OUT+	GPO1 positive end/Positive end of flash output	Default to flash output
	2	Yellow	GPO1-/STRB_OUT-	negativeterminalGPO1negative end/Flash output	Default to flash output
	3	White	GPI1+/TRIG_IN+	GPI1positive end/Trigger input positive end	Default to trigger input
	4	Brown	GPI1-/TRIG_IN-	GPI1 negative end/Trigger input negative terminal	Default to trigger input
	5	Red	PWR+	Camera power input positive terminal	12~24V
	6	Black	PWRGND	Camera power input negative terminal	
Port B	1	Green	GPO2+	GPO2 positive end output	
	2	Yellow	GPO3+	GPO3 positive end output	
	3	White	GPO4+	GPO4 positive end output	
	4	Brown	GPI2+	GPI2 positive input	
	5	Red	GPI3+	GPI3 positive input	
	6	Black	GPIO_COM	GPIO common negative terminal	

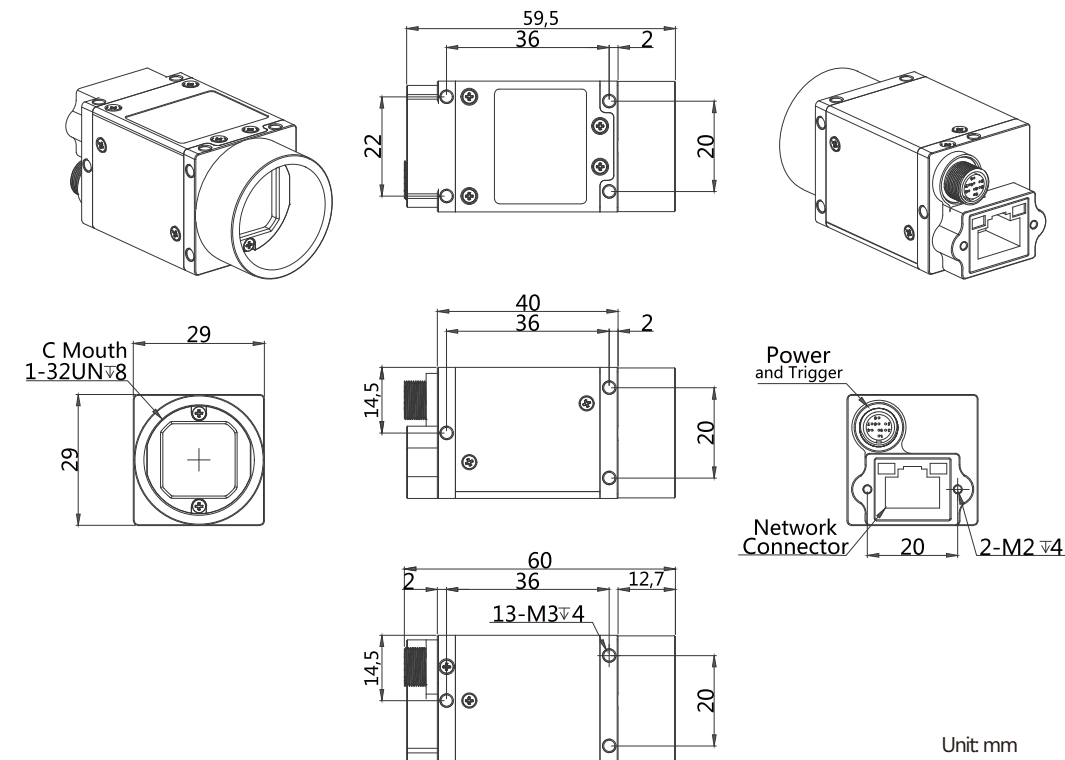
C-Mount/CS-Mount Camera Drawing



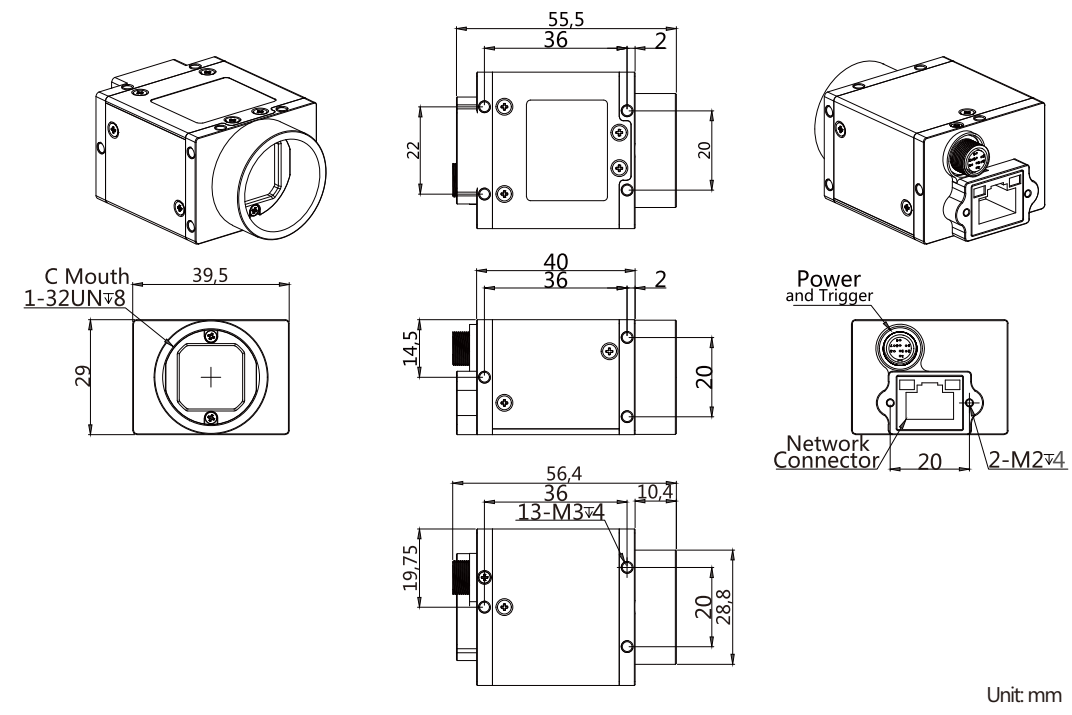
GE Series Dimension Drawing



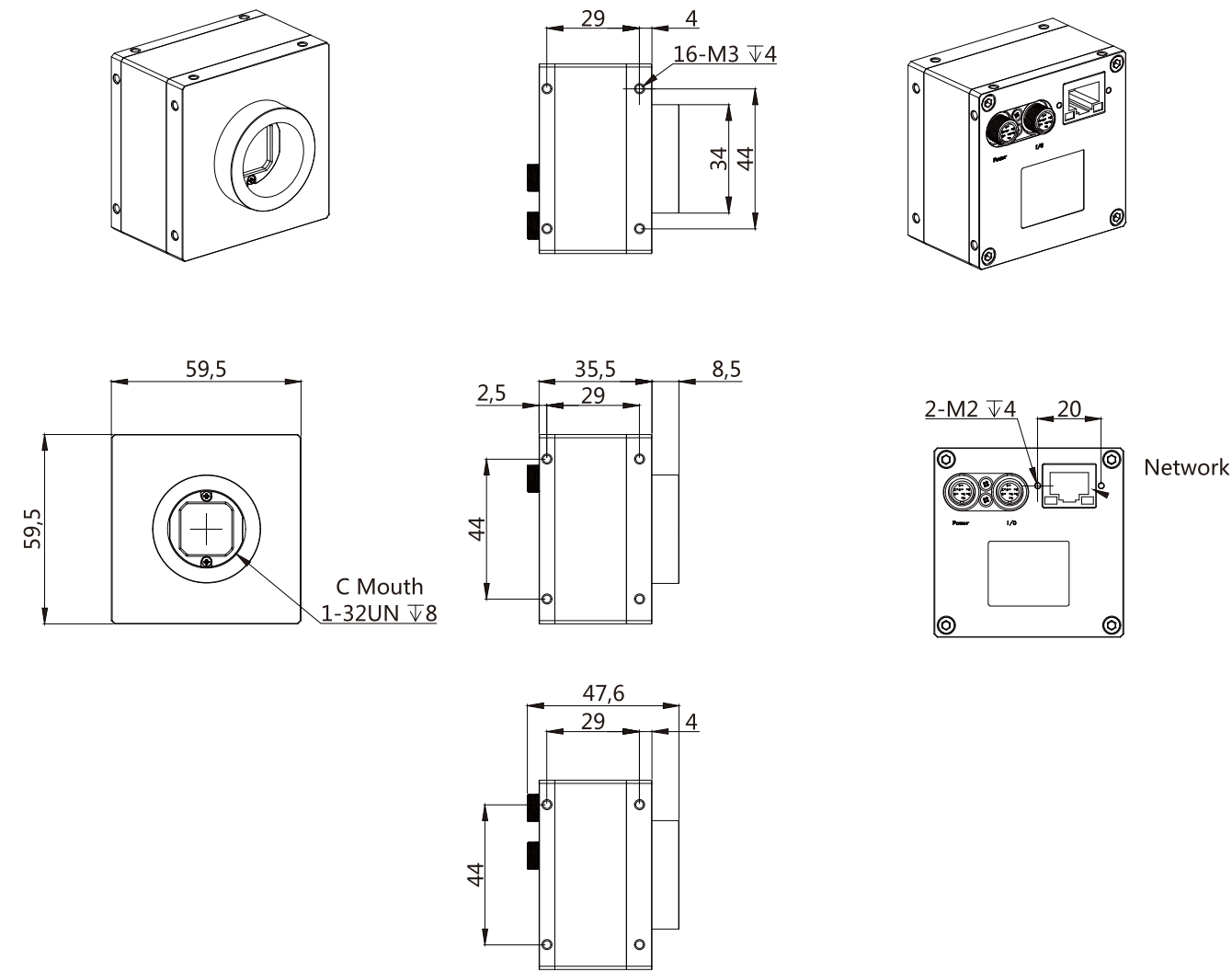
GED Series Dimension Drawing



GEF Series Dimension Drawing

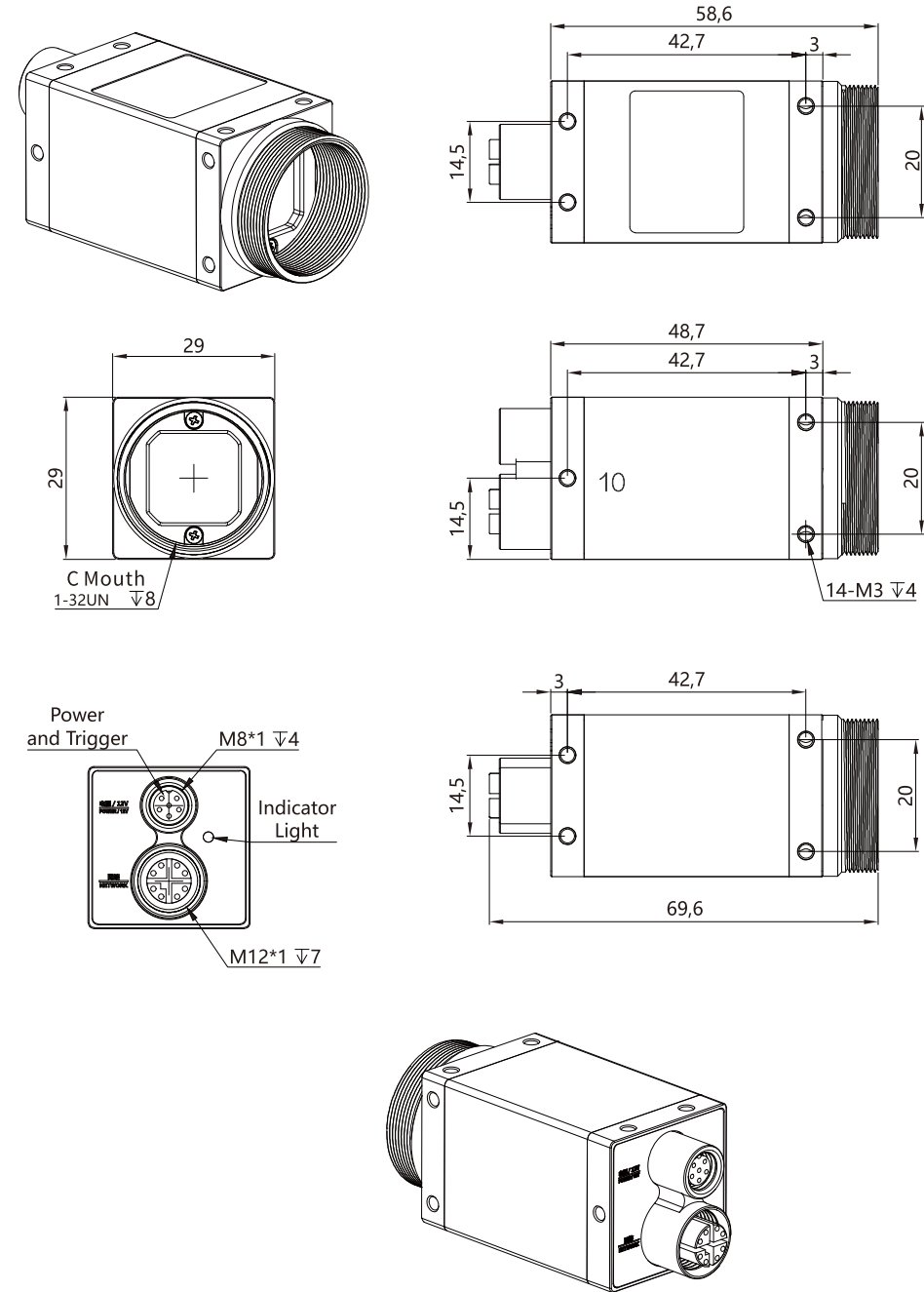


GEC Series Dimension Drawing



Unit: mm

Waterproof Series Dimension Drawing



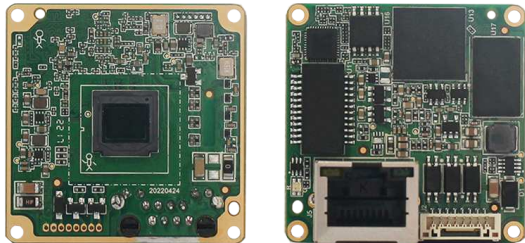
Unit: mm

GigE Board Level Camera / Module

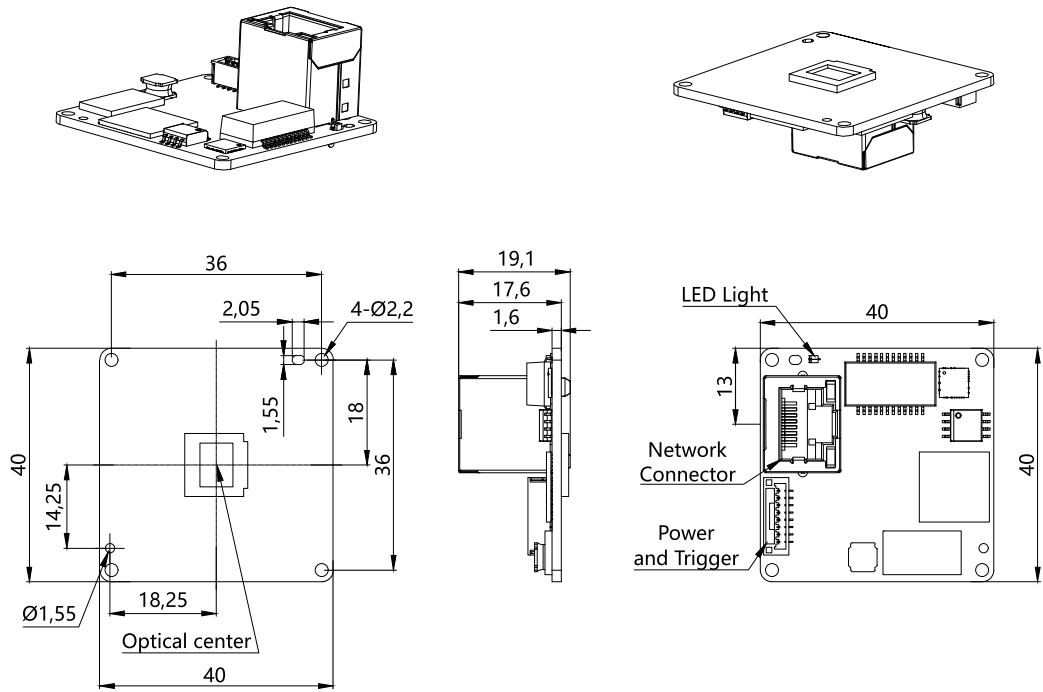
This series of Gigabit Ethernet camera module design is compatible with GigE Vision and genicam standards. Fully compatible with LabVIEW, Halcon, visionpro and other mainstreamVisual software. It is suitable for electrical semiconductors, automation, food and beverage, pharmaceutical packaging and image measurement.

Product Features

- Single board design, small size, ultra-thin size
- Embedded system integration is more convenient and integrated
- Support 16bit gray and 48bit color lossless format output
- Multiple options are available: single board /c interface /m12 interface
- The camera has its own programmable IO, which supports external trigger flash synchronous photography
- Support PC Linux system and arm Linux system, which can be integrated into embedded devices
- Compatible with vision protocol, seamlessly compatible with Halcon, visionproand LabVIEWOther visual software



Single Board Module Dimension Drawing



Unit: mm

Technical Parameter

Model	GEM34GC/M	GEM134GC/M	GEM2000C
Resolution@frame rate	640x480@387FPS	1280x1024@91FPS	5488×3672@6FPS
Shutter mode	Global Shutter	Global Shutter	Rolling Shutter
Effective pixel	0.3MP	1.3MP	20MP
Pixel size	4.8μm×4.8μm	4.8μm×4.8μm	2.4μm×2.4μm
Sensor	1/4" CMOS PYTHON	1/2" CMOS PYTHON	1" CMOS IMX183
Dynamic range	60dB	60dB	/
Ad Width	10	10	12
Video output format	Color : Bayer8/Bayer12 Mono : Mono8/Mono12	Color : Bayer8/Bayer12 Mono : Mono8/Mono12	Mono8/Mono12
Maximum gain (multiple)	16.5	16.5	22
Exposure time range (ms)	0.0040~524.3	0.008~1048	0.044~11534.3
Pixel bit depth	10bit	10bit	12bit

Model	GEM1201M	GEM1202C
Resolution@frame rate	4000x3000@9.75FPS	4000x3000@9.5FPS
Shutter mode	Rolling Shutter	Rolling Shutter
Effective pixel	12MP	12MP
Pixel size	1.85μm×1.85μm	1.65μm×1.65μm
Sensor	1/1.7" CMOS IMX226	1/2" CMOS AR1202
Dynamic range	66dB	48dB
Ad Width	12	10
Video output format	Mono8	Raw8
Maximum gain (multiple)	22	64
Exposure time range (ms)	0.001~1310.7	0.034~9792
Pixel bit depth	10bit	8bit

GigE Line Scan Industrial Camera Series

To meet the needs of line scan cameras for large-format inspection and high-speed assembly line inspection in industrial production, Huatengvision independently developed industrial line scan cameras HT-GEL21M, HT-GEL41M, HT-GELM44M/C; using domestic chips, GigE interface; 4K high resolution, up to 28KHz line frequency, greatly improving the production efficiency and detection accuracy in the field of industrial inspection; integrating a variety of ISP image algorithms and functions, Support line trigger, frame trigger and other external trigger modes; HT-GEL21M, HT-GEL41M and HT-GELM44M/C are economical products with high performance and low cost, which can enhance product competitiveness for customers! Suitable for printing, textile, railway, logistics, metallurgy, food, pharmaceuticals, material sorting, etc.

Product Features

- The line frequency can reach up to 48K, the software can be set freely, the color line scan camera supports RGB three-line true color output
- Support Sensor hardware 2-line SUM, doubles the brightness and reduces the requirements for light sources
- Support multiple trigger modes such as encoder synchronous trigger, photoelectric sensor and encoder combination synchronous trigger
- Support the frequency division and multiplication of the encoder signal, and accurately match the motion platform
- Hardware ISP supports Gamma, contrast, lens shadow correction, etc.
- Gigabit Ethernet port, the maximum transmission distance is 100 meters



Product Selection Table

Product Selection table	Effective Pixel	Sensor Type	Shutter Mode	Pixel Size	Line Frequency	Target size	TDI	Minimum Exposure	Sensor Model	Colour
HT-GEL21M	2048x2	CMOS	Global shutter	14μm	48K	28.6mm	/	0.003ms	Huatengvision	Mono
HT-GEL41M	4096x2	CMOS	Global shutter	7μm	28K	28.6mm	/	0.003ms	Huatengvision	Mono
HT-GELM41M	4096x2	CMOS	Global shutter	7μm	28K	28.6mm	/	0.003ms	Huatengvision	Mono
HT-GELM44C	4096x6	CMOS	Global shutter	7μm	9K	28.6mm	/	0.003ms	vision	Colour
HT-GELM44M	4096x8	CMOS	Global shutter	7μm	28K	28.6mm	4	0.003ms	Huatengvision	Mono

General parameters

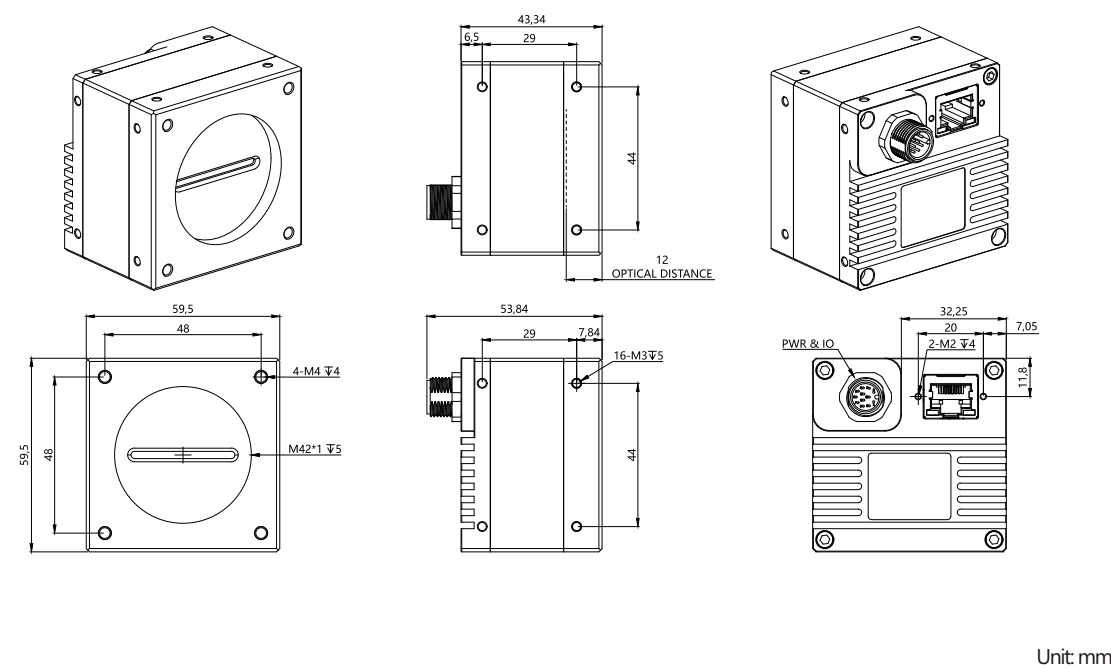
Programming language pack	C/C++/C#/VB6/VB. NET/Delphi/BCB/Python/Java
Operating system	WIINXP, WIN7/8/10 32@64 bit system, Linux and ARM Linux driver, Android platform driver, MAC OS system
Driver	Directshow component Halcon special component Labview special driver OCX component TWAIN component
Data interface	10G copper cable 10GBase-T, compatible with 100M/1G/2.5G/5G
Vision Standard Agreement	GigE Vision V1.2、GenIcam
Filter	The black-and-white camera is equipped with double-sided ar antireflection film as standard, and the color camera is equipped with 650nm infrared cut-off filter as standard
Working temperature @humidity	0~50℃ @ 20%~80%(No condensation)
Storage temperature @humidity	-30~60℃ @ 20%~95%(No condensation)
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

Technical parameter

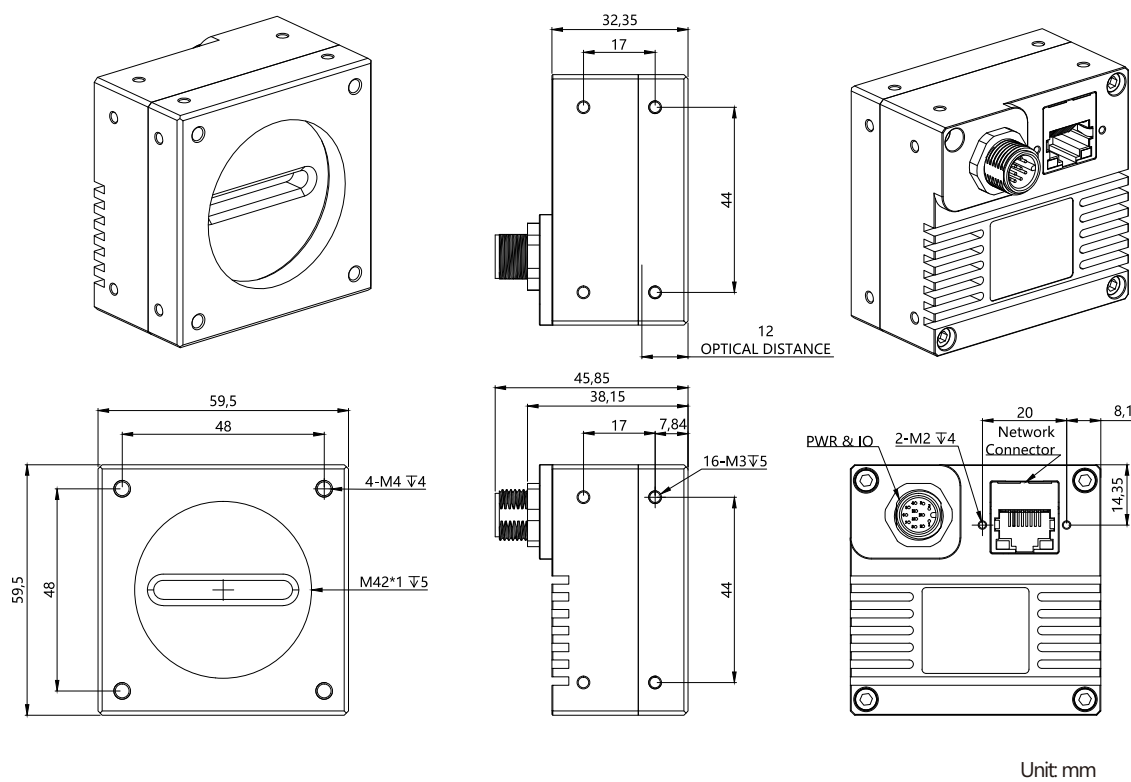
Model Parameter	HT-GEL21M	HT-GEL41M
Resolution	2048x2	4096x2
Frame rate	48K	28K
Pixel size	14μm×7μm	7μm×7μm
Pixel bit depth	8bit	8bit
Acquisition mode	Continuous/soft trigger/hard trigger	Continuous/soft trigger/hard trigger
Maximum gain (multiple)	8	8
Exposure time range (ms)	0.003~5	0.003~5
Frame buffer	128M Bytes	128M Bytes
User-defined data area	2K Bytes	2K Bytes
Video output format	Mono8	Mono8
TDI	/	/
Sensor Size	28.6mm	28.6mm
Lens Mount	M42×1.0(Flange distance12mm)	M42×1.0(Flange distance12mm)
Power supply	12~24V	12~24V
Power	<6W	<6W
Dimensions	59.5x59.5x43.3mm(Without back shell interface)	59.5x59.5x43.3mm(Without back shell interface)
Weight	<500g	<500g

Model Parameter	HT-GELM41M	HT-GELM44C/M
Resolution	4096x2	Color:4096x6 Mono:4096x8
Frame rate	28K	Color: 9K Mono:28K
Pixel size	7μm×7μm	
Color	Mono	Color / Mono
Data format	Mono	RGB/Mono
Sensor size	28.6mm(Photosensitive region)	
Lens Mount	M42×1.0, Flange distance12mm	
TDI	/	4
Exposure time range (ms)	0.003~5	
Dimensions	59.5x59.5x32.35mm (Without back shell interface)	
Data bit width	8bits (camera output to the user data width)	
Transmission mode	GigE Vision vl.2、GenIcam	
Transmission interface	Gigabit network	
Power	<6W	
Power supply	12~24V	
Weight	<500g	

GEL Dimension Drawing



GELM Dimension Drawing



10 Gigabit Network Array Industrial Camera-General Series

With the widely application of machine vision in industries, high-speed and high-precision visual inspection has a higher requirements on the resolution and frame rate of industrial cameras. In order to meet the market demand, HuatengVision launched 10GigE 8K line scan industrial camera to adapt to various line scan cameras scenario application requirements. It is widely used in printing, textile, railway, logistics, metallurgy, food, pharmaceutical, material sorting, etc.

Product Features

- Support flat-field correction function
- With multi-exposure automatic switching function
- Gain and Offset programmable settings are supported
- The interface is matched with the capture card, which greatly reduces the cost
- Adopting standard 10 Gigabit network interface design, and stable communication can be realized by super six network cables
- Compared with the CameraLink interface with a capture card, the cost is greatly reduced
- The farthest transmission distance can reach 100 meters, and the industrial field wiring is no longer restricted and backward compatible with Gigabit Ethernet
- The effective bandwidth is 1200MByte, which is 10 times that of dry meganet, which greatly reduces the image transmission time and delay
- Support GigEVision. GenICam standard and the same SDK as Gigabit camera, shorten customer development cycle



Product Selection Table

Model Number	Resolution	Sensor type	Shutter method	Pixel size	Line frequency	Target size	TDI	Minimum exposure	Sensor model	Colour
HT-XGL83C	8192×3(RGB)	CMOS	Global	7μm	50KHz	57mm	/	0.004ms	Huatengvision	Color
HT-XGL83M	8192×2	CMOS	Global	7μm	105KHz	57mm	2	0.004ms	Huatengvision	Mono
HT-XGL163C	16384×2(Bayer)	CMOS	Global	3.5μm	72KHz	57mm	/	0.004ms	Huatengvision	Color
HT-XGL163M	16384×2	CMOS	Global	3.5μm	72KHz	57mm	2	0.004ms	Huatengvision	Mono
HT-XGLC83C	8192×3	CMOS	Global	7μm	50KHz	57mm	/	0.004ms	Huatengvision	Color
HT-XGLC83M	8192×2	CMOS	Global	7μm	105KHz	57mm	2	0.004ms	Huatengvision	Mono
HT-XGLC83BM	8192×4	CMOS	Global	7μm	52KHz	57mm	4	0.004ms	Huatengvision	Mono
HT-XGL164C	16384×6	CMOS	Global	5μm	25KHz	81.92mm	2	0.004ms	Huatengvision	Color
HT-XGL164M	16384×1	CMOS	Global	5μm	72KHz	81.92mm	/	0.004ms	Huatengvision	Mono

Technical Parameter

Model Parameter	HT-XGL83M/C	HT-XGL163M/C	HT-XGL164M/C
Color specification	Mono/color	Mono/color	Mono/color
Data format	Mono/RGB	Mono/Bayer	Mono/Bayer
Line frequency	Mono: 106k(8bit)/70k(12bit); Color: 50k	Color:72KHz Mono:72k (8bit) 48k (12bit)	Mono:8bit (72K) Color:8bit (25K)
Resolution ratio	8192×2/8192×3(RGB)	16384×2/16384×2(Bayer)	16384×1/16384×6
Sensor size	57mm(photosensitive area)	57mm(photosensitive area)	81.92mm(photosensitive area)
TDI	Mono:2TDI		Mono:2TDI Color:2TDI
Lens interface	M72×0.75, flange distance 12mm		M90×1.0,flange distance 12mm
Exterior dimensions	80×80×69.29mm(excluding lens holder and rear shell interface)		110×110×69.26mm (excluding lens holder and rear shell interface)
AD width	12bit (refers to the width supported by sensor)	12bit (refers to the width supported by sensor)	12bit (refers to the width supported by sensor)
Pixel size	7μm×7μm	3.5μm×3.5μm	5μm×5μm
Data bit width	8/12bits(width of data output by camera to user)		
Dynamic range	65db		
Transmission mode	GigE Vision v1.2、GenIcam		
Transmission interface	10 Gigabit network port (10 gigabit copper cable 10GBase-T, compatible with 100M/1G/25/5G)		
Power consumption	Color<15W/Mono<10W	<11W	<15W
Working temperature	0~50℃		
Working humidity	20~80%(without condensation)		
Storage temperature	-30~60℃		
Storage humidity	20~95%(without condensation)		
Power supply	12V± 10%		
Weight	<800g	<1300g	<1500g

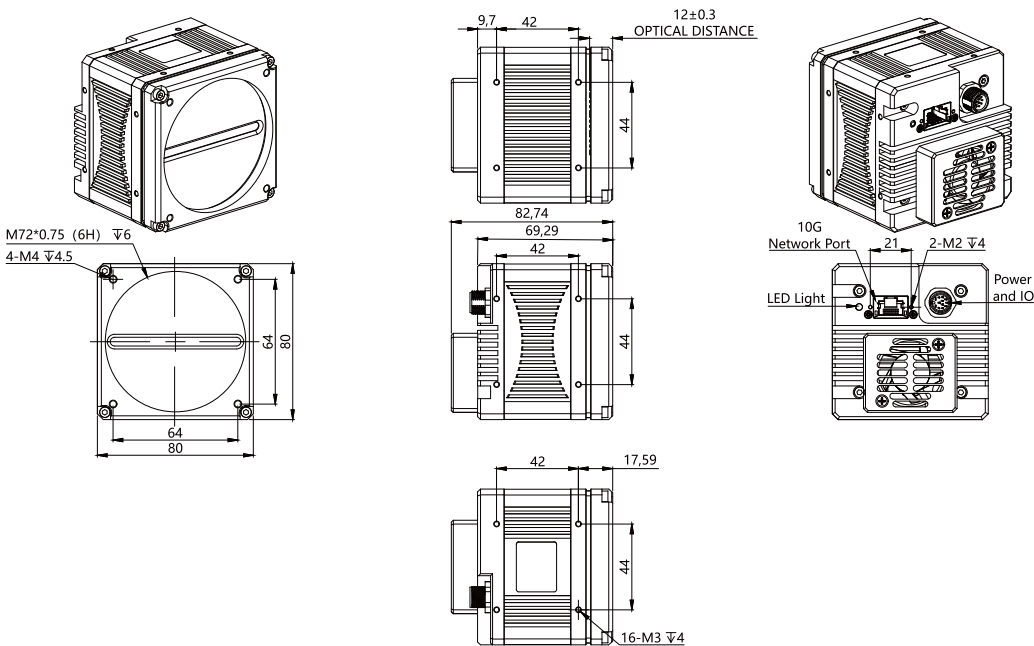
Technical Parameter

Specifications	Parameter		Specifications	Parameter	
Model	HT-XGLC83C/M	HT-XGLC83BM	Frame buffer	1GB	
Pixel size	7μm×7μm		Video output format	Mono:Mono8/12 Color:RGB8	Monø Mono8/12
Pixel bit depth	8/12bit		Lens Mount	M72×0.75,Flange distance 12mm	
Line frequency	105KHz	52KHz	Power supply	12V± 10%	
Resolution ratio	8192×2/8192×3(RGB)	8192×4	Power	<10W	
Maximum gain (multiple)	4		Overall dimension	80×80×56.9mm(excluding lens holder and rear shell interface)	
Exposure time range (ms)	0.002~5.0	0.004~5.0	Weight	<800g	

General parameters

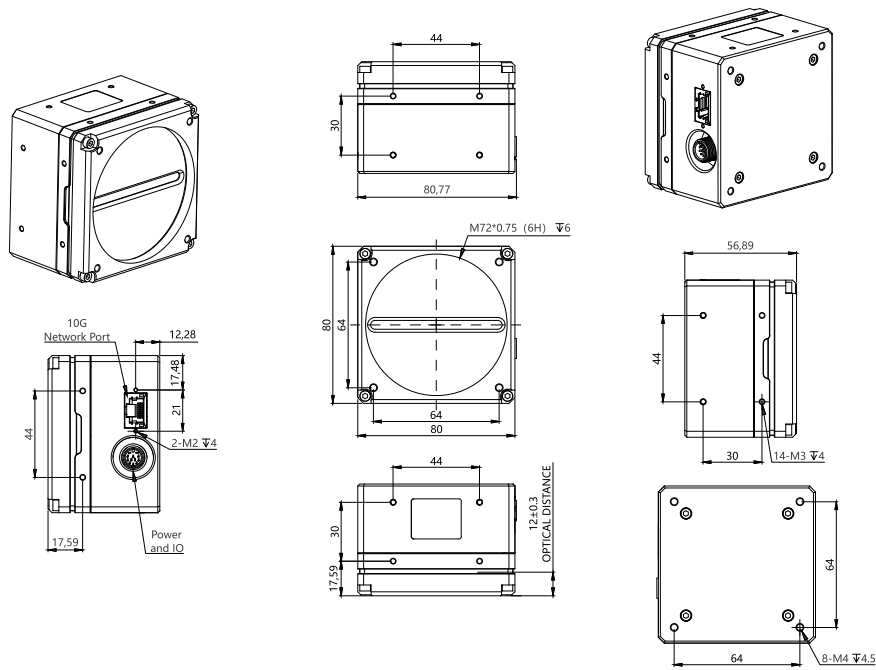
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Operating system	WIINXP, WIN7/8/10 32@64 bit system, Linux and ARM Linux driver, Android platform driver, MAC OS system
Driver	Directshow component Halcon special component Labview special driver OCX component TWAIN component
Data interface	10G copper cable 10GBase-T, compatible with 100M/1G/2.5G/5G
Vision Standard Agreement	GigE Vision V1.2、GenIcam
Filter	Black and white cameras come standard with double-sided AR antireflection film, and color cameras come standard with 650nm infrared cut filter
Working temperature @humidity	0~50℃ @ 20%~80%(No condensation)
Storage temperature @humidity	-30~60℃ @ 20%~95%(No condensation)
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

XGL83&XGL163 Dimension Drawing



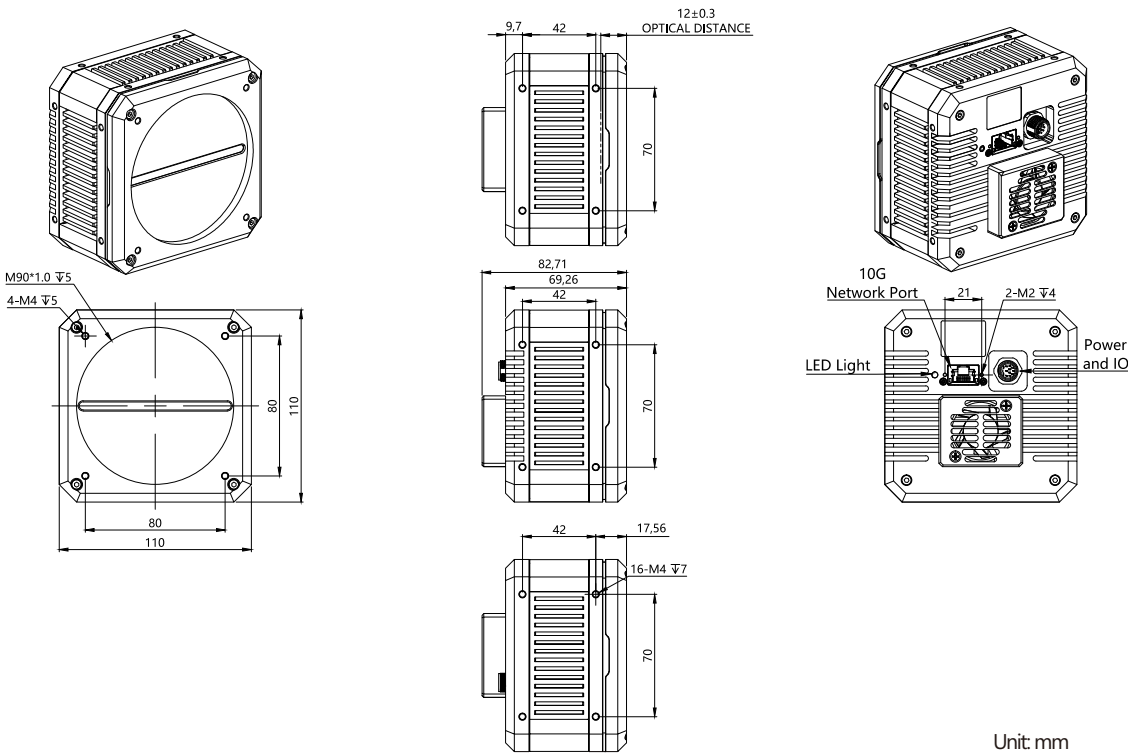
Unit: mm

XGLC Dimension Drawing



Unit: mm

XGL164 Dimension Drawing



Unit: mm

10 Gigabit Network Array Industrial Camera -256 TDI Series

At the end of 2022, Huatengvision was the first in China to launch a 256-class TDI 10-gigabit network array industrial camera, which made up for the domestic gap. This series adopts domestic linear array chips. At present, there are altogether three specifications in this series: HT-XGL42M-T2/HT-XGL82M-T2/HT-XGL162M-T2. The line pixels are 4096/8192/16384 respectively, and the highest line frequency can reach 250K/144K/72K, meeting the requirements of customers for high speed and stability.



Product Features

- 5um pixel, 10 Gigabit network interface
 - 26-grade TDI, reducing the demand of light source
 - Sensor line frequency up to 250k/200k(sensor output pixel width 10bits/12bits)
 - Support 256-line area array focusing mode
- Gain and Offset programmable settings are supported, and flat-field correction is supported.
 - Support encoder synchronous trigger, photoelectric sensor and encoder combination multiple trigger modes

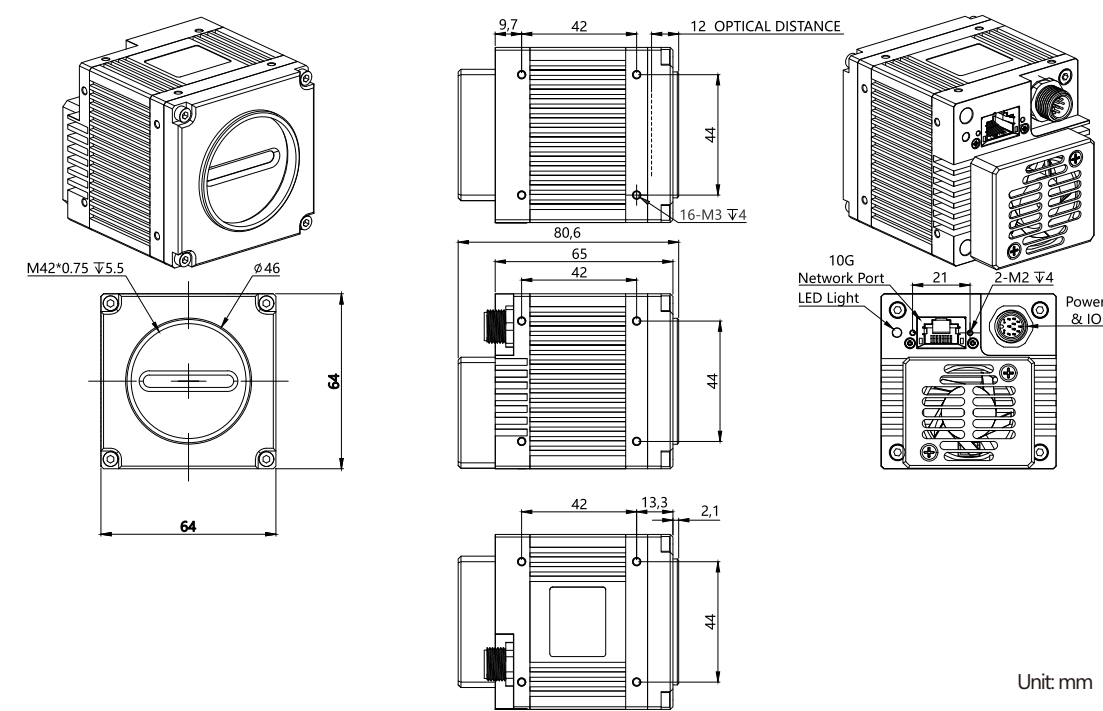
Product Selection Table

Model Number	Resolution	Sensor type	Shutter method	Pixel size	Line frequency	Target size	Minimal exposure	TDI	Sensor model	Colour
HT-XGL42M	4096×256	CMOS-TDI	Global	5 μm	250/200KHZ	21.177mm	0.004ms	256	Huatengvision	Mono
HT-XGL82M	8192×256	CMOS-TDI	Global	5 μm	145/96KHZ	41.673mm	0.004ms	256	Huatengvision	Mono
HT-XGL162M	16384×256	CMOS-TDI	Global	5 μm	72/48KHZ	82.666mm	0.004ms	256	Huatengvision	Mono

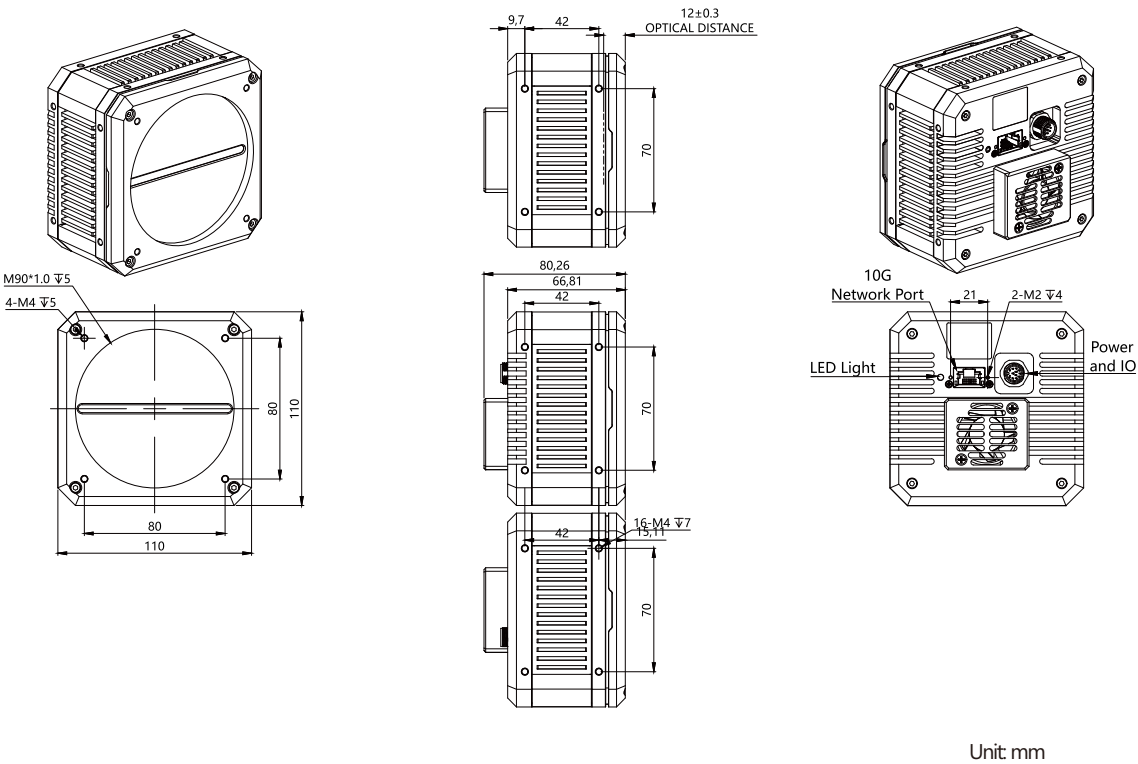
Technical Parameter

Model Parameter	HT-XGL42M	HT-XGL82M	HT-XGL162M
Resolution	4096×256	8192×256	16384×256
Sensor size	21.177mm (photosensitive area)	41.637mm(photosensitive area)	82.666mm(photosensitive area)
Lens interface	M42×0.75, flange distance 12mm	M58×0.75, (flange distance 12mm)	M90×1.0,(flange distance 12mm)
Exterior dimensions	64×64×65mm(excluding lens holder and rear shell interface)	80×80×66.24mm (excluding lens holder and rear shell interface)	110×110×66.81mm(excluding lens holder and rear shell interface)
weight	<550g	<800g	<1500g
Camera line frequency	250k(8bit)/200k(12bit)	145k(8bit)/96k(12bit)	72k(8bit)/48k(12bit)
Sensor line frequency	250 k/200 k (sensor output pixel width 10bits/12bits)		
Ad width	12bit (refers to the width supported by sensor)		
Pixel size	5um X5um		
Data bit width	8/12bits (camera output data width to user)		
Data format	Mono		
Color specification	Mono		
TDI	256TDI		
Dynamic range	low sensitivity mode:80 DN/nJ/cm2 @530nm High sensitivity mode:340 DN/nJ/cm2 @530nm		
Sensitivity	340DN/nJ/cm2		
Transmission mode	GigEVision v1.2xGenIcam		
Transmission interface	10 gigabit network port (10 gigabit copper cable 10GBase-T, compatible with 100M/1G/2.5/5G)		
Colour	Mono		
Power	<12W		
Working temperature	0-50℃		
Working humidity	20-80%(without condensation)		
Storage temperature	-30-60℃		
Storage humidity	20-95%(without condensation)		
Power supply	12V± 10%		

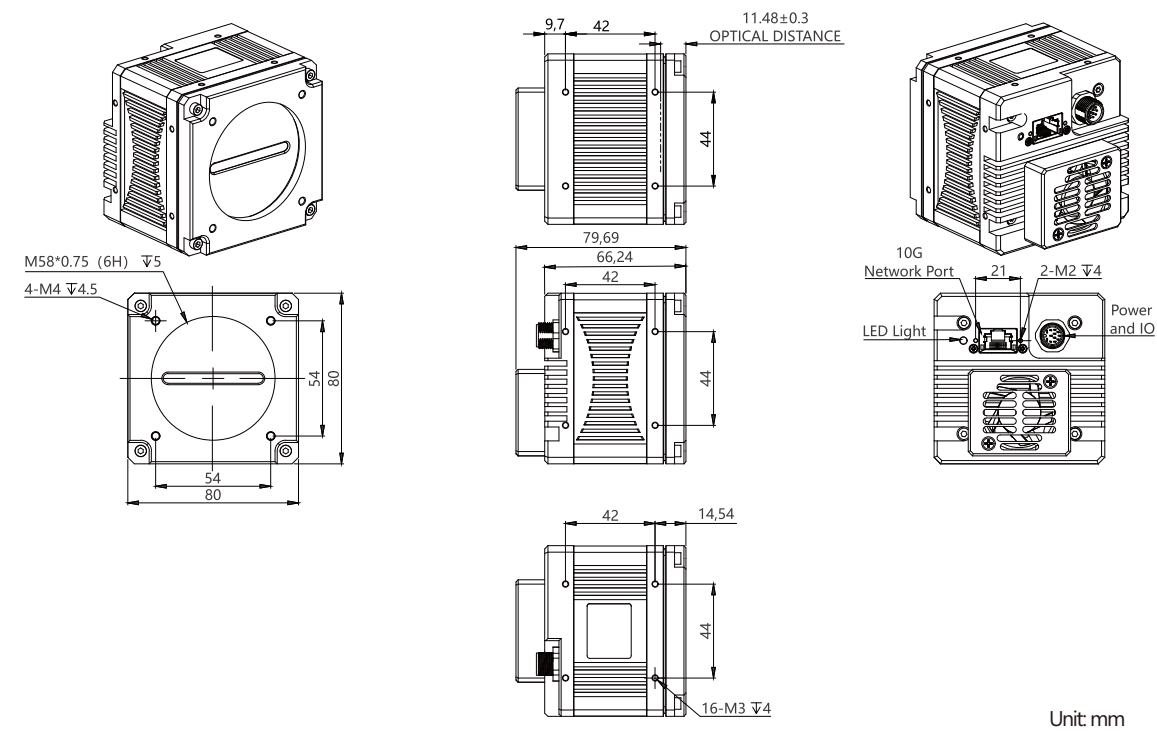
XGL42M Dimension Drawing



XGL162M Dimension Drawing



XGL82M Dimension Drawing



10GigE Area Scan Industrial Camera

With the widely application of machine vision in industries, high-speed and high-precision visual inspection has a higher requirements on the resolution and frame rate of industrial cameras. HuatengVision launched a number of new products, includes 10Gigabit network camera with 0.5MP-65MP pixels,to meet customers' requirements for high frame rate, high resolution requirements . It is widely used in PCB, screen detection, lithium battery and railway.

Product Features

- Reserve a fan interface to effectively control the camera temperature
- 0.5MP to 65 MPpixels with a maximum frequency of 1594FPS
- Adopting standard 10 Gigabit network interface design, and stable communication can be realized by super six network cables
- Compared with the CameraLink interface with a capture card, the cost is greatly reduced
- The farthest transmission distance can reach 100 meters, and the industrial field wiring is no longer restricted and backward compatible with Gigabit Ethernet
- The effective bandwidth is 1200MByte, which is 10 times that of the gigabit network, which greatly reduces the image transmission time and delay
- Support GigEVision. GenICam standard and the same SDK as the dry network camera, shorten the customer development cycle



Product Selection Table

Model Number	Pixel	Sensor type	Shutter method	Maximum resolution	Pixel size	Frame rate (FPS)	Target size	Minimum exposure	Sensor model	Colour
HT-XG51GC/M	0.51MP	CMOS	Global	816X624	9.0µm	1594	1/1.7"	0.0008ms	IMX426	Color/Mono
HT-XG170GC/M	1.7MP	CMOS	Global	1600X1100	9.0µm	662	1.1"	0.001ms	IMX425	Color/Mono
HT-XG280GC/M	2.8MP	CMOS	Global	1936X1464	4.5µm	409	2/3"	0.001ms	IMX421	Color/Mono
HT-XG402GC/M	4MP	CMOS	Global	2048X2048	5.5µm	176	1"	0.0026ms	CMV4000	Color/Mono
HT-XG503GC/M	5MP	CMOS	Global	2592X2160	2.5µm	120	1/2"	0.003ms	GMAX2505	Color/Mono
HT-XG903GC/M	9MP	CMOS	Global	4208X2160	2.5µm	121	2/3"	0.003ms	GMAX2509	Color/Mono
HT-XG1205GC/M	12MP	CMOS	Global	4096X3072	3.2µm	90	1"	0.003ms	XGS12000	Color/Mono
HT-XG1803GC/M	18MP	CMOS	Global	4512X4096	2.5µm	64	1"	0.003ms	GMAX2518	Color/Mono
HT-XG2500GC/M	25MP	CMOS	Global	5120X5120	2.5µm	42	1.1"	0.004ms	GMAX0505	Color/Mono
HT-XG2600C/M	26MP	CMOS	Rolling	6240X4168	3.76µm	45	1.8"	0.005ms	IMX571	Color/Mono
HT-XG3100GC/M	31MP	CMOS	Global	6480X4860	3.45µm	35	27.9MM (APS-C)	0.0056ms	IMX342	Color/Mono
HT-XG4701C/M	47MP	CMOS	Rolling	8240X5628	2.315µm	24	1.4"	0.007ms	IMX492	Color/Mono
HT-XG6500GC/M	65MP	CMOS	Global	9344X7000	3.2µm	15.5	29.9mmx22.4mm	0.015ms	GMAX3265	Color/Mono
HT-XGC2500GC/M	25MP	CMOS	Global	5120X5120	2.5µm	42	1.1"	0.004ms	GMAX0505	Color/Mono

Technical Parameters

Model	HT-XG51GC/M	HT-XG170GC/M	HT-XG280GC/M	HT-XG402GC/M
Resolution@ frame rate	816X624@1594FPS	1600X1100@650FPS	1936X1464@408FPS	2048X2048@172FPS
Pixel size	9.0µmX9.0µm	9.0µmX9.0µm	4.5µmX4.5µm	5.5µmX5.5µm
Pixel bit depth	12bit	12bit	12bit	12bit
Sensitivity	4010mV 1/30s	4010mV 1/30s	1677mV 1/30s	8.5 V/lux*s
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	125	125	125	16.5
Exposure time range (ms)	0.0008~838.9	0.0011~1153.4	0.001~1572.9	0.0026~1874.3
Frame buffer	256M Bytes			
User-defined data area	2K Bytes			
Video output format	Color : Bayer8/Bayer12 bit Mono : Mono8/Mono12 bit			Color : Bayer8/Bayer12 bit Mono : Mono8/Mono12 bit
GPIO	2 inputs and 2 outputs, 1 configurable input and output, support trigger and flash sync mode			
Lens Mount	C-Mount			
Power supply	12V			
Power	< 12W			
Dimensions	64x64x61.7mm (Without lens mount and rear shell interface)			
Weight	< 550g			

Model	HT-XG503GC/M	HT-XG903GC/M	HT-XG1205GC/M	HT-XG1803GC/M
Resolution@ frame rate	2592X2160@120FPS	4208X2160@121	4096X3072@90FPS	4512X4096@64
Pixel size	2.5µmX2.5µm	2.5µmX2.5µm	3.2µmX3.2µm	2.5µmX2.5µm
Pixel bit depth	8bit	8bit	12bit	8bit
Sensitivity	1.03x10 ⁻⁷ e ⁻ /((W/m ²) · s) @500nm	1.03x10 ⁻⁷ e ⁻ /((W/m ²) · s) @500nm	/	1.05x107e ⁻ / ((W/m ²) · s)
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	20	20	8	20
Exposure time range (ms)	0.003~393.2	0.003~393.2	0.003~196.6	0.003~393.2
Frame buffer	1GB			
User-defined data area	2K Bytes			
Video output format	Color : Bayer8/Bayer12 bit Mono : Mono8/Mono12 bit		Color : Bayer8/Bayer12 bit Mono : Mono8/Mono12 bit	Color : Bayer8/Bayer12 bit Mono : Mono8/Mono12 bit
GPIO	2 inputs and 2 outputs, 1 configurable input and output, support trigger and flash sync mode			
Lens Mount	C-Mount			
Power supply	12V			
Power	< 12W			
Dimensions	64x64x61.7mm (Without lens mount and rear shell interface)			
Weight	< 500g			

Technical Parameters

Model Parameter	HT-XG2500GC/M	HT-XG2600C/M	HT-XG3100GC/M
Resolution@ frame rate	5120×5120@42FPS	6240×4168@45FPS	6480×4860@35FPS
Pixel size	2.5μm×2.5μm	3.76μm×3.76μm	3.45μm×3.45μm
Pixel bit depth	10bit	12bit	12bit
Sensitivity	1.03×10 ⁷ e-/((W/m ²)-s) @500nm	5630mV 1/30s	915mV 1/30s
Acquisition mode	Continuous/soft trigger/hard trigger		
Maximum gain (multiple)	16.5	64	125
Exposure time range (ms)	0.004~1048.6	0.005~5242	0.0056~5872
Frame buffer	1GB	256M B	1GB
User-defined data area	2K Bytes		
Video output format	Color:Bayer GB 8/Bayer12bit Mono:Mono8/Mono12bit	Color:Bayer8/Bayer12bit Mono: Mono8/Mono12bit	Color:Bayer8/Bayer12bit Mono: Mono8/Mono12bit
GPIO	2 inputs and 2 outputs, 1 configurable input and output, support trigger and flash sync mode		
Lens Mount	C-Mount	M58 (F-port adapter ring are optional)	
Power supply	12V		
Power	<12W		
Dimensions	64×64×61.7mm (Without lens mount and rear shell interface)	64×64×59.5mm (Without lens mount and rear shell interface)	64×64×59.5mm (Without lens mount and rear shell interface)
Weight	<500g	<500g	

Model	HT-XG4701C/M	HT-XG6500GC/M
Resolution@ frame rate	8240×5628@24FPS	9344×7000@15.5FPS
Pixel size	2.315μm×2.315μm	3.2μm×3.2μm
Pixel bit depth	12bit	12bit
Sensitivity	87.69mV l/30s	1.03×10 ⁷ e ⁻ /((W/m ²).s) @500nm
Acquisition mode	Continuous/soft trigger/hard trigger	
Maximum gain (multiple)	22	6
Exposure time range (ms)	0.0072-5142.9	0.015-10000
Frame buffer	256M Bytes	
User-defined data area	2K Bytes	
Video output format	Color:Bayer8/Bayer12bit Mono:Mono8/Mono12bit	
GPIO	2 inputs and 2 outputs, 1 configurable input and output, support trigger and flash sync mode	
Lens Mount	M58(F-port adapter ring are optional)	
Power supply	12V	
Power	<12W	
Dimensions	64×64×59.5mm (Without lens mount and rear shell interface)	
Weight	<500g	

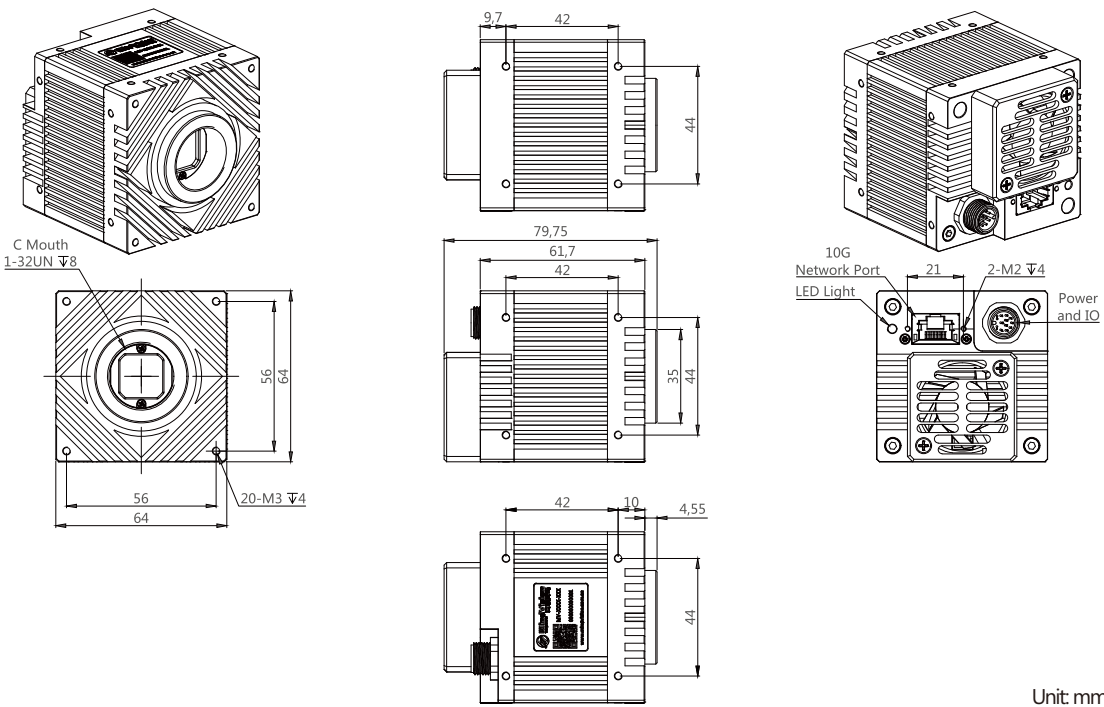
Technical Parameters

Model	HT-XGC2500GC/M
Resolution@ frame rate	5120×5120@42FPS
Pixel size	2.5μm×2.5μm
Pixel bit depth	8bit
Sensitivity	1.03×10 ⁷ e ⁻ /((W/m ²).s) @500nm
Acquisition mode	Continuous/soft trigger/hard trigger
Maximum gain (multiple)	16.5
Exposure time range (ms)	0.0040-1048.6
Frame buffer	1GB
User-defined data area	2KB
Video output format	Mono; Mono8/12
GPIO	2 inputs and 2 outputs, 1 configurable input and output, support trigger and flash sync mode
Lens Mount	M58, Flange pitch11.48(F-port adapter ring are optional)
Power supply	12V
Power	<12W
Dimensions	64×64×60.9mm (Without lens mount and rear shell interface)
Weight	<500g

General parameters

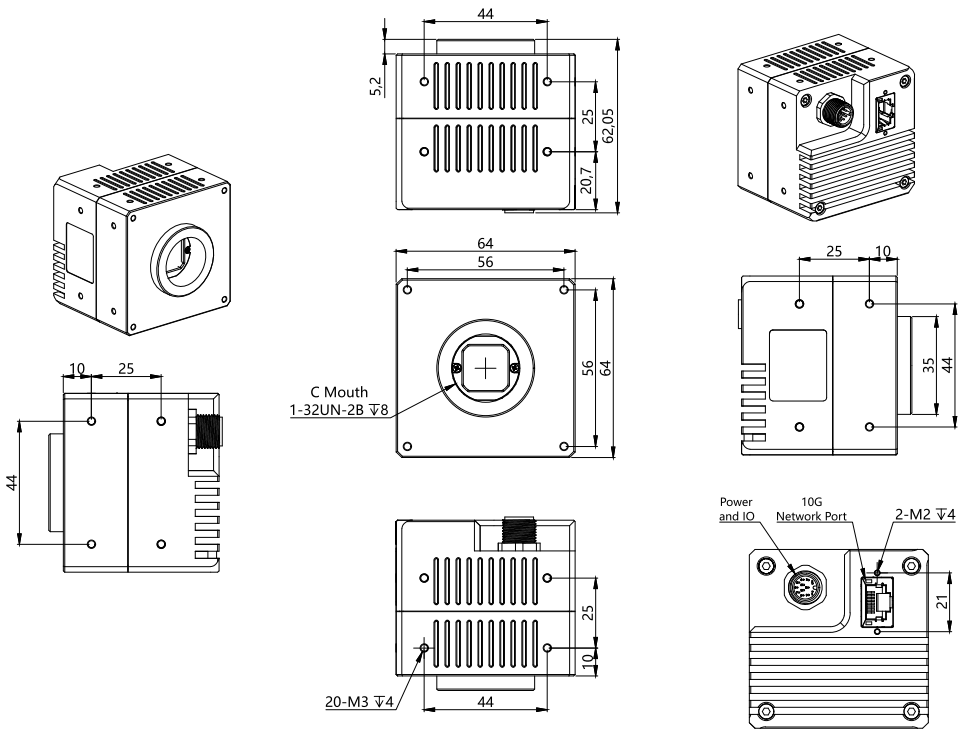
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Operating system	WIINXP, WIN7/8/10 32@64 bit system, Linux and ARM Linux driver, Android platform driver, MAC OS system
Driver	Directshow component Halcon special component Labview special driver OCX component TWAIN component
Data interface	10G copper cable 10GBase-T, compatible with 100M/1G/2.5G/5G
Vision Standard Agreement	GigE Vision V1.2、GenIcam
Filter	Black and white cameras come standard with double-sided AR antireflection film, and color cameras come standard with 650nm infrared cut filter
Working temperature @humidity	0~50℃ @ 20%~80%(No condensation)
Storage temperature @humidity	-30~60℃ @ 20%~95%(No condensation)
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

C-Mount Dimension Drawing



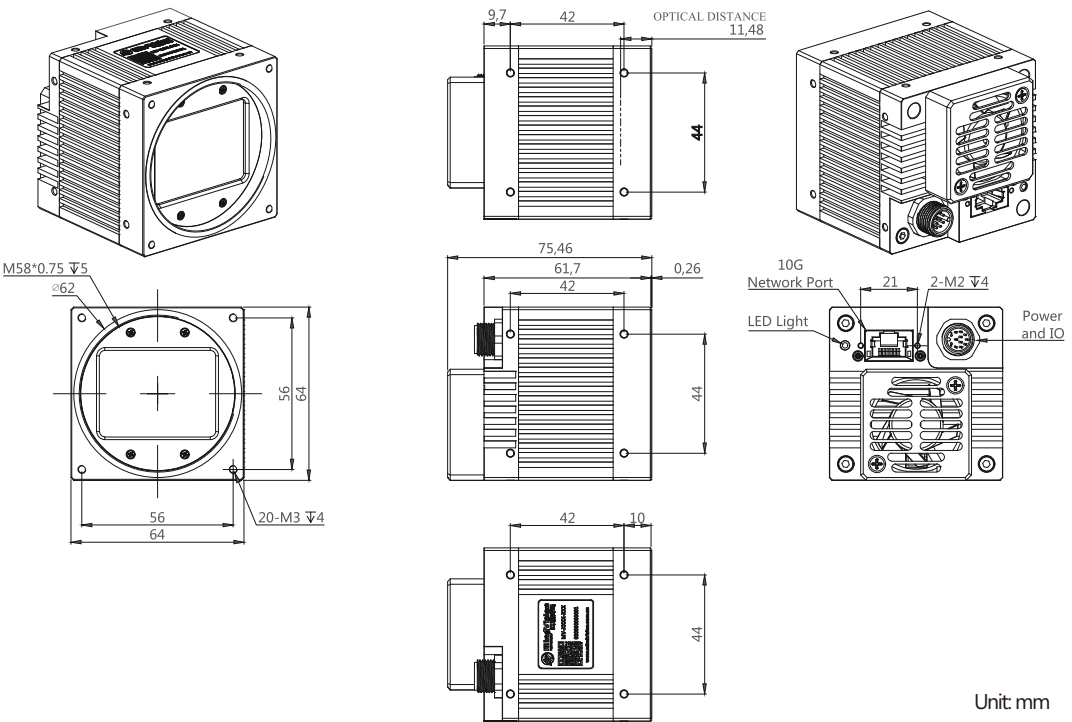
Unit: mm

C-Mount Dimension Drawing



Unit: mm

M58-Mount Dimension Drawing



Unit: mm

USB3.0 Industrial Camera

The transmission speed of USB3.0 is significantly higher than USB2.0 and GigE cameras. With 5Gbps transmission and bandwidth and high-speed sensor, The camera of 0.3MP and 800 fps can achieve a minimum transmission time of 1.2 ms, and a 20MP pixel camera can also reach 18fps; Due to the abundant transmission bandwidth, a computer can guarantee high frame rate compatible with multiple cameras; The USB3.0 series has a low CPU utilization. DMA transmission is adopted. The 5Gbps data transmission part hardly occupies CPU resources. It is widely used in electronic 3C, automatic semiconductor, logistics code scanning, intelligent transportation, biomedicine, packaging and printing, etc.



SUA Series



SUF Series

Product features

- The camera has its own programmable IO, which supports external trigger flash synchronous photography
- Support 16bit gray and 48bit color lossless format output
- 0.3MP to 20MP CMOS, multiple global shutter cameras are optional
- One computer can be connected to multiple cameras, and it can work stably for a long time without dropping frames
- Fully compatible with GigE camera SDK, seamless replacement
- It supports PC Linux system and ARM Linux system , and can be integrated into embedded devices
- Support Vision protocol, seamless compatible with Halcon, Vision Pro, Labview and other vision software

Product features

Applicable to electronic 3C, automated semiconductors, logistics code scanning, intelligent transportation, biomedical, packaging and printing, etc.

Product Selection Table

Product number	Pixels	Sensor type	Shutter method	Maximum resolution	Pixel size	Frame rate (FPS)	Target size	Minimum exposure	Sensor model	Colour
HT-SUA33GC/M	0.3MP	CMOS	Global	640×480	4.0μm	791	1/5.6"	0.0025ms	SmartSens	Color/Mono
HT-SUA34GC/M	0.3MP	CMOS	Global	640×480	4.8μm	454	1/4"	0.0045ms	PYTHON	Color/Mono
HT-SUA50GC/M	0.48MP	CMOS	Global	800×600	4.8μm	118	1/3.6"	0.0098ms	PYTHON480	Color/Mono
HT-SUA133GC/M	1.3MP	CMOS	Global	1280×1024	4.0μm	241.5	1/2.7"	0.004ms	SC130GS	Color/Mono
HT-SUA134GC/M	1.3MP	CMOS	Global	1280×1024	4.8μm	216.25	1/2"	0.0045ms	PYTHON	Color/Mono
HT-SUA40GC/M	0.4MP	CMOS	Global	720×540	6.9μm	523.75	1/2.9"	0.0032ms	IMX287	Color/Mono
HT-SUA300GC/M	3MP	CMOS	Global	2048×1536	3.45μm	55.6	1/1.8"	0.0114ms	IMX265	Color/Mono
HT-SUA505GC/M	5MP	CMOS	Global	2592×2048	3.2μm	42.75	2/3"	0.0113ms	XGS5000	Color/Mono
HT-SUA200GC/M	2MP	CMOS	Global	1600×1200	4.5μm	60	1/1.8"	0.0157ms	EV76C570	Color/Mono
HT-SUA202GC/M	2MP	CMOS	Global	1920×1200	4.8μm	165.5	2/3"	0.0039ms	PYTHON2000	Color/Mono
HT-SUA230GC/M	2.3MP	CMOS	Global	1920×1200	5.86μm	165.25	1/1.2"	0.0049ms	IMX174	Color/Mono
HT-SUA231GC/M	2.3MP	CMOS	Global	1920×1200	5.86μm	40	1/1.2"	0.0201ms	IMX249	Color/Mono
HT-SUA501GC/M	5MP	CMOS	Global	2448×2048	3.45μm	40	2/3"	0.018ms	IMX264	Color/Mono
HT-SUA502C/M	5MP	CMOS	Rolling	2592×1944	2.2μm	59	1/2.5"	0.0084ms	AR0522	Color/Mono
HT-SUA630C/M	6.3MP	CMOS	Rolling	3088×2064	2.4μm	59.5	1/1.8"	0.008ms	IMX178	Color/Mono
HT-SUA800C/M	8MP	CMOS	Rolling	3840×2160	2.0μm	45.75	1/1.8"	0.02ms	OEM	Color/Mono
HT-SUA1000C/M	10MP	CMOS	Rolling	3664×2748	1.67μm	8	1/2.3"	0.0545ms	MT9J003	Color/Mono
HT-SUA1201C/M	12MP	CMOS	Rolling	4000×3000	1.85μm	32	1/1.7"	0.0102ms	IMX226	Color/Mono
HT-SUA1600C/M	16MP	CMOS	Rolling	4608×3456	1.34μm	12	1/2.3"	0.0233ms	IMX206	Color/Mono
HT-SUA2000C/M	20MP	CMOS	Rolling	5488×3672	2.4μm	19.5	1"	0.0138ms	IMX183	Color/Mono
HT-SUF401GC/M	4MP	CMOS	Global	2048×2048	5.5μm	88	1"	0.0054ms	OEM	Color/Mono
HT-SUF890GC/M	8.9MP	CMOS	Global	4096×2160	3.45μm	32	1"	0.141ms	IMX267	Color/Mono
HT-SUF1200C/M	12MP	CMOS	Rolling	4080×3072	3.1μm	30.75	1"	0.0201ms	RJ5DY1BA0LT	Color/Mono
HT-SUF1200GC/M	12MP	CMOS	Global	4096×3000	3.45μm	23.5	1.1"	0.0141ms	IMX304	Color/Mono
HT-SUF1205GC/M	12MP	CMOS	Global	4096×3072	3.2μm	28.5	1"	0.0113ms	XGS12000	Color/Mono

Technical Parameters

Model Parameter	HT-SUA33GC/M	HT-SUA34GC/M	HT-SUA50GC/M	HT-SUA133GC/M	HT-SUA134GC/M
Resolution@ frame rate	640×480@791FPS	640×480@454FPS	800×600@118FPS	1280×1024@245FPS	1280×1024@211FPS
Pixel size	4.0μm×4.0μm	4.8μm×4.8μm	4.8μm×4.8μm	4.0μm×4.0μm	4.8μm×4.8μm
Pixel bit depth	8bit	8bit	10bit	8bit	8bit
Sensitivity	8 V/lux-s 540nm	7.3 V/lux-s 540nm	7.7 V/lux.s 550nm	8 V/Lux.s	7.3 V/lux-s 540nm
Acquisition mode	Continuous/soft trigger/hard trigger				
Maximum gain (multiple)	8	16.5	8	6	16.5
Exposure time range (ms)	0.0025~20.4	0.0045~584.4	0.0098~640.9	0.004~32.4	0.0045~584.4
Frame buffer	128M Bytes				
User-defined data area	2K Bytes				
Video output format	Color:Bayer8bit Mono:Mono8bit	Color:Bayer8bit Mono:Mono8bit	Color:Bayer8/Bayer10bit Mono:Mono8/Mono10bit	Color:Bayer8bit Mono:Mono8bit	Color:Bayer8bit Mono:Mono8bit
GPIO	1 optocoupler isolated input, 1 optocoupler isolated output, 1 non-isolated input and output				
Lens Mount	C-Mount (optional for CS port)				
Power supply	5V, USBbus power supply				
Power	<3W				
Dimensions	29×29×32.7mm(Without lens mount and rear shell interface)				
Weight	<75g				

Model Parameter	HT-SUA40GC/M	HT-SUA300GC/M	HT-SUA505GC/M
Resolution@ frame rate	720×540@523.75FPS	2048×1536@55.75FPS	2592×2048@42.75FPS
Pixel size	6.9μm×6.9μm	3.45μm×3.45μm	3.2μm×3.2μm
Pixel bit depth	12bit	12bit	12bit
Sensitivity	3660mV 1/30s	146mV 1/30s	/
Acquisition mode	Continuous/soft trigger/hard trigger		
Maximum gain (multiple)	32	64	8
Exposure time range (ms)	0.0032~1955.4	0.0114~11966.6	0.0113~740.3
Frame buffer	128M Bytes		
User-defined data area	2KB		
Video output format	Color:Bayer8/Bayer12bit Mono:Mono8/Mono12bit		Color:Bayer RG 8/Bayer12bit Mono:Mono8/Mono12bit
GPIO	1 optocoupler isolated input, 1 optocoupler isolated output, 1 non-isolated input and output		
Lens Mount	C-Mount (optional for CS port)		
Power supply	5V, USBbus power supply		
Power	<3W		
Dimensions	29×29×32.7mm(Without lens mount and rear shell interface)		
Weight	<75g		

Technical Parameters

Model Parameter	HT-SUA200GC/M	HT-SUA202GC/M	HT-SUA230GC/M	HT-SUA231GC/M
Resolution@ frame rate	1600×1200@60FPS	1920×1200@165FPS	1920×1200@165FPS	1920×1200@40FPS
Pixel size	4.5μm×4.5μm	4.8μm×4.8μm	5.86μm×5.86μm	5.86μm×5.86μm
Pixel bit depth	10bit	10bit	8bit	12bit
Sensitivity	7.4V/lux-s 3200K	7.5 V/lux.s 550nm	Color:1016mV 1/30s Mono:825mV 1/30s	Color:1016mV 1/30s Mono:825mV 1/30s
Acquisition mode	Continuous/soft trigger/hard triggerh			
Maximum gain (multiple)	8	8	50	250
Exposure time range (ms)	0.0157~95.8	Low speed:0.0039~256.1 High speed:0.0039~5	0.0049~10188.6	0.0201~42112.8
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color:Bayer BG 8/Bayer12bit Mono:Mono8/Mono12bit		Color:Bayer RG 8bit Mono:Mono8bit	Color:Bayer RG 8/Bayer12bit Mono:Mono8/Mono12bit
GPIO	1 optocoupler isolated input, 1 optocoupler isolated output, 1 non-isolated input and output			
Lens Mount	C-Mount (optional for CS port)			
Power supply	5V, USBbus power supply			
Power	<3W			
Dimensions	29×29×32.7mm(Without lens mount and rear shell interface)			
Weight	<75g			

Model Parameter	HT-SUA501GC/M	HT-SUA502C/M	HT-SUA630C/M	HT-SUA800C/M
Resolution@ frame rate	2448×2048@40	2592×1944@59	3088×2064@60	3840x2160@14.25
Pixel size	3.45μm×3.45μm	2.2μm×2.2μm	2.4μm×2.4μm	2μm×2μm
Pixel bit depth	12bit	10bit	8bit	12bit
Sensitivity	Color:1146mV 1/30s Mono:915mV 1/30s	Color:18.8 ke-/lux*sec Mono:36ke-/lux*sec	425mV 1/30s	2200mV/lux.s
Acquisition mode	Continuous/soft trigger/hard triggerh			
Maximum gain (multiple)	249.5	15	32	15
Exposure time range (ms)	0.0133~13924	0.0084~553.4	0.008~10284.8	0.02~655.4
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color:Bayer BG 8/Bayer12bit Mono:Mono8/Mono12bit	Color:Bayer GR 8 Mono:Mono8bit		Color:Bayer RG 8/Bayer12bit Mono:Mono8/Mono12bit
GPIO	1 optocoupler isolated input, 1 optocoupler isolated output, 1 non-isolated input and output			
Lens Mount	C-Mount (optional for CS port)			
Power supply	5V, USBbus power supply			
Power	<3W			
Dimensions	29×29×32.7mm (Without lens mount and rear shell interface)			
Weight	<75g			

Technical Parameters

Parameter \ Model	HT-SUA1000C/M	HT-SUA1201C/M	HT-SUA1600C/M	HT-SUA2000C/M
Resolution@ frame rate	3664×2748@8FPS	4000×3000@32FPS	4608×3456@12FPS	5488×3672@19.5FPS
Pixel size	1.67μm×1.67μm	1.85μm×1.85μm	1.34μm ×1.34μm	2.4μm×2.4μm
Pixel bit depth	12bit	8bit	12bit	8bit
Sensitivity	0.31V/lux-sec 550nm	250mV l/30s	142mV l/30s	Color: 462mV1/30s Mono: 388mVl/30s
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	8	22	16	22
Exposure time range (ms)	0.0545~951.7	0.0102~2665.1	0.0233~7741.4	0.0138~3608.7
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color: Bayer RG 8/Bayer12bit Mono: Mono8/Mono12bit	Color: Bayer8bit Mono: Mono8bit	Color: Bayer8/Bayer12bit Mono: Mono8/Mono12bit	Color: Mono8bit Mono: Mono8bit
GPIO	1 trigger input, 1 flash control output; 1 non-isolated GPIO input and output			
Lens Mount	C-Mount (optional for CS port)			
Power supply	5V, USBbus power supply			
Power	<3W			
Dimensions	29X29X32.7mm (Without lens mount and rear shell interface)			
Weight	<75g			

Parameter \ Model	HT-SUF401GC/M	HT-SUF890GC/M	HT-SUF1200C/M	HT-SUF1200GC/M
Resolution@ frame rate	2048×2048@88FPS	4096×2160@32FPS	4080×3072@30.5FPS	4096×3000@23FPS
Pixel size	5.5μm×5.5μm	3.45μm×3.45μm	3.1μm ×3.1μm	3.45μm×3.45μm
Pixel bit depth	8bit	12bit	8bit	12bit
Sensitivity	7V/lux*s 540nm	Color: 1146mV Mono:915mV	Color: 1420 mV/lxs @3200k Mono:2340 mV/lxs @3200k	Color: 1146mV Mono:915 mV
Acquisition mode	Continuous/soft trigger/hard trigger			
Maximum gain (multiple)	16.5	250	64	250
Exposure time range (ms)	0.0054~704.5	0.0141~29546.9	0.0201~2635.2	0.0141~2954.9
Frame buffer	128M Bytes			
User-defined data area	2K Bytes			
Video output format	Color: Bayer8bit Mono: Mono8bit	Color: Bayer GB 8bit Mono: Mono8bit		
GPIO	1 trigger input, 1 flash control output; 1 non-isolated GPIO input and output			
Lens Mount	C-Mount (optional for CS port)			
Power supply	5V, USB bus power supply			
Power	<2W			
Dimensions	29×39.5×32.7mm (Without lens mount and rear shell interface)			
Weight	<100g			

Technical Parameters

Parameter \ Model	HT-SUF1205GC/M
Resolution@ frame rate	4096×3072@28.5FPS
Pixel size	3.2μm×3.2μm
Pixel bit depth	8bit
Sensitivity	/
Acquisition mode	Continuous/soft trigger/hard trigger
Maximum gain (multiple)	8
Exposure time range (ms)	0.0113~740.3
Frame buffer	128M Bytes
User-defined data area	2K Bytes
Video output format	Color: Bayer GR 8/Bayer12bit Mono Mono8/Mono12bit
GPIO	1 trigger input, 1 flash control output; 1 non-isolated GPIO input and output
Lens Mount	C-Mount (optional for CS port)
Power supply	5V, USB bus power supply
Power	<2W
Dimensions	29X29X32.7mm (Without lens mount and rear shell interface)
Weight	<100g

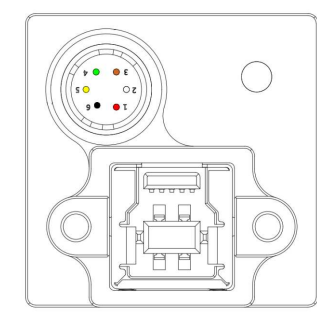
General Parameters

Vision Standard Agreement	USB3Vision 1.0, GenICam
Data interface	USB3.0 TYPE B
Operating system	WIINXP, WIN7/8/10 32@64-bit system, Linux and ARM Linux driver (customizable), Android platform driver (customizable)Linux and ARM Linux driver (customizable), Android platform driver (customizable)
Driver	Directshow component Halcon special component Labview special driver OCX component TWAIN component
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Working temperature @humidity	0~50℃ @ 20%~80%(No condensation)
Storage temperature @humidity	-30~60℃ @ 20%~95%(No condensation)
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

SUA/SUF Model selection table of camera tail patch

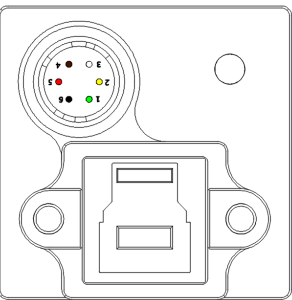
Function Suffix	Aircraft head line sequenceDefinitio n diagram	12V Power supply	Power supply shrapnel typeAircraft head	Lens interface C-mount	Lens interface CS-mount	State
-T1V-C	1	●	●	●		Recommend
-T	2		●		●	Planned shutdown
-T-C	2		●	●		Planned shutdown
-TV-C	3	●	●	●		Planned shutdown

-T1V-C Line sequence definition1



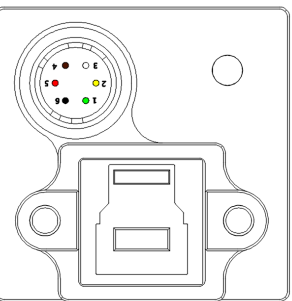
Port	Pin number	Line color	Signal name	Signal description	Remarks
Port A	1	Red	PWR+	Camera power input positive terminal	
	2	White	GPI1+/TRIG_IN+	GPI1 Positive end / trigger input positive end	Default to trigger input
	3	Brown	GPI2/GPO2	GPIO2 Input / output	Non isolated bidirectional IO
	4	Green	GPO1+/STRB_OUT+	GPO1Positive end / flash output positive end	Default to flashoutput
	5	Yellow	GPO1-/STRB-OUT-/TRIG_IN-	GPO1 Negative terminal / flash output negative terminal / trigger input negative	GPIO Common negative terminal
	6	Black	PWRGND	Camera power input negative terminal	

-T、-T-C Line sequence definition 2



Pin number	Line color	Signal name	Signal description	Remarks
1	Green	GPO1+/STRB_OUT+	GPO1Positive end / flash output positive end	Default to flash output
2	Yellow	GPO1-/STRB_OUT-	GPO1Negative terminal / flash output negative terminal	Default to flash output
3	White	GPI1+/TRIG_IN+	GPI1Positive end / trigger input positive end	Default to trigger input
4	Brown	GPI1-/TRIG_IN-	GPI1Negative end / trigger input negative end	Default to trigger input
5	Red	GPO2+	GPO2Positive end output	
6	Black	GPO2-	GPO2Negative end output	

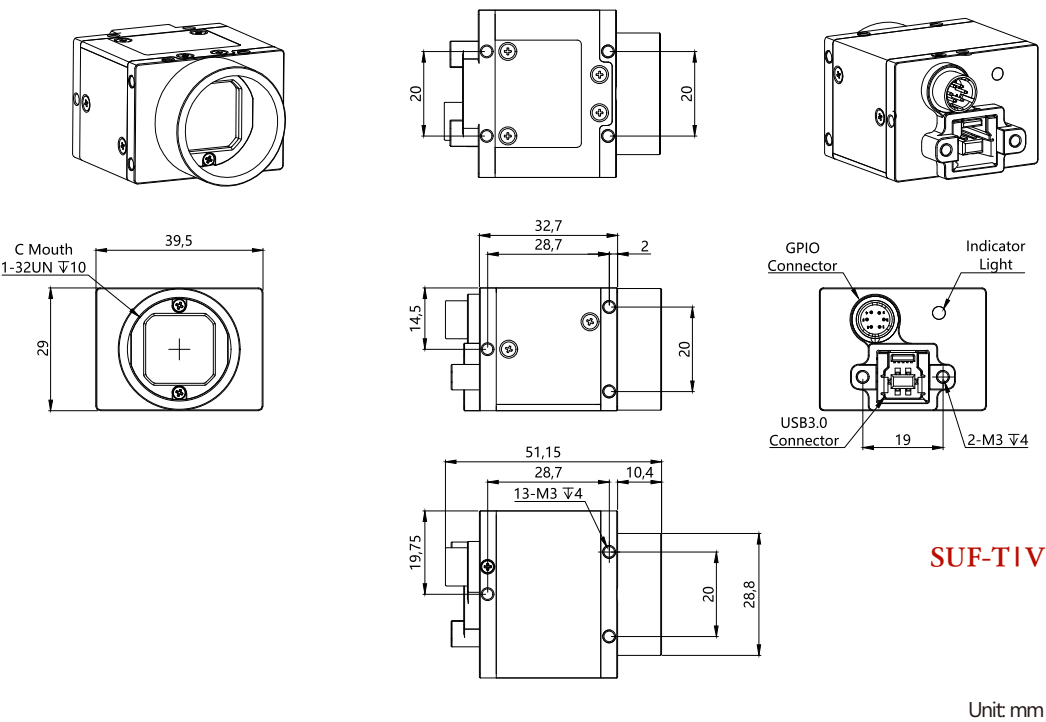
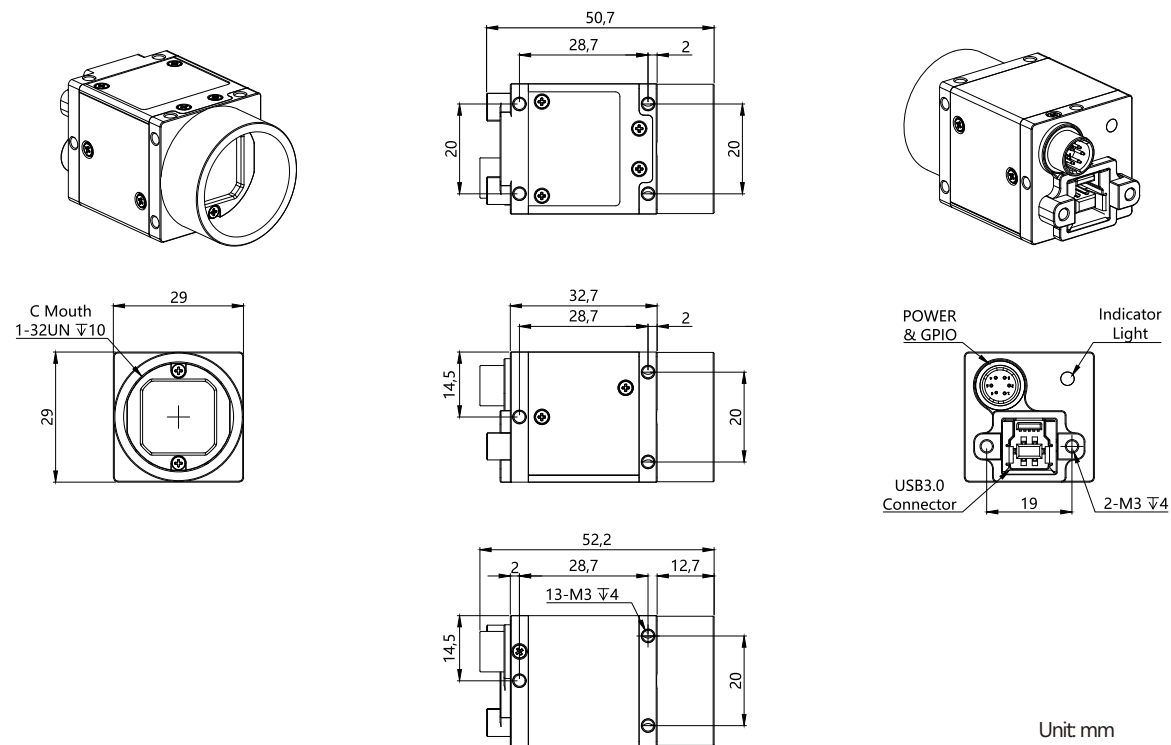
-TV-C Line sequence definition 3



Port	Pin numbe	Line color	Signal name	Signal description	Remarks
Port A	1	Green	GPO1+/STRB_OUT+	GPO1Positive end / flash output positive end	Default to flash output
	2	Yellow	GPO1-/STRB_OUT-	GPO1Negative terminal / flash output negative terminal	Default to flash output
	3	White	GPI1+/TRIG_IN+	GPI1Positive end / trigger input positive end	Default to trigger input
	4	Brown	GPI1-/TRIG_IN-	GPI1Negative end / trigger input negative end	Default to trigger input
	5	Red	PWR+	Camera power input positive terminal	
	6	Black	PWRGND	Camera power input negative terminal	

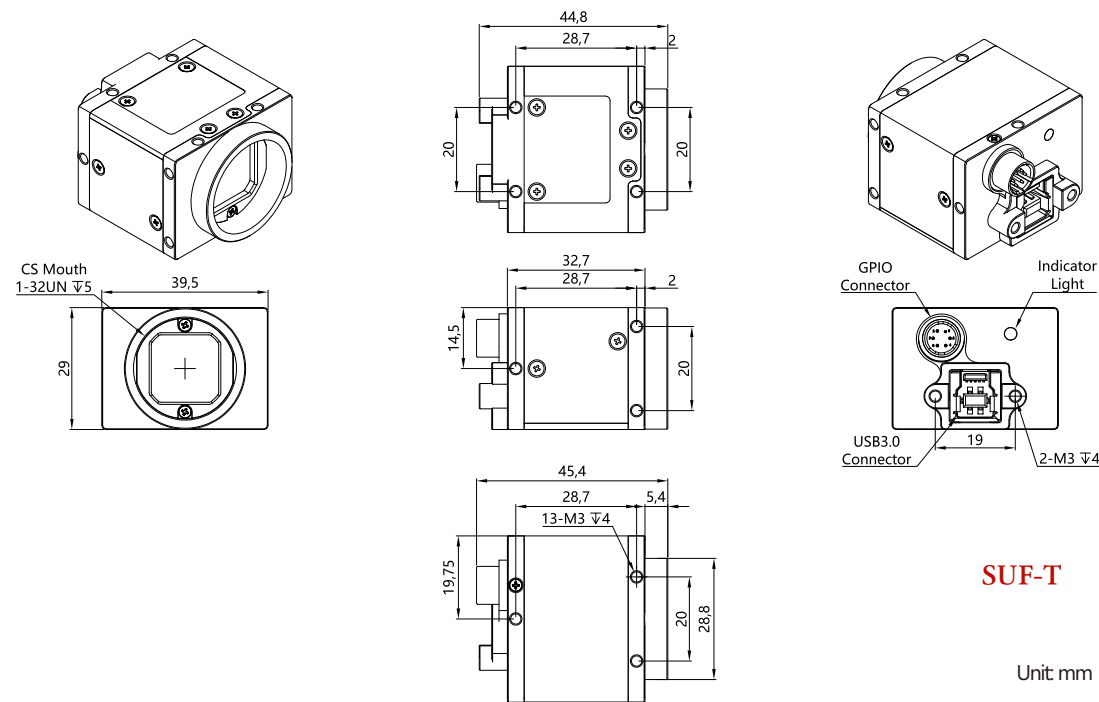
SUA Series Dimension Drawing

SUF Series Dimension Drawing



SUF-T1V

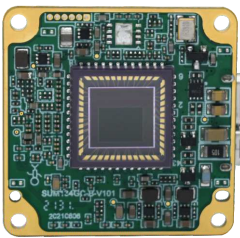
SUF Series Dimension Drawing



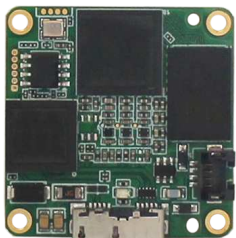
SUF-T

USB3.0 Board Level Industrial Camera/Module

The series of USB3.0 single board modules are designed to be compatible with USB3.0 Vision and GenICam Standard. Besides , L Laview,Halcon,VisionPro and other mainstream vision software fully compatible with it . It is applicable to electronics,semi-conductor,factory automation,food and beverage,medical packaging and image measurement.



Front of board module



Back of board module



C-Mount



M12-Mount

Product Features

- Single board design, small size, ultra-thin size
- Embedded system integration is more convenient and integrated
- Support 16bit gray and 48bit color lossless format output
- Multiple options are available: single board /c interface /m12 interface
- The camera has its own programmable IO, which supports external trigger flash synchronous photography
- Support PC Linux system and arm linux system, which can beintegrated into embedded devices
- Compatible with vision protocol, seamlessly compatible with Halcon, visionpro and LabVIEWOther visual software

Technical Parameter

Specification	HT-SUM33GC/GM	HT-SUM133GC/GM	HT-SUM134GC/M
Model			
Resolution@ frame rate	640X480@790FPS	1280X1024@221.25FPS	1280X1024@210FPS
Shutter method	Global	Global	Global
Pixels	0.3MP	1.3MP	1.3MP
Pixel size	4.0μmX4.0μm	4.0μmX4.0μm BSI	4.8μmX4.8μm
Sensor type	1/5.6" CMOS	1/2.7" CMOS	1/2" CMOS
Pixel bit depth	10bit	8bit	8bit
Maximum gain (multiple)	8	6	16.5
Exposure time range (ms)	0.0027~22.3	0.00043~35.2	0.0045~584.4
Sensitivity	8 V/lux.s	8 V/Lux.s	7.3 V/1ux-s 540nm
GPIO	2 birectional IO, each IO has input or output function, which can be configured as input/output/ trigger/flash in IO control		

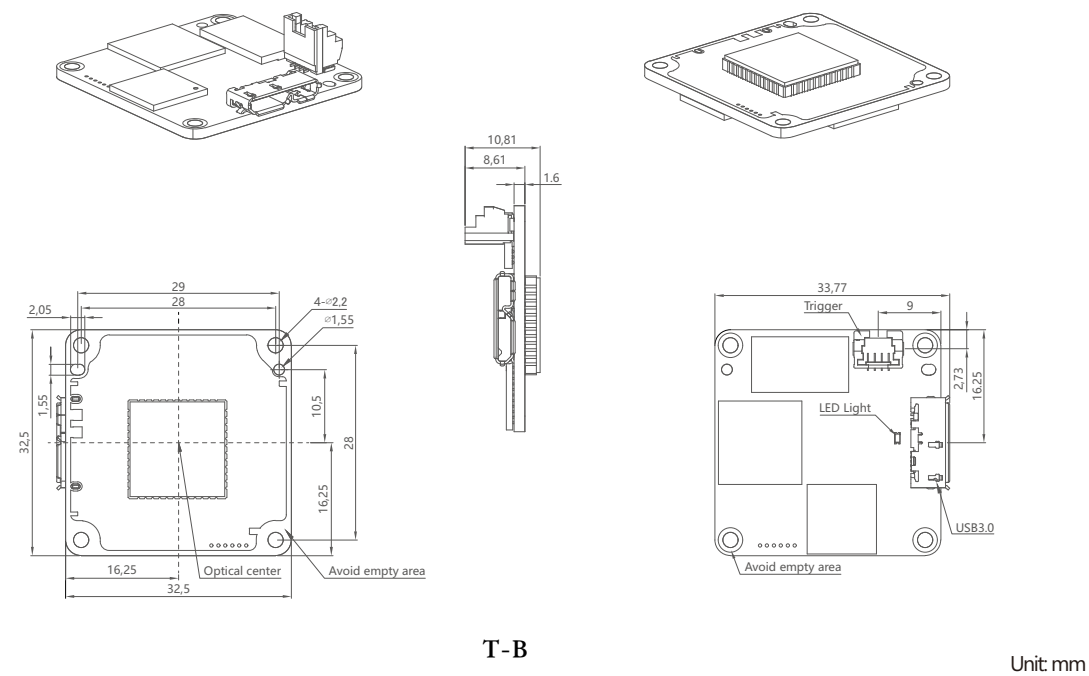
General Parameter

Acquisition mode	Continuous / soft trigger / hard trigger
Frame cache	128MB
User defined data area	2KB
Lens interface	Veneer/C-Mount/M12
Power supply	5V
Weight	Veneer : <10g C-Mount/M12-Mount : <30g
Overall dimension	Veneer : 32.5x32.5x1.6mm (Without sensor and external interface) C-Mount/M12-Mount : 35x35x8.6mm (Without lens holder and external interface)
Visual standards protocol	USB3.0 Vision 1.0, GenICam
Data interface	Micro B
Operating system	WIINXP、WIN7/8/10 32@64 Bit system, Linux and arm linux driver, Android platform driver, Mac OS System
Driver	DirectShow component Halcon special component LabVIEW special driver OCX component Twain component
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Operating temperature @ humidity	0~50°C @ 20%~80%(No condensation)
Storage temperature @humidity	-30~60°C @ 20%~95%(No condensation)
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead center coordinate correctionISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc

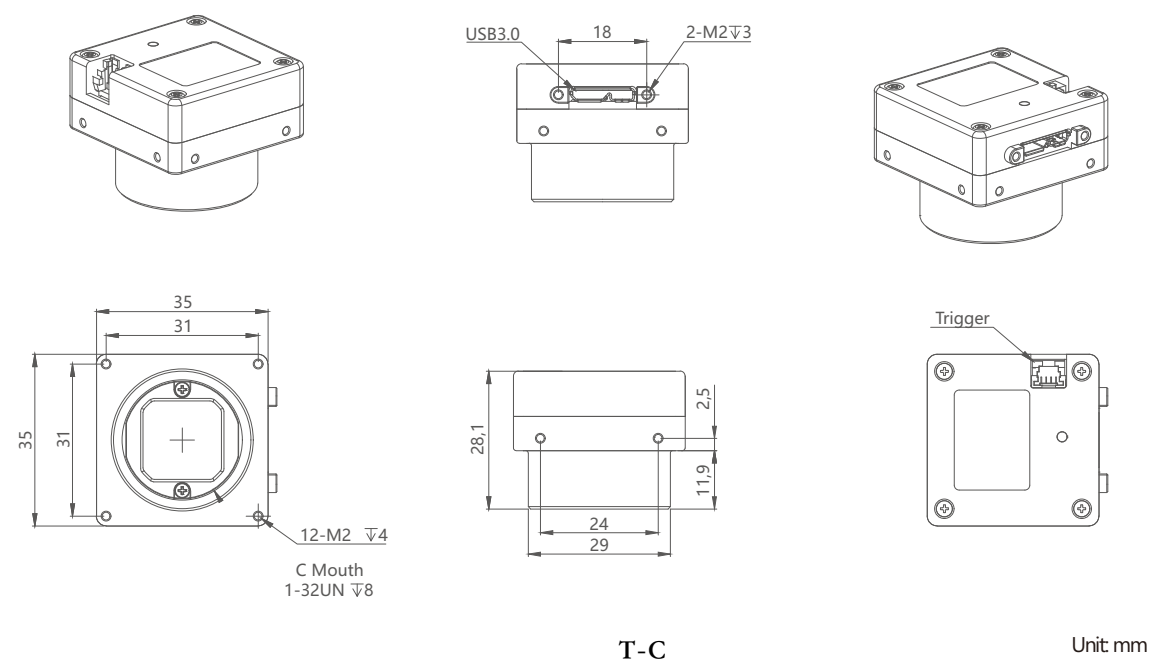
SUM Camera Suffix Selection Table

Function	Module	M12-Mount	C-Mount	Status
Suffix				
-T-B	●			Recommended
-T-C			●	Recommended
-T-M		●		Recommended

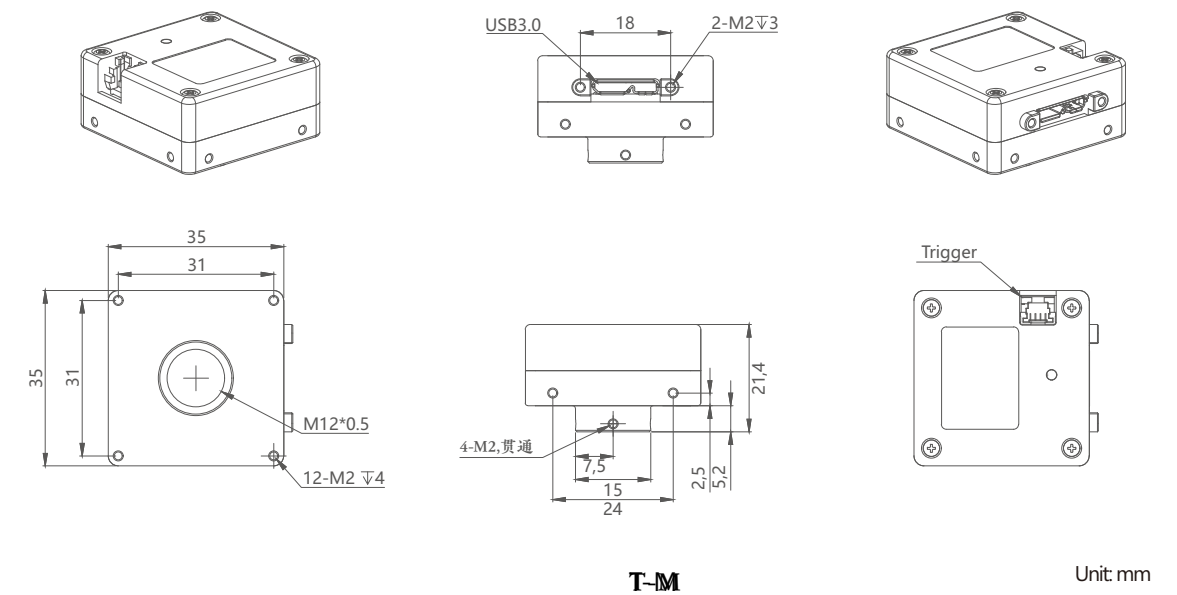
Single Board Module Dimension Drawing



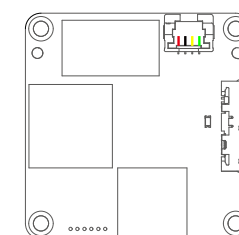
C-Mount Dimension Drawing



M12-Mount Dimension Drawing



Line Sequence Definition



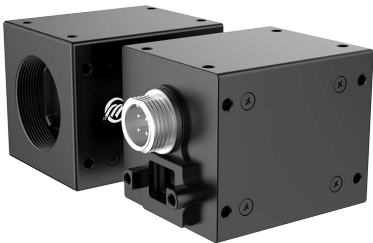
Port	Pin Number	Line Color	Signal Number	Signal Description	Remark
Port A	1	Red	NC		
	2	Black	GND	GPIO common port	
	3	Yellow	GPIO1+	Bidirectional IO 1	Default trigger
	4	Green	GPIO2+	Bidirectional IO 2	Default flash mode

USB2.0 Industrial Camera

USB2.0 Industrial Camera is simple in structure and has multiple single board modules with small size .This feature can effectively save product size. It is suitable for small instruments and equipment; no need for external power supply, easy to use,most cost-effective choice. It is applicable to factory automation, university experiment, logistics code scanning, packaging and printing and textile.



UBS Series



UBD Series

Product Features

- Support external trigger and flash synchronization
 - Custom data can be written in the camera
 - 0.3MP to 10MP CMOS, CCD Sensor can be chosen
 - Support 16bit gray and 48bit color lossless format output
- USB2.0 interface, no need for external power supply, plug and play
 - With hardware frame buffer, it supports more than 32 cameras working at the same time
 - Support windows, Linux, ARM Linux and Android systems

Product selection table

Model Number	Pixel	Sensor type	Shutter method	Maximum resolution	Pixel size	Frame rate (FPS)	Target size	Minimum exposure	Sensor model	Colour
HT-UBS130RC	1.2MP	CMOS	Rolling	1280X960	3.75µm	38	1/3"	0.0325ms	AR0130	Color
HT-UBS130M	1.3MP	CMOS	Rolling	1280X1024	5.2µm	30	1/2"	0.032ms	MT9M001	Mono
HT-UBS402C	4MP	CMOS	Rolling	2688X1520	2.9µm	9	1/1.8"	0.0238ms	SC450AI	Color
HT-UBS100M	1MP	CMOS	Rolling	1344X760	5.8µm	31	1/1.8"	0.0238ms	OEM	Mono
HT-UBS502C/M	5MP	CMOS	Rolling	2592X1944	2.2µm	8	1/2.5"	0.0238ms	Ar0522	Color/Mono
HT-UBS1000C/M	10MP	CMOS	Rolling	3664X2748	1.67µm	4.7	1/2.3"	0.064ms	MT9J003	Color/Mono
HT-UBD32C/M	0.3MP	CCD	Global	640X480	7.4µm	140	1/3"	0.0054ms	SHARP	Color/Mono
HT-UBD130C/M	1.3MP	CCD	Global	1280X960	3.75µm	35	1/3"	0.0063ms	SHARP	Color/Mono

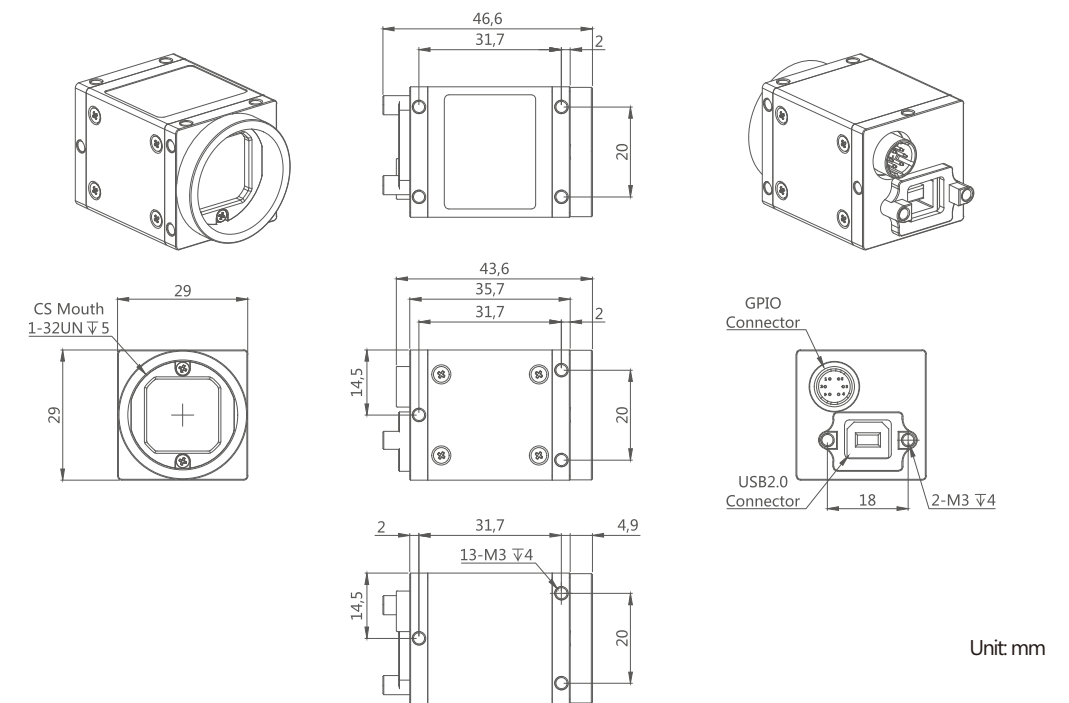
Technical Parameters

Model Parameter	HT-UBS130RC	HT-UBS130M	HT-UBS402C
Resolution@ frame rate	1280X960@38	1280X1024@30	2688X1520@8
Pixel size	3.75µmX3.75µm	5.2µmX5.2µm	2.9µmX2.9µm
Pixel bit depth	12bit	10bit	10bit
Sensitivity	5.5V/lux-sec@550nm	2.1V/lux-sec550nm	7072mV/lux*s
Acquisition mode	Continuous / Soft trigger / Hard trigger	Continuous / Soft trigger / Hard trigger	Continuous / Soft trigger / Hard trigger
Maximum gain (multiple)	8	15	35
Exposure time range (ms)	0.0325~62.4	0.032~524.6	0.023~1507.3
Frame buffer	32M Bytes	32M Bytes	16M Bytes
User-defined data area	2K Bytes	2K Bytes	2K Bytes
Video output format	Color : Bayer8/Bayer12 bit Mono : Mono8/Mono12 bit		Bayer12 bit
GPIO	Aviation head connector, 1 channel trigger input, 1 channel flash control output; 1 channel GPO output		
Lens Mount	CS interface of fuselage, factory pre installed removable 5mmc/cs adapter ring, compatible with C-Port lens		
Power supply	5V, USB Bus power supply		
Power	< 2W		
Dimensions	29X29X35.7mm (Without lens holder and rear housing interface)		29X29X32mm (Without lens holder and rear housing interface)
Weight	< 75g	< 75g	< 60g

Model Parameter	HT-UBS100M	HT-UBS502C/M	HT-UBS1000C/M
Resolution@frame rate	1344X760BIN2@31	2592X1944@8	3664X2748@4.7
Pixel size	5.8µmX5.8µm	2.2µmX2.2µm	1.67µmX1.67µm
Pixel bit depth	8bit	12bit	12bit
Sensitivity	7072mV/lux*s	Color : 18.8ke-/lux*sec Mono : 36ke-/lux*sec	0.31V/lux-s 550nm
Acquisition mode	Continuous / Soft trigger / Hard trigger	Continuous / Soft trigger / Hard trigger	Continuous / Soft trigger / Hard trigger
Maximum gain (multiple)	35	15	15.88
Exposure time range (ms)	0.0238~786.4	0.0238~1556.5	0.064~1407
Frame buffer	16M Bytes	16MBytes	32M Bytes
User-defined data area	2K Bytes	2K Bytes	2K Bytes
Video output format	Bayer12 bit	Color : Bayer GR 8 bit Mono : Mono 8 bit	Color : Bayer8/Bayer12 bit Mono : Mono8/Mono12 bit
GPIO	Aviation head connector, 1 channel trigger input, 1 channel flash control output; 1 channel GPO output		
Lens Mount	CS interface of fuselage, factory pre installed removable 5mmc/cs adapter ring, compatible with C-Port lens		
Power supply	5V, USB Bus power supply		
Power	< 2W		
Dimensions	29X29X32mm (Without lens holder and rear housing interface)		29X29X35.7mm (Without lens holder and rear housing interface)
Weight	< 60g	< 120g	< 75g

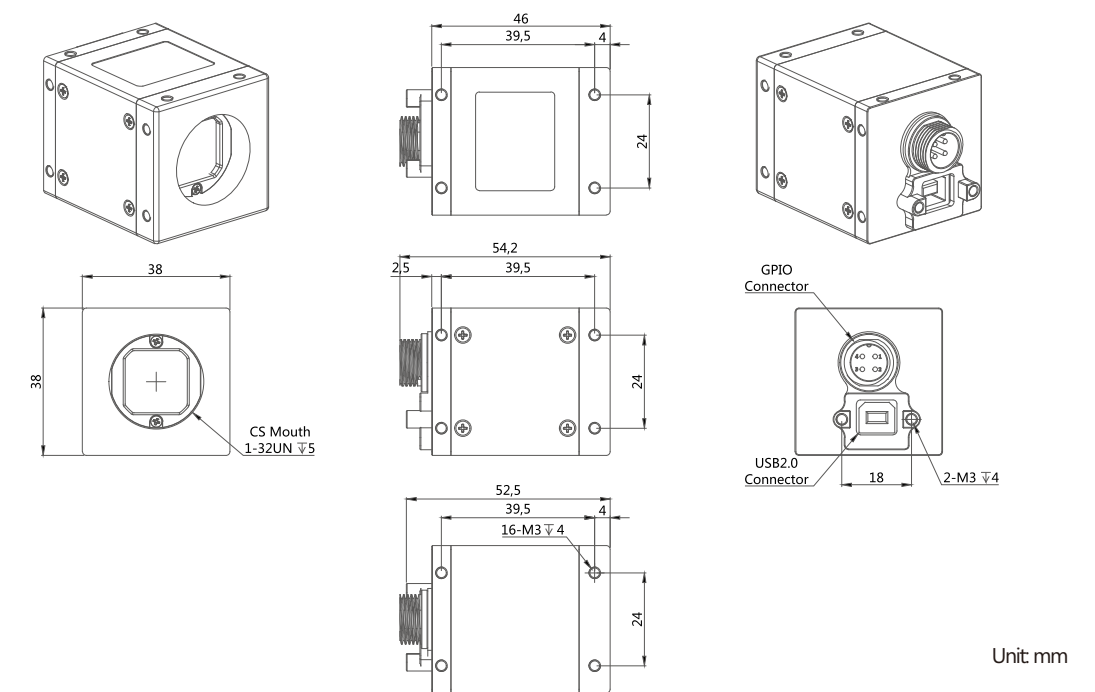
UBS Interface Dimension Drawing

Model		
Parameter	HT-UBD32C/M	HT-UBD130C/M
Resolution@ frame rate	640X480@140	1280X960@35
Pixel size	7.4μmX7.4μm	3.75μmX3.75μm
Pixel bit depth	12bit	12bit
Sensitivity	Color : 3000mV 1/30s Mono : 4500mV 1/30s	Color : 950mV 1/30s Mono : 1430mV 1/30s
Acquisition mode	Continuous / soft trigger / hard trigger	
Maximum gain (multiple)	64	64
Exposure time range (ms)	0.0054~500	0.0063~4615
Frame buffer	32M Bytes	
User-defined data area	2K Bytes	
Video output format	Color : Bayer8/Bayer12 Mono : Mono8/Mono12	
GPIO	Aviation head connector, 1-channel optical isolation input, 1-channel optical isolation output	
Lens Mount	CS interface of fuselage, factory pre installed removable 5mmc/cs adapter ring, compatible with C-Port lens	
Power supply	5V, USB Bus power supply	
Power	< 2W	
Dimensions	38X38X46mm (Without lens holder and rear housing interface)	
Weight	< 120g	



UBD Interface Dimension Drawing

Visual Standards Protocol	Dshow Twain
Data Interface	USB 2.0 TYPE B
Operating system	WIINXP、WIN7/8/10 32@64-bit system, Linux and ARM Linux driver (customizable), Android platform driver (customizable)Linux and ARM Linux driver (customizable), Android platform driver (customizable)
Driver	Directshow component Halcon special component Labview special driver OCX component TWAIN component
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Working temperature @humidity	0~50°C @ 20%~80%(No condensation)
Storage temperature @humidity	-30~60°C @ 20%~95%(No condensation)
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.



CoaXPress High-speed Camera

HuaTengVision PX high-speed industrial camera series adopt CoaXPress-12 standard interface, containing different types with pixels from 2 megapixels to 150megapixels. It is mainly used for machine vision scenesrequiring high frame rate and high pixels, especially applicable to 3D line laser detection, PCB detection, biomedicine, aerospace, military industry, scientific research and other applications.

Product Features

- CXP-12 interface, maximum effective data transmission bandwidth 4700Mbyte, 4 times faster than that of 10gigabit network interface, 12times faster than that of USB 3.0 interface.
- Cameras are designed with excellent SDK, users are no need to install acquisition card, which eliminates tedious configuration, using processesas simple as using a gigabit netwrk industrial camera.
- Support flexible combination of 1, 2, and 4 channels.
- Adopt HD-BNC to achieve standardized industrial cable connection.
- The cable length can be up to 30 meters, and a coaxial cable can realize signal triggering and power supply.
- Low power consumption and no fan design ensure the stability of the flying bat.



Product selection table

Model Number	Pixel	Sensor type	Shutter method	Maximum resolution	Pixel size	Frame rate (FPS)	Target size	Minimum exposure	Sensor model	Colour
HT-PX200GC/M	2MP	CMOS	Global	2048×1216	4.5um	2350	9.21mm×5.47mm	0.001ms	GSPRINT4502	Color/Mono
HT-PX1000GC/M	10MP	CMOS	Global	4608×2176	4.5um	470	20.74mm × 9.79mm	0.001ms	GSPRINT4510	Color/Mono
HT-PX2100GC/M	21MP	CMOS	Global	5120×4096	4.5um	222	23.04mm × 18.43mm	0.004ms	GSPRINT4521	Color/Mono
HT-PX2500GC/M	25MP	CMOS	Global	5120×5120	2.5um	150	1.1"	0.004ms	GMAX0505	Color/Mono
HT-PX6500GC/M	65MP	CMOS	Global	9344×7000	3.2um	71	29.9mm × 22.4mm	0.015ms	GMAX3265	Color/Mono
HT-PX15000GC/M	150MP	CMOS	Global	16556×9200	3.2um	16	53.0mm × 29.4mm	0.015ms	GMAX32152	Color/Mono

Technical Parameters

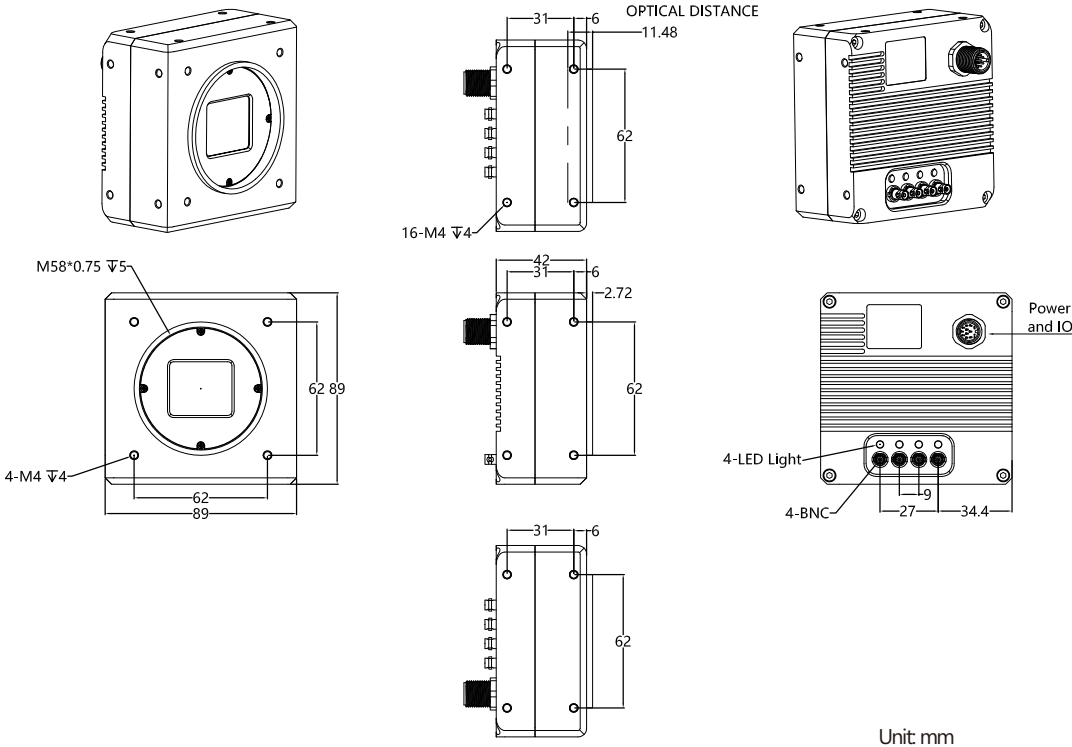
Model Parameter	HT-PX200GC/M	HT-PX1000GC/M	HT-PX2100GC/M
Resolution@ frame rate	2048×1216@2350FPS	4608×2176@470FPS	5120×4096@222FPS
Pixel size	4.5um×4.5um	4.5um × 4.5um	4.5um × 4.5um
Pixel bit depth	10bit	10bit	10bit
Sensitivity	<-86dB	<-86dB	<-86dB
Acquisition mode	Continuous / Soft trigger / Hard trigger		Continuous / Soft trigger / Hard trigger
Maximum gain (multiple)	4	3.973x	3.973x
Exposure time range (ms)	0.001~100	0.001~100	0.004~100
User-defined data area	2KB		
Video output format	Color:Bayer8/Bayer10/Bayer12 Mono:Mono8/Mono10/Mono12		Color:Bayer8/Bayer10/Bayer12 Mono:Mono8/Mono10/Mono12
GPIO	2 optocoupler isolated input, 1 optocoupler isolated output, 1non-isolated bidirectional IO		
Lens Mount	C-Mount	M58×0.75	
Power supply	POCXP/DV12V		
Power	<30W		
Dimensions	89×89×42mm(Without lens holder and rear housing interface)		
Weight	<600g		

Model Parameter	HT-PX2500GC/M	HT-PX6500GC/M	HT-PX15000GC/M
Resolution@ frame rate	5120×5120@150FPS	9344×7000@71FPS	16556×9200@16FPS
Pixel size	2.5um × 2.5um	3.2um × 3.2um	3.2um×3.2um
Pixel bit depth	8bit	12bit	12bit
Sensitivity	1.03×10 ^{^7} e-/((W/m ^{^2})·s) @500nm	1.03×10 ^{^7} e-/((W/m ^{^2})·s) @500nm	1.79×10 ⁷ e-/ ((W/m ²)·s) @510nm
Acquisition mode	Continuous / Soft trigger / Hard trigger		
Maximum gain (multiple)	16.5	6	5.2
Exposure time range (ms)	0.0040~1048.6	0.015~10000	0.015~100
User-defined data area	2KB		
Video output format	Color:Bayer8/Bayer12 Mono:Mono8/Mono12	Color:Bayer8/Bayer12 Mono:Mono8/Mono12	Color:Bayer12 Mono:Mono12
GPIO	2 optocoupler isolated input, 1 optocoupler isolated output, 1non-isolated bidirectional IO		
Lens Mount	C-Mount	M58×0.75	
Power supply	POCXP/DV12V		
Power	<30W		
Dimensions	89×89×42mm(Without lens holder and rear housing interface)		
Weight	<600g		

General Parameters

Operating System	WIINXP.WIN7/8/10 32@64-bit system, Linux and ARM Linux driver (customizable), Android platform driver (customizable), Mac OS system
Data interface	CoaXPress
Working temperature	0~50°C
Working humidity	20~80%(No condensation)
Storage temperature	-30~60°C
Storage humidity	20~95%(No condensation)
Driver	DirectShow component Halcon special component LabVIEW special driver OCX component Twain component
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

CoaXPress High Speed Camera Dimensions



CoaXPress Line-scan Camera

HuatengVision PXL high-speed linear array industrila cameras series adopt CoaXPress-12 standard interface, The line frequency of RGB true color output can reach 40KHz in the mode of 16K resolution2TSI mode.

Product Features

- The camera is with 16K resolution; RGB true color supports 2TDI mode, and line frequency is 40KHz.
- CXP-12 interface, maximum effective data transmission bandwidth 4700Mbyte, 4 times faster than that of 10gigabit network interface, 12times faster than that of USB 3.0 interface.
- Cameras are designed with excellent SDK, users are no need to install acquisition card, which eliminates tedious configuration, using processas simple as using a gigabit netwrk industrial camera.
- Support flexible combination of 1, 2, and 4 channels.
- Adopt HD-BNC to achieve standardized industrial cable connection.
- The cable length can be up to 30 meters, and a coaxial cable can realize signal triggering and power supply.



Product selection table

Model Number	Pixel	Sensor type	Shutter method	Pixel size	Line frequency	Target size	Minimum exposure	TDI	Sensor model	Colour
HT-PXL164C	16384×6 _(RGB)	CMOS	Global	5um	72KHz	81.92mm	0.003ms	2	Huatengvision	Color
HT-PXL164M	16384×1	CMOS	Global	5um	25KHz	81.92mm	0.003ms	2	Huatengvision	Mono

Technical Parameters

Model	HT-PXL164M	HT-PXL164C
Resolution@ frame rate	16384×1	16384×6 _(RGB)
Horizontal Frequency	72KHz	25KHz
Pixel size	5um × 5um	5um × 5um
Pixel bit depth	8bit	8bit
Sensitivity	/	
Acquisition mode	Continuous/software trigger/frame trigger/line trigger/conditional line trigger	
Maximum gain (multiple)	4	
Exposure time range (ms)	0.003~5	
User-defined data area	2KB	
Video output format	Color:Bayer10/Bayer12 Mono:Mono10/Mono12	
Lens Mount	M90	
Power supply	POCXP/DV12V	
Power	<30W	

General Parameters

Model Parameter	HT-PXL164C/M
Dimensions	110×110×69.26mm (Without lens holder and rear housing interface)
Weight	<1500g
Operating System	WIINXP,WIN7/8/10 32@64-bit system, Linux and ARM Linux driver (customizable), Android platform driver (customizable), Mac OS system
Data interface	CoaXPress
Working temperature	0~50°C
Working humidity	20~80% (No condensation)
Storage temperature	-30~60°C
Storage humidity	20~95% (No condensation)
Driver	DirectShow component Halcon special component LabVIEW special driver OCX component Twain component
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

X86 Smart Industrial Camera

ITA/ ITAE is a smart camera product series based on X86 platform of HuatengVision (not including vision software, SDK software provided for image capturing , with pixels from 0.3 to 20 megapixels.This series not only generates low heat, but also have high integration and high cost performance. Standard Wndows and Linux operating systems are built into the camera. Applications can be directly copied ro run inside the camera. It can extend mouse, keyboard and monitor, reserving USB dongle position inside, no need for external computer, very applicable to single workstation visual inspection, such as code scanning, measurement, products recognition on the assembly linepositioning and other needs.

Product Features

- ITA:CPU Intel Atom Z8350 (1.44-1.92GHZ), 2G/32G 4G/64G (Memory/Disk)
- ITAE:CPU Intel N4120 (1.1-2.4GHZ), 4G/32G 4G/64G (Memory/Disk)
- Display HDMI X1supports 1920x1080 resolution, Ethernet 1000mx1, and GigE vision camera
- ITA:Support external USB2.0×2ports,ITAE:USB2.0×1 USB3.0×1
- Support external USB2.0 ports x 2, trigger x1, flash x1, RS232 x1
- M12 12core threaded aviation connector, supporting 12v/1A power input
- 1-channel input and 2-channel output interfaces can set the output delay without processing in the program
- Support Windows 10 64bit and Ubuntu16.04 64bit systems
- Built in USB2.0 interface, easily disassembled and can be connected to the dongl
- Integrated light source controller, optional LED lighting components, supporting stroboscopic synchronization functione



Product Selection Table

Model Number	Effective Pixels	Sensor type	Shutter method	Maximum resolution	Pixel size	Frame rate (FPS)	Target size	Minimum exposure	Sensor model	Colour
HT-ITA(E)33GC/M	0.3MP	CMOS	Global	640×480	4.0μm	790	1/5.6"	0.00124ms	SmartSens	Color/Mono
HT-ITA(E)133GC/M	1.3MP	CMOS	Global	1280×1024	4.0μm	245	1/2.7"	0.00194ms	SC130GS	Color/Mono
HT-ITA(E)134GC/M	1.3MP	CMOS	Global	1280×1024	4.8μm	211	1/2"	0.0045ms	PYTHON	Color/Mono
HT-ITA(E)200GC/M	2MP	CMOS	Global	1600×1200	4.5μm	60	1/1.8"	0.015ms	EV76C570	Color/Mono
HT-ITA(E)202GC/M	2.3MP	CMOS	Global	1920×1200	4.8μm	165	2/3"	0.0039ms	PYTHON2000	Color/Mono
HT-ITA(E)230GC/M	2.3MP	CMOS	Global	1920×1200	5.86μm	165	1/1.2"	0.049ms	IMX174	Color/Mono
HT-ITA(E)231GC/M	2.3MP	CMOS	Global	1920×1200	5.86μm	40	1/1.2"	0.0201ms	IMX249	Color/Mono
HT-ITA(E)402GC/M	4MP	CMOS	Global	2048×2048	5.5μm	88	1"	0.0054ms	CMV4000	Color/Mono
HT-ITA(E)501GC/M	5MP	CMOS	Global	2448×2048	3.45μm	40	2/3"	0.0133ms	IMX264	Color/Mono
HT-ITA(E)502C/M	5MP	CMOS	Rolling	2592×1944	2.2μm	59	1/2.5"	0.0084ms	AR0522	Color/Mono
HT-ITA(E)630C/M	6.3MP	CMOS	Rolling	3088×2064	2.4μm	60	1/1.8"	0.008ms	IMX178	Color/Mono
HT-ITA(E)890GC/M	8.9MP	CMOS	Global	4096×2160	3.45μm	32	1"	0.034ms	IMX267	Color/Mono
HT-ITA(E)1000C/M	10MP	CMOS	Rolling	3664×2748	1.67μm	8	1/2.3"	0.049ms	MT9J003	Color/Mono
HT-ITA(E)1201C/M	12MP	CMOS	Rolling	4000×3000	1.85μm	32	1/1.7"	0.0102ms	IMX226	Color/Mono
HT-ITA(E)1600C/M	16MP	CMOS	Rolling	4608×3456	1.34μm	12	1/2.3"	0.0233ms	IMX206	Color/Mono
HT-ITA(E)2000C/M	20MP	CMOS	Rolling	5488×3672	2.4μm	19.5	1"	0.0138ms	IMX183	Color/Mono
HT-ITSI890GC/M	8.9MP	CMOS	Global	4096×2160	3.45μm	32	1"	0.0652ms	IMX267	Color/Mono

Technical Parameters

Model Parameter	HT-ITA(E)33GC/M	HT-ITA(E)133GC/M	HT-ITA(E)134GC/M	HT-ITA(E)200GC/M
Sensor type	1/5.6" CMOS	1/2.7" CMOS	1/2" CMOS	1/1.8" CMOS
Color	Color:/Mono	Color:/Mono	Color:/Mono	Color:/Mono
Effective Pixels	0.3MP	1.3MP	1.3MP	2MP
Pixel size	4.0μm×4.0μm	4.0μm×4.0μm	4.8μm×4.8μm	4.5μm×4.5μm
Shutter type	Global	Global	Global	Global
Sensitivity	8 V/lux-s 540nm	8 V/Lux.s	7.3 V/lux-s 540nm	7.4V/lux-s 3200K
Pixel bit depth	8bit	8bit	8bit	10bit
Maximum resolution	640×480	1280×1024	1280×1024	1600×1200
Frame rate	790fps	245fps	211fps	60fps
Maximum gain (multiple)	8	6	16.5	8
Exposure time range (ms)	0.00124~20.7	0.00194~31	0.0045~584	0.015~91
Video output format	Color:Bayer8 Mono:Mono8	Color:Bayer8 Mono:Mono8		Color:Bayer8/Bayer12 Mono:Mono8/Mono12
CPU	ITA:Intel Z8350(1.44-1.92GHZ)/ITAE:Intel N4120 (1.1-2.4GHZ)			
RAM	ITA:2G(Optional 4G)/ITAE:4G			
Disk	32G(Optional 64G)			

Model Parameter	HT-ITA(E)202GC/M	HT-ITA(E)230GC/M	HT-ITA(E)231GC/M	HT-ITA(E)402GC/M
Sensor type	2/3"CMOS	1/1.2" CMOS	1/1.2" CMOS	1" CMOS
Color	Color/Mono	Color/Mono	Color/Mono	Color/Mono
Effective Pixels	2.3MP	2.3MP	2.3MP	40MP
Pixel size	4.8μm×4.8μm	5.86μm×5.86μm	5.86μm×5.86μm	5.5μm5×.5μm
Shutter type	Global	Global	Global	Global
Sensitivity	7.5 V/lux.s 550nm	Color:1016mV 1/30s Mono:825mV 1/30s	Color:1016mV 1/30s Mono:825mV 1/30s	8.5 V/lux-s 540nm
Pixel bit depth	10bit	8bit	12bit	8bit
Maximum resolution	1920×1200	1920×1200	1920×1200	2048×2048
Frame rate	165fps	165fps	40fps	88fps
Maximum gain (multiple)	8	64	249.9	16.5
Exposure time range (ms)	Low speed 0.0039~256.1 High speed 0.0039~5	0.049~20000	0.0201~42112	0.0054~704
Video output format	Color:Bayer8/Bayer12 Mono:Mono8/Mono12	Color:Bayer8/Bayer12 Mono:Mono8/Mono12	Color:Bayer8 Mono:Mono8	Color:Bayer8 Mono:Mono8
CPU	ITA:Intel Z8350(1.44-1.92GHZ)/ITAE:Intel N4120 (1.1-2.4GHZ)			
RAM	ITA:2G(Optional 4G)/ITAE:4G			
Disk	32G(Optional 64G)			

Technical Parameter

Model Parameter	HT-ITA(E)501GC/M	HT-ITA(E)502C/M	HT-ITA(E)630C/M	HT-ITA(E)890GC/M
Sensor type	2/3" CMOS	1/2.5" CMOS	1/1.8" CMOS	1" CMOS
Color	Color/Mono	Color/Mono	Color/Mono	Color/Mono
Effective Pixels	50MP	50MP	6.3MP	8.9MP
Pixel size	3.45μm×3.45μm	2.2μm×2.2μm	2.4μm×2.4μm	3.45μm×3.45μm
Shutter type	Global	Rolling	Rolling	Global
Sensitivity	Color:1146mV 1/30s Mono:915mV 1/30s	Color:18.8ke-/lux*sec Mono:36ke-lux*sec	425mV 1/30s	Color:1146mV 1/30s Mono:915mV 1/30s
Pixel bit depth	12bit	10bit	8bit	12bit
Maximum resolution	2448×2048	2592×1944	3088×2064	4096×2160
Frame rate	40fps	59fps	60fps	32fps
Maximum gain (multiple)	249.9	15	32	249.9
Exposure time range (ms)	0.0133~13924	0.0084~553.4	0.008~5000	0.034~29546.9
Video output format	Color:Bayer8/Bayer12 Mono:Mono8/Mono12	Color:Bayer8 Mono:Mono8	Color:Bayer8 Mono:Mono8	Color:Bayer8/Bayer12 Mono:Mono8/Mono12
CPU	ITA:Intel Z8350(1.44-1.92GHZ)/ITAE:Intel N4120 (1.1-2.4GHZ)			
RAM	ITA:2G(Optional 4G)/ITAE:4G			
Disk	32G(Optional 64G)			

Model Parameter	HT-ITA(E)1000C/M	HT-ITA(E)1201C/M	HT-ITA(E)1600C/M	HT-ITA(E)2000C/M
Sensor type	1/2.3" CMOS	1/1.7" CMOS	1/2.3" CMOS	1" CMOS
Color	Color/Mono	Color/Mono	Color/Mono	Color/Mono
Effective Pixels	10MP	12MP	16MP	20MP
Pixel size	1.67μm×1.67μm	1.85μm×1.85μm	1.34μm×1.34μm	2.4μm×2.4μm
Shutter type	Rolling(GRR mode supported)	Rolling	Rolling(GRR mode supported)	Rolling(GRR mode supported)
Sensitivity	0.31V/lux-s 550nm	250mV 1/30s	142mV 1/30s	Color:462mV 1/30s Mono:388mV 1/30s
Pixel bit depth	12bit	8bit	12bit	8bit
Maximum resolution	3664×2748	4000×3000	4608×3456	5488×3672
Frame rate	8fps	32fps	12fps	19.5fps
Maximum gain (multiple)	8	22	16	22
Exposure time range (ms)	0.049~951	0.0102~2665.1	0.0233~7741	0.0138~3608
Video output format	Color:Bayer8/Bayer12 Mono:Mono8/Mono12	Color:Bayer8 Mono:Mono8	Color:Bayer8/Bayer12 Mono:Mono8/Mono12	Color:Bayer8 Mono:Mono8
CPU	ITA:Intel Z8350(1.44-1.92GHZ)/ITAE:Intel N4120 (1.1-2.4GHZ)			
RAM	ITA:2G(Optional 4G)/ITAE:4G			
Disk	32G(Optional 64G)			

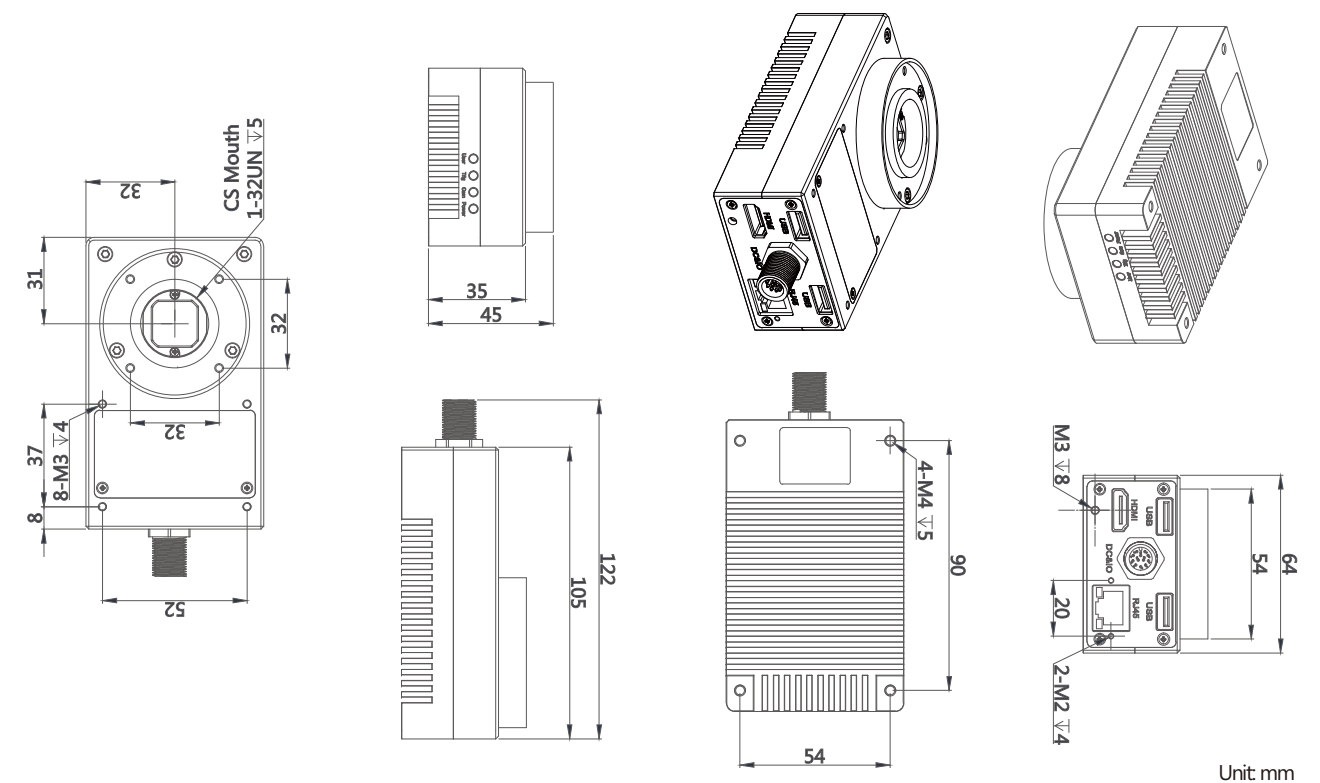
Technical Parameter

Model	Parameter	HT-ITS1890GC/M
Sensor type		1" CMOS
Color		Color/Mono
Effective Pixels		8.9MP
Pixel size		3.45μm×3.45μm
Shutter type		Global shutter
Sensitivity		915mV
Pixel bit depth		12bit
Maximum resolution		4096×2160
Frame rate		32fps
Maximum gain (multiple)		4
Exposure time range (ms)		0.0652~59.2
CPU		Intel N4120(1.10-2.60GHZ)
RAM		4G
Disk		32G (Optional 64G)
System		Windows10 64bit、Ubuntu 64bit
Interface		USB2.0 portx1 USB3.0 port x1(circuit 1 built-in USB2.0), Gigabit Ethernet port x1,HDMI port x1, RS232x1, RTC clock
I/O port		1 trigger input, 1 flash output, 1optical isolation input, 2 optical isolation output
Power supply		12V-24V
Power		≤18W
Lens Mount		C/CS-Mount, pre-installed adapter ring provided
Dimensions		136X63X50.5mm(Without lens mount and rear shell interface)
Weight		<500g
Operating temperature		0~50℃
Working humidity		20~80% (No condensation)
Storage temperature		-30~60℃
Storage humidity		20~95% (No condensation)
Driver		Directshow component, Halcon specialized component, Labview special driver, OCX component and TWAIN component
Programming language pack		C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Other functions		Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

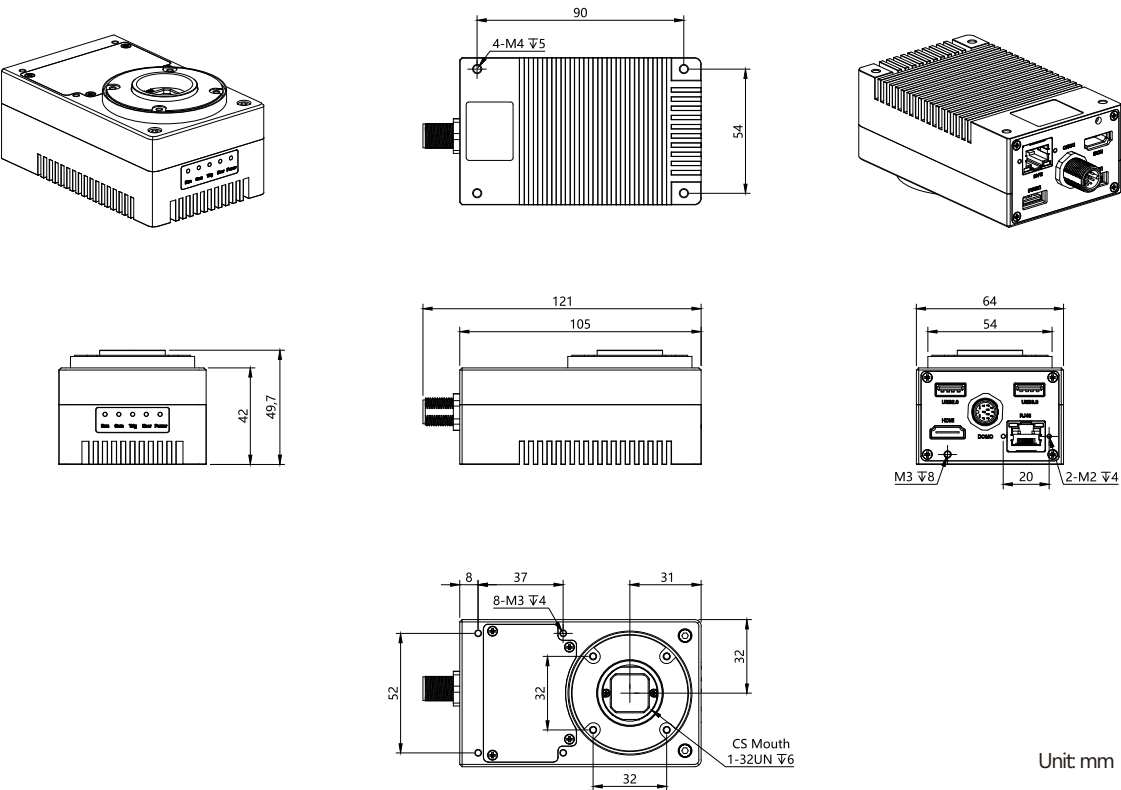
ITA General Parameters

System	Win10 64-bit system is pre-installed by default, Win10 32-bit system or Ubuntu 64bit system can be optionally installed
Interface	External USB2.0 portx2, Gigabit Ethernet port x1, HDMI port x1, built-in USB2.0 port x1, light source power supply port x1, RS232x1, RTC clock
I/O port	1 optical isolation trigger input, 1 optical isolation flash output, 1 optical isolation input, 2 optical isolation outputs
Power supply	12V~24V
Power	≤18W
Lens Mount	CS-Mount, pre-installed adapter ring, compatible with C-mount lens, optional M12 Lens Mount
Dimensions	64X105X35mm(Without lens mount and rear shell interface)
Weight	<500g
Operating temperature	0~50℃
Working humidity	20~80% (No condensation)
Storage temperature	-30~60℃
Storage humidity	20~95% (No condensation)
Driver	Directshow component, Halcon specialized component, Labview special driver, OCX component and TWAIN component
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black levelcorrection, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

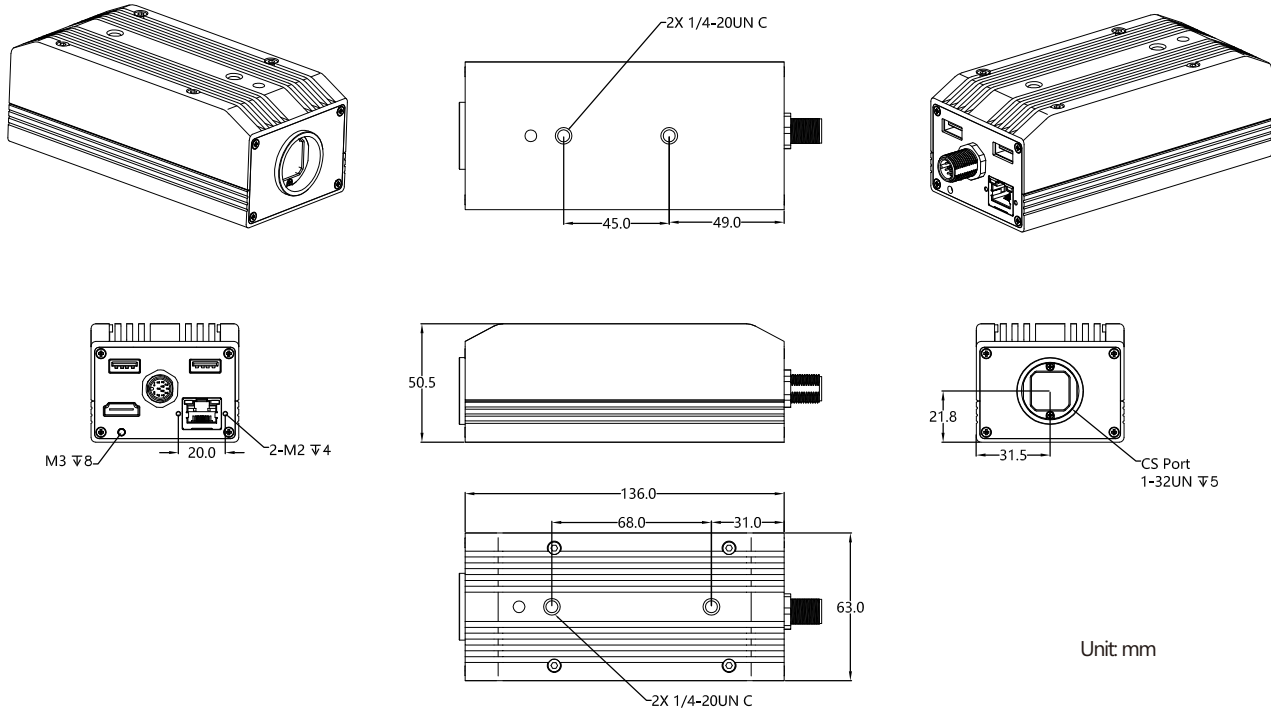
ITA Dimension Drawing



TAE Smart Camera Dimension Drawing



ITSI Dimension Drawing



ARM Smart Industrial Camera

ITE Smart Camera aims at creating a compact and practical smart camera and provides an ideal hardware platform for engineers with visual development ability. Engineers can use the embedded ITE camera for identification, detection and other applications. It is integrated into the compact mini smart camera with vision system, providing customers with more simple, practical and reliable visual scheme. The extremely compact size and wide-angle optical features can provide the best plan for carrying out close-up machine vision work. In a single compact vision scheme effectively and stably carry out inspection, color matching, code reading and OCR, so as to give full play to the advantages of the product. It is widely used in reading one-dimensional and two-dimensional codes and optical character recognitionClassification, dynamic part positioning, inspection, product screening, assembly verification, etc.



Product Features

- Monochrome or color sensor, 0.3MP~2 MP can be chosen
- Open platform, supporting secondary development
- It supports master-slave mode and multi machine linkage control
- Compact size, ultra-low power consumption and small heat
- M12 lens with different focal length can be selected to meet different needs of customers
- Quad core arm 64 bit Cortex-A53 architecture, fast response speed, stable andreliable performance
- Highly integrated, with built-in camera SDK, light source and M12 lens, which can realize fastInstallation and use

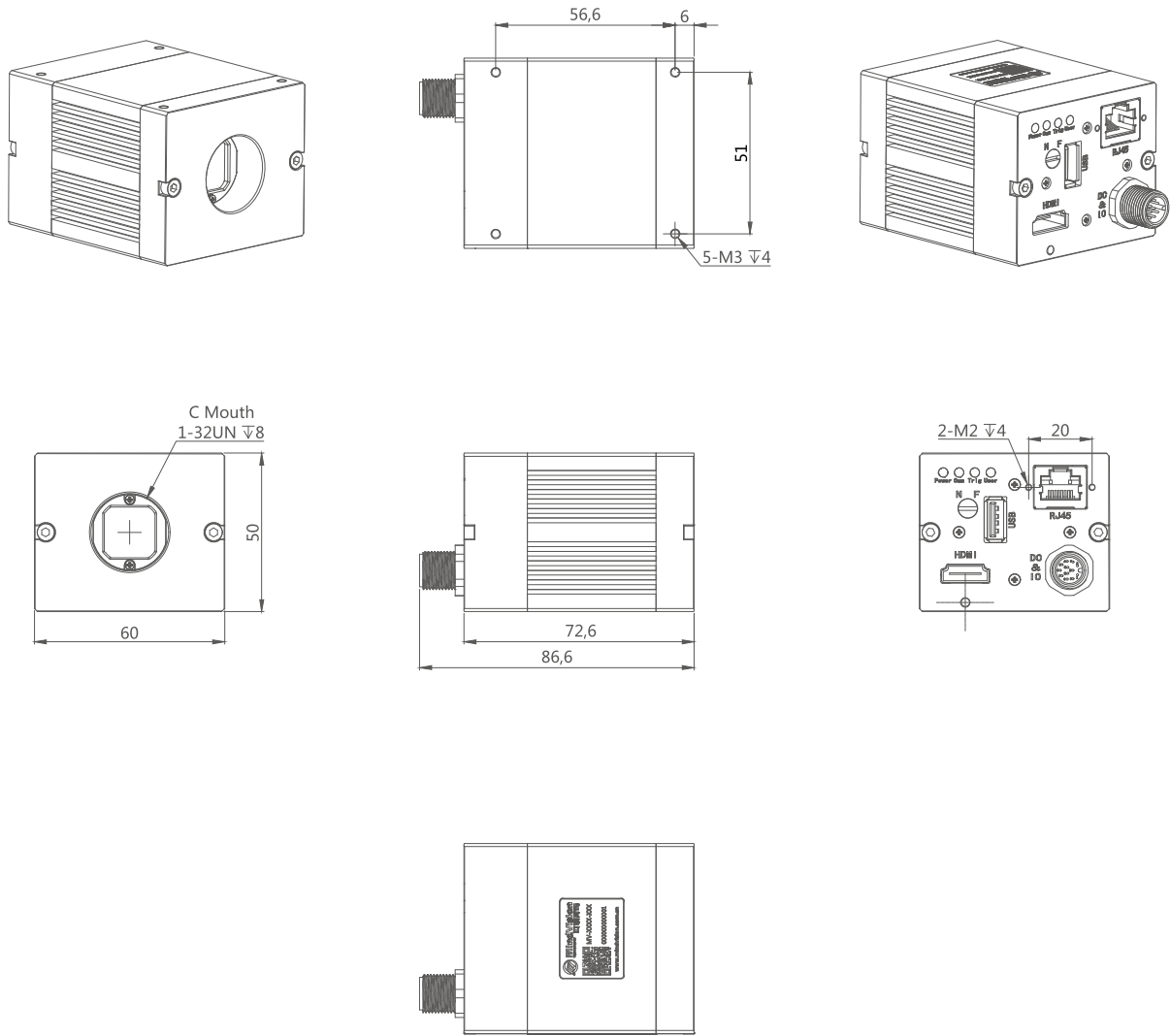
Product Selection Table

Model Number	Effective pixels	Sensor type	Shutter method	Maximum resolution	Pixel size	Frame rate (FPS)	Target size	Minimum exposure	Sensor model	Colour
HT-ITE31GC/M	0.36MP	CMOS	Global	752×480	6.0μm	108	1/3"	0.0652ms	MT9V024	Colour/Mono
HT-ITE130GM	1.2MP	CMOS	Global	1280×960	3.75μm	62	1/3"	0.0167ms	AR0135	Mono
HT-ITE131GC	1.2MP	CMOS	Global	1280×960	3.75μm	60	1/3"	0.014ms	AR0134	Colour
HT-ITE200GC/M	2MP	CMOS	Global	1600×1200	4.5μm	46	1/1.8"	0.015ms	EV76C570	Colour/Mono

Technical Parameters

Model	HT-ITE31GC/M	HT-ITE130GM	HT-ITE131GC	HT-ITE200GC/M
Parameter				
Sensor type	1/3" CMOS	1/3" CMOS	1/3" CMOS	1/1.8" CMOS
Colour	Colour/Mono	Colour/Mono	Colour/Mono	Colour/Mono
Effective Pixels	36万	120万	120万	200万
Pixel size	6.0μm×6.0μm	3.75μm×3.75μm	3.75μm×3.75μm	4.5μm×4.5μm
Shutter type	Global shutter			
Sensitivity	4.8V/lux-s 550nm	6.1V/lux-s 550nm	5.3 V/lux-s 550nm	7.4V/lux-s 3200K
Pixel bit depth	12bit	12bit	12bit	10bit
Maximum resolution	752×480	1280×960	1280×960	1600×1200
Frame rate	108FPS	62FPS	60FPS	46FPS
Maximum gain (multiple)	4	4	4	8
Exposure time range (ms)	0.0652~59	0.0167~32	0.014~26	0.015~91
CPU frequency	Quad-core ARM 64 bit Cortex-A53 architecture main frequency 1.8g			
RAM	2G			
Disk	16G			
System	Ubuntu			
Video output format	Colour:Bayer8 Mono:Mono8			
Interface	External USB2.0 portx1,100m network port x1, HDMI interface x1, and built-in light source (light source is optional)			
I/O port	1-channel optocoupler isolated external trigger input, 1-channel optocoupler isolated flash output, 1-channel optocoupler isolated input programmable IO, 3-channel optocoupler isolated output programmable IO			
Power supply	M12 aviation plug-in 12-24V			
Power	C mouth : 4W/M12 : 7W			
Lens interface	Port C/ M12 (optional focal length)			Port C
Dimensions	60X50X72.6mm (Without lens and rear housing connector)			
Weight	<280g			
Operating temperature	0~50℃			
Working humidity	20~80% (No condensation)			
Storage temperature	-30~60℃			
Storage humidity	20~95% (No condensation)			
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead center coordinate correctionISP image processing acceleration, 3D noise reduction, custom LUT, frame rate adjustment, custom camera name, etc			

ITE Series Dimension Drawing



Unit: mm

Note: the above models can be equipped with IP67. Please consult our sales engineer for more detailed parameters

Smart Code Scanning Industrial Camera

HT-CRDC series is an industrial area-array intelligent code scanning camera with high integration density, integrated design, easy installation and fast speed code reading, 2M/5M/10M/12M/20M CMOS sensors to be selected. HT-CRDC series camera includes LED stroboscopic light source, large depth of field industrial lens, high-speed and high-resolution image capture components, embedded processors and other components, and integrates the functions of image acquisition, data processing and result output; Bar code identification based on ZYNQ UltraScale + MP SOC, The algorithm adopts the completely self-developed Fine Decode technology, with built-in technologies such as "horizontal and vertical bidirectional bar code splicing", "super-resolution reconstruction" and "pixel lossless correction", which can read the bar code at a high speed and easily deal with the problems of bar code fouling, wrinkle, deformation, tilt, blur, etc.

Product Features

- Field width up to 1200mm
- Depth of field up to 1000mm
- Bar code recognition rate reaches over 99%
- Optional red, white and infrared LED light sources are optional
- High degree of integration, collection, processing and output
- Identify diversified codes, multiple one-dimensional code and two dimensional codes supported
- Humanized design is easy to install, and one installer can complete it in 10 minutes

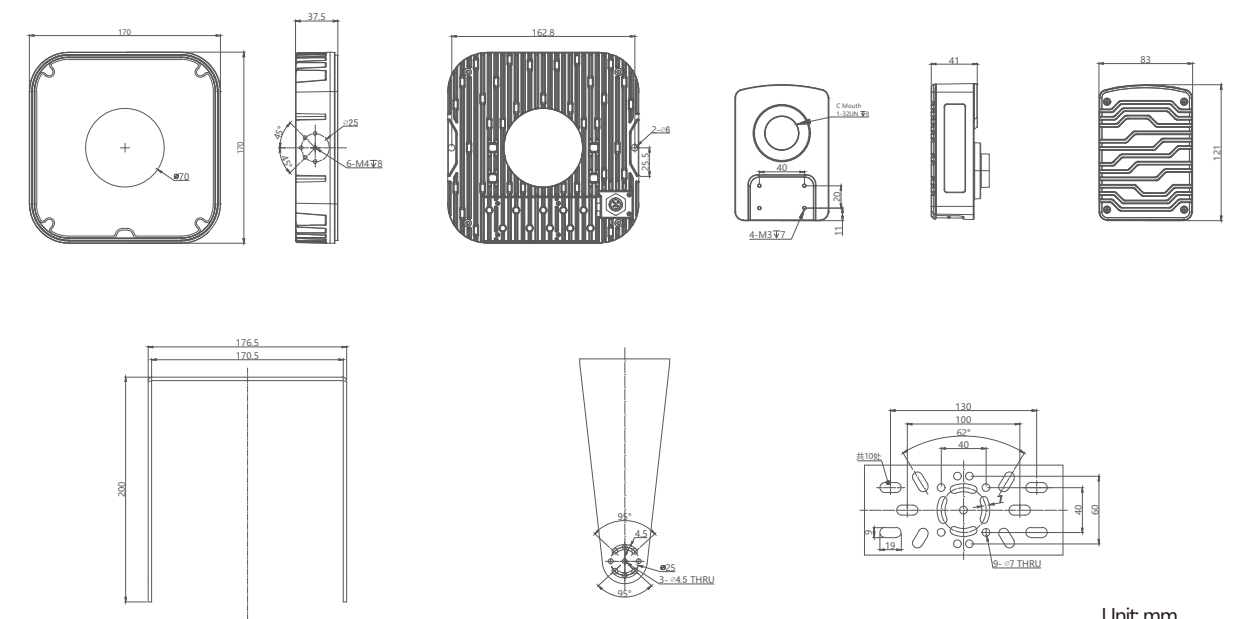
Technical Parameter

Specifications	Parameter	Specifications	Parameter
Model	HT-CRDC1000	Barcode density	≥0.2mm
Image Resolution	10MP(4096X2432)	Image output	JPEG
Shutter exposure	Global exposure	Recognition rate	≥99%
Frame rate	60 Frames/Sec	Lens type	C Interface
Conveyor speed	0~3m/s	Communication Interface	Ethernet, RS485
Code reading speed	10Times/Sec	Power Specifications	DC24V/4A
Multi-code reading	Support 10 at a time	Protection level	IP65
Width of field of view	800mm	LED light source	uilt in bright LED light source, visible in dark environment
Field Width	0~700mm	Secondary development	Support (C + +/C/C) SDK development
Barcode type	code128、EAN13、code39、code93、Cross 25. QR QR code		

Technical Parameter

Model Parameter	HT-CRDC230	HT-CRDC500	HT-CRDC1200	HT-CRDC2000
Image Resolution	2.3MP(1920X1200)	5MP(2576X2048)	12MP(4096X3072)	20MP(5488X3672)
Shutter exposure type	Global shutter	Global shutter	Global shutter	Roller shutter
Frame rate	200Frames/Sec	85Frames/Sec	60Frames/Sec	18Frames/Sec
Conveyor speed	0~3m/s	0~3m/s	0~3m/s	0~3m/s
Code reading speed	15Times/Sec	15Times/Sec	10Times/Sec	10Times/Sec
Multi-code reading	Support 10 at a time			
Width of field of view	500mm	600mm	900mm	1200mm
Item height	0~600mm	0~600mm	0~1000mm	0~1000mm
Barcode type	Code128, EAN13, code39, code93, Cross 25, QR code			
Barcode density	≥0.2mm	≥0.2mm	≥0.15mm	≥0.15mm
Image output	JPEG			
Recognition rate	≥99%			
Lens type	C-Mount			
Communication Interface	Ethernet, RS485			
Power Specifications	DC24V/4A			
Protection level	IP65			
LED light source	Ethernet has its own bright LED light source ,visible in dark environment			
Secondary development	Support (C++/C/C) SDK development			

CRDC Series Dimension Drawing



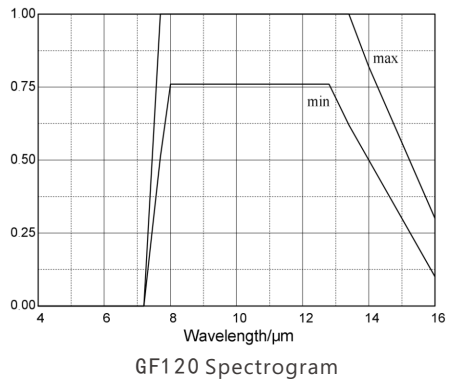
Unit: mm

Thermal Imaging Industrial Camera

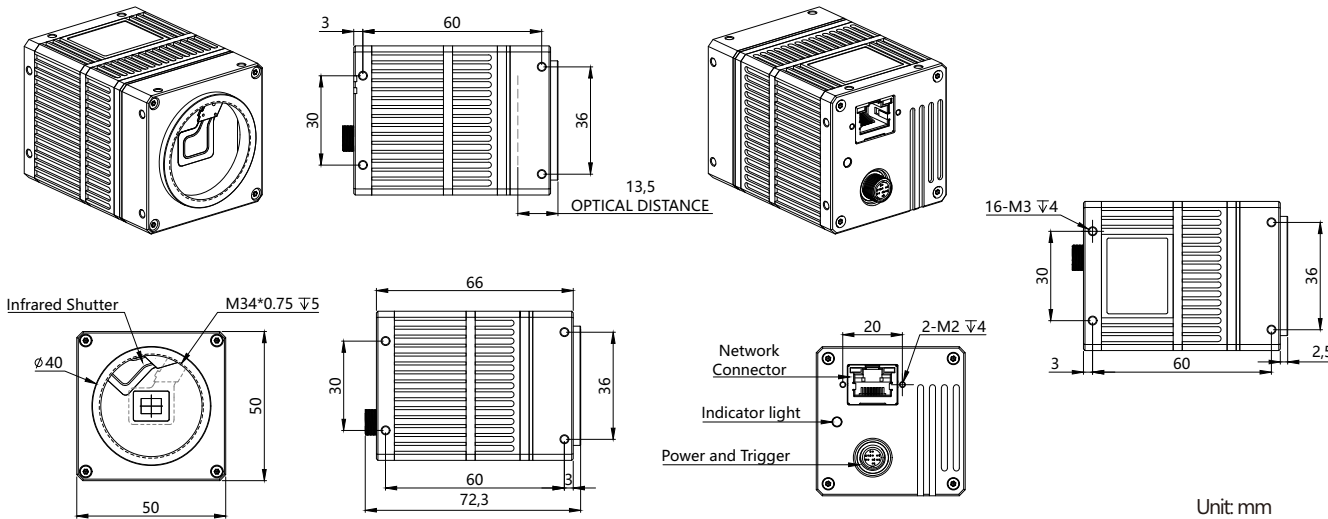
HT-GFF120 is an industrial far-infrared thermal imager that can measure the surface temperature of objects and display real-time thermal images. It can provide temperature measurement in different ranges measurement range, multiple color modes. User can set the emissivity, capture data per frame, video recording, provided with complete secondary development SDK and demo routines, and support C/C++/VB/Delphi/Python and other programming languages. Emissivity setting, central temperature display, high and low temperature point temperature display and user interest area Domain temperature display, horizontal image, vertical image, distance correction, humidity correction, user- defined temperature compensation, high and low temperature alarm, user interested temperature range, color temperature degree contrast, image sharpening level setting, blackbody furnace calibration etc. Three formats of data: temperature of each point, gray value of each point and pseudo color of each point.

Product Features

- High sensitivity and high resolution infrared detectors provide clear Infrared imagery and temperature measurement accuracy
 - Rj45 Gigabit Ethernet interface is adopted for 100m long-distance stable transmission
 - Support POE power supply, plug and play driver design, simple and easy to use, with the stability of network communication and the convenience of USB device
 - 400x300 resolution, up to 50Hz frame rate
- Thermal sensitivity≤55mk, thermal response time < 15ms
 - Support multiple cameras working at the same time, unlimited number, arbitrary networking, long operating time without dropping lines or losing frames
 - Built-in hardware accelerates image processing and reduces CPU utilization at the host end
 - Designed with GigE vision1.2 visual standard protocol, it supports Halcon,Visionpro and other software



GF120 Camera Dimension Drawing



Technical Parameter

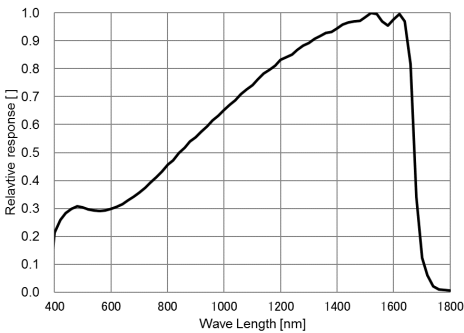
Model	HT-GF120		
Parameter			
Optical size	6.8mmx5.1mm	Sensor type	Thermal Imaging
Sensor model	GST417M	Effective pixels	0.12MP
Pixel size	17um	Exposure method	Automatic
Signal to noise ratio	75dBFS	Dynamic Range	80dBc
Spectral range	6.8μm -5.1μm	Thermal sensitivity	NETD≤55mK@f/1,300K
Pixel bit depth	14bit	Default resolution and frame rate	400x300 50Hz
Video output format	RGB24/Mono16bit	Maximum gain (magnification)	0~7
Exposure time range (ms)	1~6	Frame buffer	128MB
User-defined data area	2KB	Vision Standard Agreement	GigE Vision V2.0 GenICam
Data interface	RJ45 Gigabit Ethernet interface, backward compatible with 100M network standard	Detector type	Unrefrigerated vanadium oxide
Spectral response	R8μm -14μm	Lens focal length	Optional
Field of view	21.7°×16.4°	Spatial resolution	0.99mrad 0.67mrad
Minimum imaging distance	0.5m	Typical power consumption	≤250mW (Without TEC)
Power supply	12V (POE optional)	Power	3W
Lens Mount	M34x0.75(Customizable)	Dimensions	66mmx50mmx50mm(Without lens)
Weight	<300g	Operating temperature	0°C-50°C
Working humidity	20%~80%(No condensation)	Storage temperature	-30°C~+60°C
Storage humidity	20%~95%(No condensation)	Image mirroring	Horizontal / Vertical
Focusing method	Manual focus	Response rate non-uniformity	≤5%
Temperature range	Gear 1 : -20°C~150°C Gear2 : 50°C~400°C, Temperature measurement accuracy ±1°C or ±2% (Take the maximum)		
Temperature measurement mode	1. Ambient temperature and human body temperature measurement (20°C-50°C); 2. Medium temperature mode (-40°C-170°C); 3. High temperature mode (-40°C-800°C)		
Color mode	Rainbow color coding (this coding mode is used by default) hot metal, white hot, black hot		
Operating system	WINXP, WIN7/8/10 32&64 bit system, Linux and ARM Linux driver, Android platform driver, MAC OS system		
Driver	Directshow component Halcon special component Labview special driver OCX component TWAIN component		
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java		
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.		

Short Wave Infrared Industrial Camera

HT-GEC130I camera is a scientific level, large target area and high-resolution InGaAs short wave infrared camera, which specifically designed for low illumination scientific research short wave infrared imaging and microscopic imaging. It is suitable for fluorescence imaging and laser spot tracking imaging, semiconductor detection, solar cell detection, hyperspectral imaging, etc.

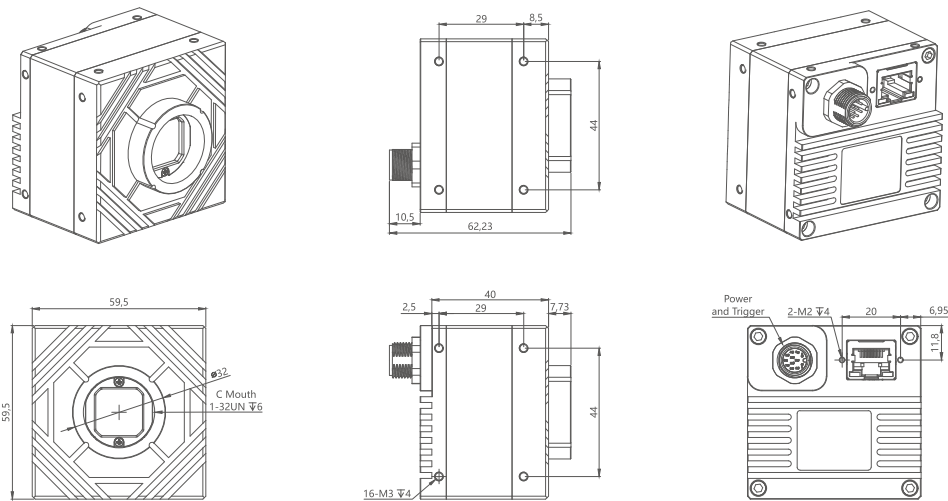
Product Features

- InGaAs detector, response wavelength 0.4-1.7um
 - Black level correction, good image uniformity
 - Gigabit network interface, compatible with vision standard
- Designed for short wave infrared imaging and microscopic imaging for low illumination scientific research
 - Resolution 1280x1024, frame rate up to 72FPS, ROI supported
 - Drive free directly supports Halcon, visionpro and other software, which is easy for system integration and development



GEC130I Spectrogram

GEC130I/ 30I Camera Dimension Drawing



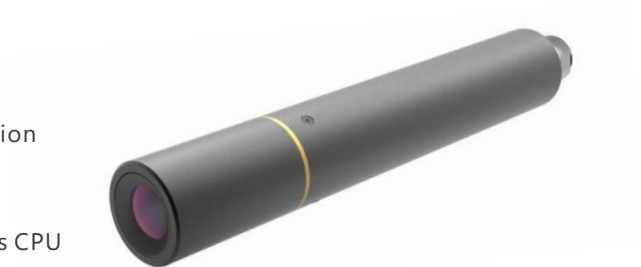
Unit: mm

Technical Parameter

Model	HT-GEC130I	HT-GEC30I
Parameter		
Sensor type	1/2" CMOS	1/4" CMOS
Camera type	Black and white	
Pixel size	5.0μm×5.0μm	
Shutter type	Global shutter	
Effective Pixels	1.3MP	0.3MP
Maximum resolution	1280×1024	640×512
Frame rate	72FPS	241FPS
Pixel bit depth	10bit	8bit
Sensitivity	121mv 1/30s	
Maximum gain (multiple)	24	
Exposure time range (ms)	0.007~2150.4	
Frame buffer	256MB	
User-defined data area	2KB	
Acquisition mode	Continuous/soft trigger/hard trigger	
Video output format	Mono8	
Vision Standards Protocol	GigE Vision V1.2/GenICam	
Data interface	RJ45Gigabit Ethernet	
GPIO	2 inputs and 2 outputs, 1 configurable input and output, support trigger and flash synchronization mode	
Power supply	12-24V(POE Standard configuration)	
Power	<12W (Including Refrigeration)	
Lens Mount	C-Mount	
Dimensions	59.5*59.5*44mm (Without lens holder and rear end interface)	
Weight	<300g	
Operating temperature	0~40℃	
Working humidity	20~80%(No condensation)	
Storage temperature	-30~60℃	
Storage humidity	20~95%(No condensation)	
Driver	Directshow component Halcon dedicated component Labview dedicated driver OCX component TWAIN component	
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java	
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.	

Pen Mini Industrial Camera

- Integrated M12 lens, small size, diameter only 20mm
- Interchangeable lens, unlimited field of vision
- Gigabit network interface, 100m long-distance stable transmission
- Support simultaneous operation of multiple cameras, unlimited number, and arbitrary networking
- The built-in hardware accelerates image processing and reduces CPU utilization at the host end
- Compatible with the vision standard, drive free and directly support Halcon, Visionpro and other software



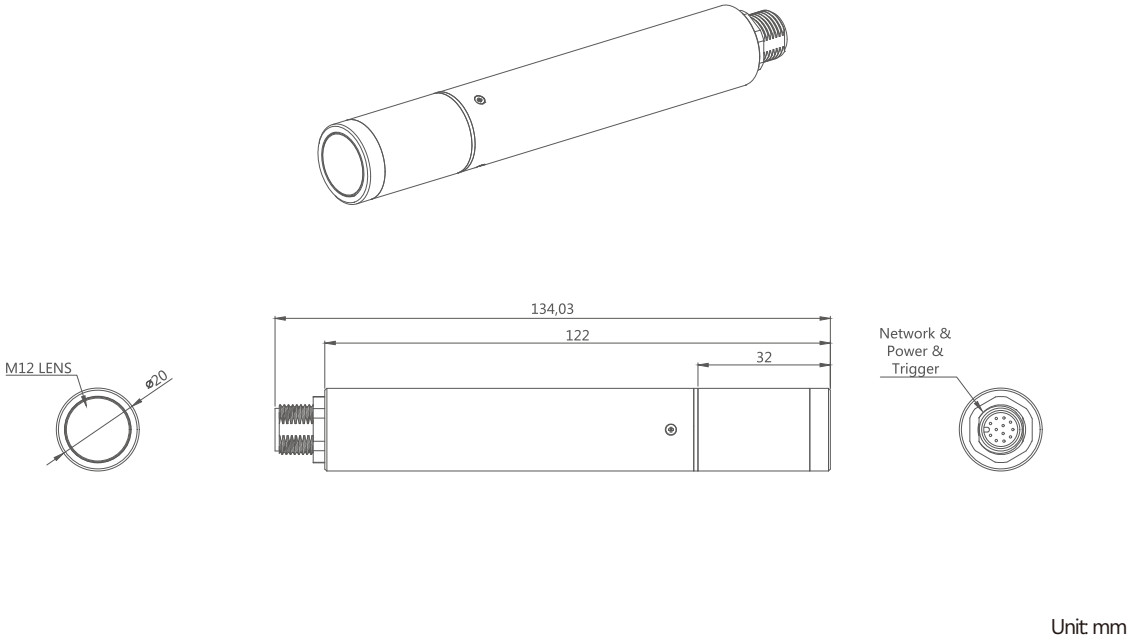
Technical Parameters

Specification	Parameter	Specification	Parameter
Model	HT-GEM232GC/M-T3-Y	Acquisition mode	Continuous/soft trigger/hard trigger
Resolution@ frame rate	1920X1200@52FPS	Frame buffer	128MB
Pixel size	3.0μmX3.0μm	User-defined data area	2KB
Pixel bit depth	8bit	GPIO	1 way flash output
Field of view (standard product)	15.8X7mm	Lens Mount	M12*0.5
Object distance (standard product)	100mm	Power supply	12V
Maximum gain (multiple)	16	Power	<2.66W
Exposure time range(ms)	0.014~917.5	Dimensions	20x122mm (Excluding back-end interface size)
Sensitivity	Color : 3.1V/lux*s Mono : 3.6V/lux*s	Weight	<100g

General parameters

Vision Standard Agreement	GigE Vision V1.2、GenIcam
Data interface	RJ45 Gigabit Ethernet interface, backward compatible with 100M network form
Operating system	WIINXP, WIN7/8/10 32@64 bit system, Linux and ARM Linux driver, Android platform driver, MAC OS system
Driver	Directshow component Halcon dedicated component Labview dedicated driver OCX component TWAIN component
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Working temperature @humidity	0~50°C @ 20%~80%(No condensation)
Storage temperature @humidity	-30~60°C @ 20%~95%(No condensation)
Other functions	Support any size of ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead point coordinate correction,ISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc.

Mini Camera Dimension Drawing



Miniature Industrial Camera

- M12 lens is supported, and the overall section size is 20x18mm
- Camera color/black and white options, lens replaceable, unlimited field of view
- Gigabit network interface, 100m long-distance stable transmission
- Support simultaneous operation of multiple cameras, unlimited number, and arbitrary networking
- The built-in hardware accelerates image processing and reduces CPU utilization at the host end
- Compatible with the vision standard, drive free and directly support Halcon, Visionpro and other software



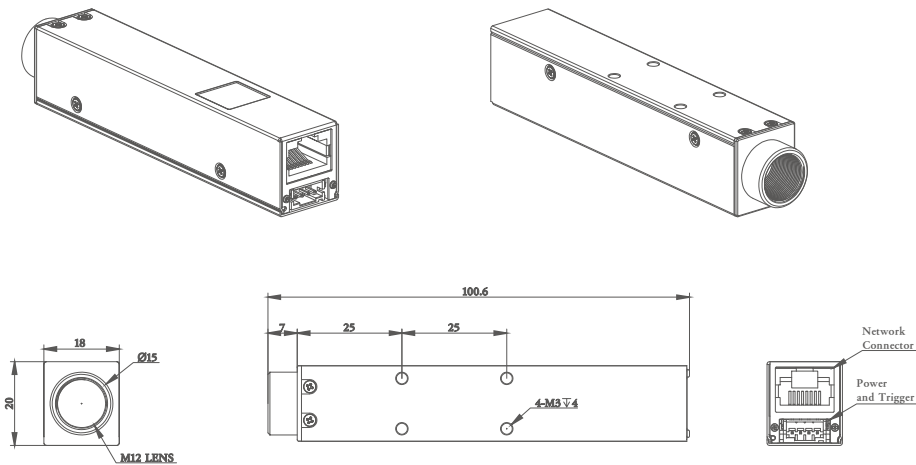
Technical Parameter

Specifications	Parameter	Specifications	Parameter
Model	HT-GEM133GC/M-T-F	Acquisition mode	Continuous / soft trigger / hard trigger
Resolution @ frame rate	1280x1024@91FPS	Frame cache	128MB
Pixel size	4μm BSI	User defined data area	2KB
Pixel bit depth	8bit	GPIO	1 way flash output
Sensor	1/2.7"CMOS	Lens Mount	M12
Video output format	raw8	Power supply	12~24V
Maximum gain (multiple)	32	Power	<2.5W
Exposure time range (ms)	0.019~79	Overall dimension	18X20X100mm(Without lens holder)
Sensitivity	8 V/Lux.s	Weight	<100g

General Parameters

Visual standards protocol	GigE Vision V1.2、GenIcam
Data Interface	RJ45 Gigabit Ethernet interface, downward compatible with 100m network form
Operating System	WIINXP、WIN7/8/10 32@64 Bit system, Linux and arm linux driver, Android platform driver, Mac OS System
Driver	DirectShow component Halcon special component LabVIEW special driver OCX component Twain component
Programming language pack	C/C++/C#/VB6/VB.NET/Delphi/BCB/Python/Java
Operating temperature @ humidity	0~50°C @ 20%~80%(No condensation)
Storage temperature @ humidity	-30~60°C @ 20%~95%(No condensation)
Other functions	Support any size ROI custom resolution, contrast and gamma adjustment, saturation adjustment, white balance correction, black level correction, custom dead center coordinate correctionISP image processing acceleration, 3D noise reduction, custom LUT table, frame rate adjustment, custom camera name, etc

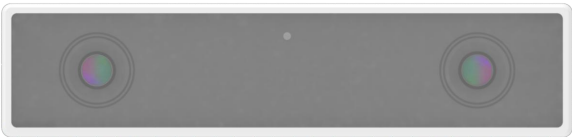
Miniature Camera Dimension Drawing



Unit: mm

Binocular Camera/Module

Binocular camera series have a higher degree of module integration compared with two ordinary cameras, automatic binocular synchronization and eliminates synchronous wiring; Reserved on board two sets of RS 2 3 2 can be used for control of other components to simplify the system and reduce the cost; Compared with the binocular module of UVC protocol on the market, this series of modules can output loss less gray scale image, omitting YUV image ,format conversion, furnishing USB3.0 interface or GigE transmission interface , and more functions are supported. Exposure, gain and other parameters can be completed manual controlled. With the special SDK, our binocular products support higher pixels, with no the constraint of the theoretical upper limit of pixels, and are more suitable for medium and advanced 3D applications.



Binocular Waterproof Camera



Binocular Module

Product Features

- FPC flexible cable length can be freely changed, and the baseline can be flexibly adjusted
 - Support lossless grayscale and raw data format output, non YUV format
 - Sensor board size can be customized according to customer requirements to match different structural requirements
 - It supports continuous soft trigger and hard trigger working modes, and the binocular images can be synchronize automatically, and the synchronization error is at the microsecond level
 - Provide special SDK and driver (non UVC), support windows, Linux and armlinux is fully compatible with Apple Mac OS and medview single camera product SDK
- 0.3MP to 65million pixels, supported by CMOS and CCD
 - Support Halcon, visionpro, opencv and other visual software
 - USB3.0 and GigE transmission interfaces are optional, and other interfaces can be customized
 - The working temperature can support -30~60 degrees, and the IP67 protective shell can be customized
 - The module reserves 2channels of RS232 interface, 1 channel of GPIO and 1 channel of flash output, 1-channel hard trigger input, custom accepted.

Product Selection Table

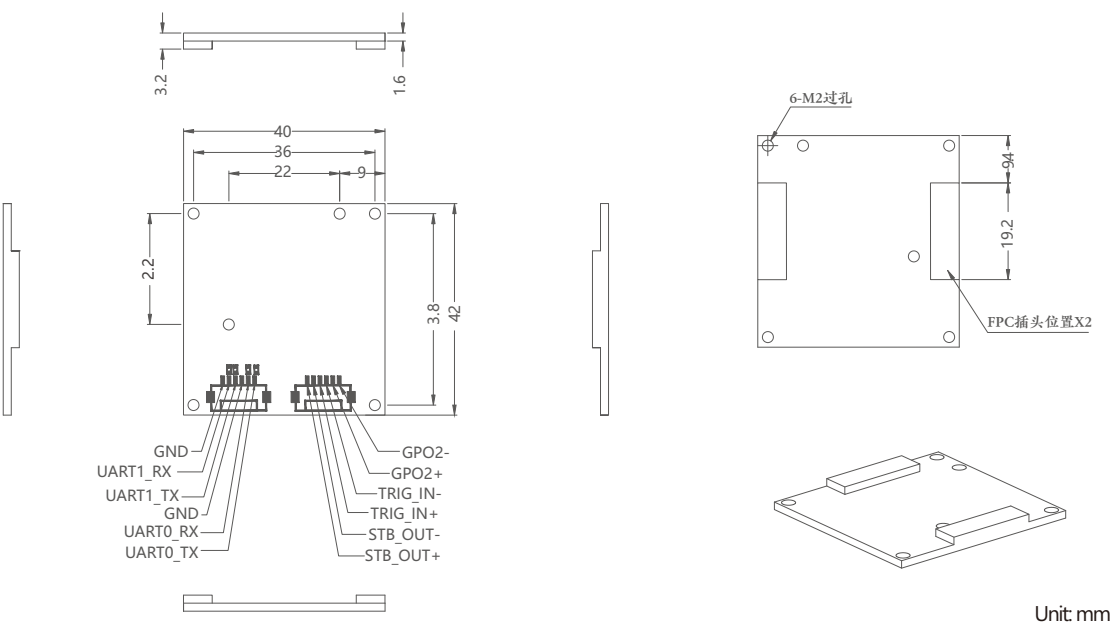
Model Number	Pixels	Sensor type	Shutter method	Maximum resolution	Pixel size	Frame rate (FPS)	Target size	Minimum exposure	Sensor model	Colour
HT-MSU33GC/M2-T	0.3MP	CMOS	Global	640X480	4.0μm	580	1/5.6"	0.00124ms	SMARTSENS	Colour/Mono
HT-MSU34GC/M2-T	0.3MP	CMOS	Global	640X480	4.8μm	380	1/4"	0.0040ms	PYTHON	Colour/Mono
HT-MSU133GC/M2-T	1.3MP	CMOS	Global	1280X1024	4.0μm	130	1/2.7"	0.00194ms	SMARTSENS	Colour/Mono
HT-MSU134GC/M2-T	1.3MP	CMOS	Global	1280X1024	4.8μm	123	1/2"	0.0076ms	PYTHON	Colour/Mono
HT-MGE34GC/M2-T	0.3MP	CMOS	Global	640X480	4.8μm	190	1/4"	0.0040ms	PYTHON	Colour/Mono
HT-MGE131GC/M2-T	1.22MP	CMOS	Global	1280X960	3.75μm	42	1/3"	0.014ms	AR0134	Colour/Mono
HT-MGE134GC/M2-T	1.3MP	CMOS	Global	1280X1024	4.8μm	45	1/2"	0.008ms	PYTHON	Colour/Mono

Technical Parameter

Model Parameter	HT-MSU33GC/M2-T	HT-MSU34GC/M2-T	HT-MSU133GC/M2-T	HT-MSU134GC/M2-T
Resolution @ frame rate	640X480@580	640X3480@380	1280X1024@130	1280X1024@123
Pixel size	4.0μmX4.0μm	4.8μmX4.8μm	4.0μmX4.0μm	4.8μmX4.8μm
Pixel bit depth	8bit	10bit	8bit	8bit
Sensitivity	8 V/lux-s 540nm	7.3V/lux-s 540nm	8V/lux-s	7.3 V/lux-s 540nm
Acquisition mode	Continuous / soft trigger / hard trigger			
Maximum gain (multiple)	8	16.5	6	16.5
Exposure time range(ms)	0.00124~20.7	0.0040~524.3	0.00194~31	0.0076~1001.8ms
Frame cache	128M Bytes			
User defined data area	2K Bytes			
Video output format	Colour : Bayer8 Mono : Mono8	Colour : Bayer8/Bayer12 Mono : Mono8/Mono12	Colour : Bayer8 Mono : Mono8	
GPIO	1 channel trigger input, 1 channel flash control output; 2-way TTL serial port; 1 channel GPIO output			
Lens Mount	M12 lens holder can be installed			
Power supply	5V, USB Bus power supply			
Power	<3W			
Overall dimension	Mainboard *1:42*42mm Sensor board *2:38*38mm Single side of flat cable: 200mm (adjustable as required)			
Weight	<75g			

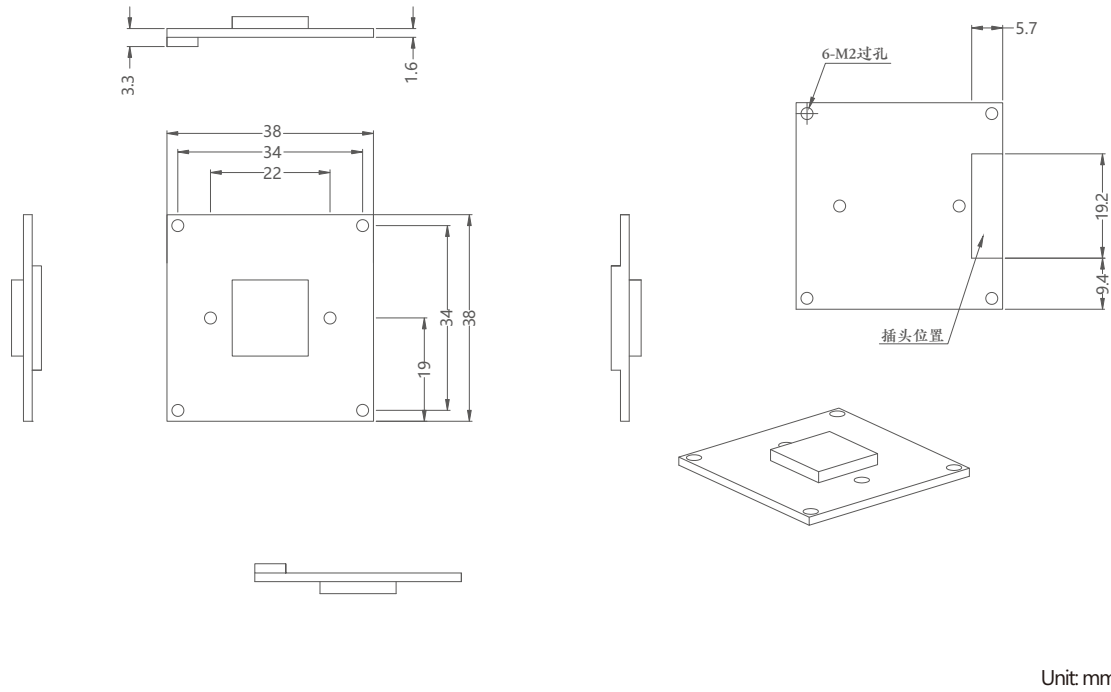
Model	HT-MGE34GC/M2-T	HT-MGE131GC/M2-T	HT-MGE134GC/M2-T
Resolution @ frame rate	640X480@190	1280X960@42	1280X1024@45
Pixel size	4.8μmX4.8μm	3.75μmX3.75μm	4.8μmX4.8μm
Pixel bit depth	10bit	12bit	10bit
Sensitivity	7.3 V/lux-s 540nm	5.3V/lux-s 550nm	7.3 V/lux-s 540nm
Acquisition mode	Continuous / soft trigger / hard trigger		
Maximum gain (multiple)	16.5	4	16.5
Exposure time range(ms)	0.0040~524.3	0.014~53	0.008~1024
Frame cache	128M Bytes		128M Bytes
User defined data area	2K Bytes		
Video output format	Colour : Bayer8/Bayer12 Mono : Mono8/Mono12		
GPIO	1 channel trigger input, 1 channel flash control output; 2-way TTL serial port; 1 channel GPIO output		
Lens Mount	M12 lens holder can be installed		
Power supply	DC9~24V		
Power	<3W		<2.5W
Overall dimension	Mainboard *1:42*42mm Sensor board *2:38*38mm Single side of flat cable: 200mm (adjustable as required)		
Weight	< 75g		

Binocular Module Dimension Drawing



Unit: mm

Senor plate dimension drawing



Unit: mm

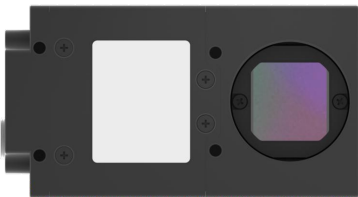
90 Degree Right Angle Industrial Camera

Huatengvision is the first company to launch a 90 degree angle industrial camera in China . Several standard machine series can greatly shorten the vertical upward space of the sensor surface by means of horizontal installation. The height in the vertical direction is only 29mm, which solves the problem of space limitation for equipment installation in the market. The performance and parameters of the 90 degree angle version camera are the same as standard machine, for only adjusting the structure. There is no difference between the camera software and the SDK. When selecting the product model, the suffix -90 means 90 degree angle camera.

At present, the cameras supporting 90 degree angle structure are HT-GE, HT-GEF, HT-GED, HT-SUA and HT-SUF series. If more models or series are required, please contact our sales.



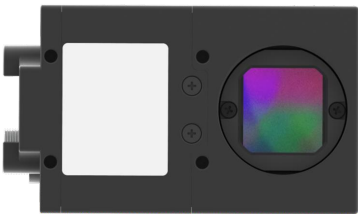
GE-90 Series



GEF-90 Series

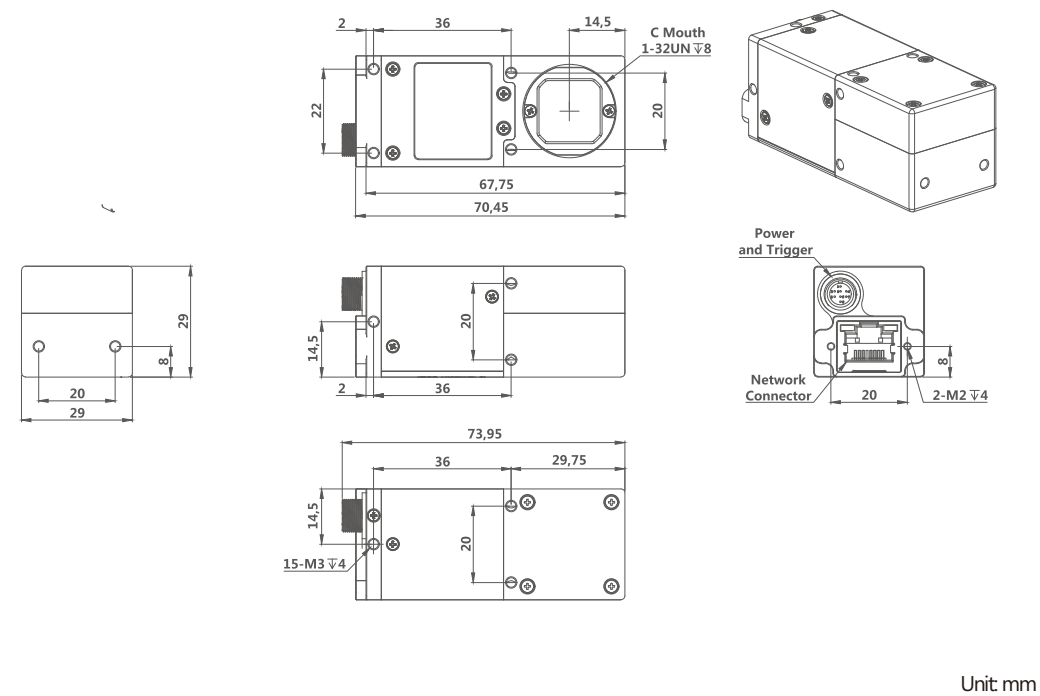


SUA-90 Series

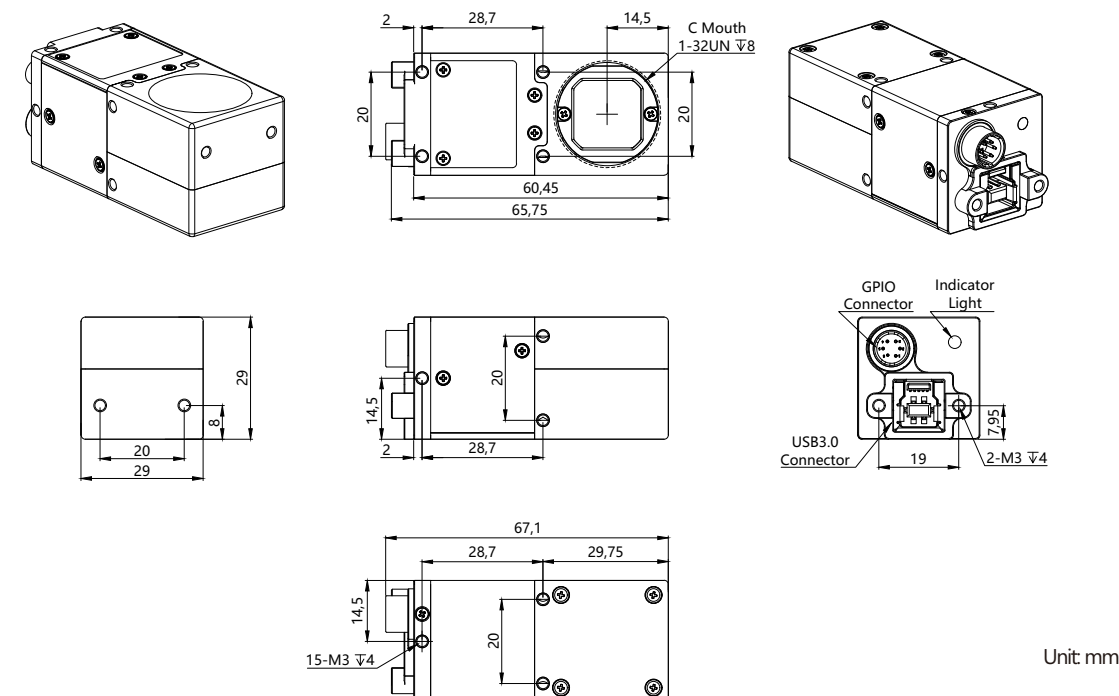


SUF-90 Series

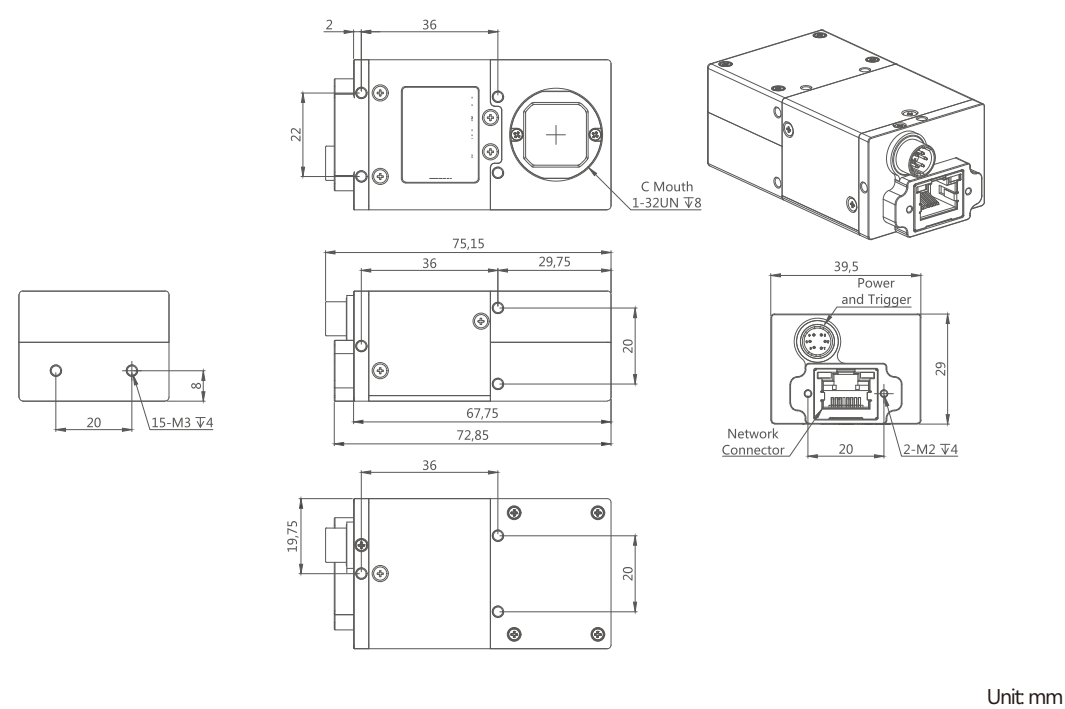
GE-90 Series Dimension Drawing



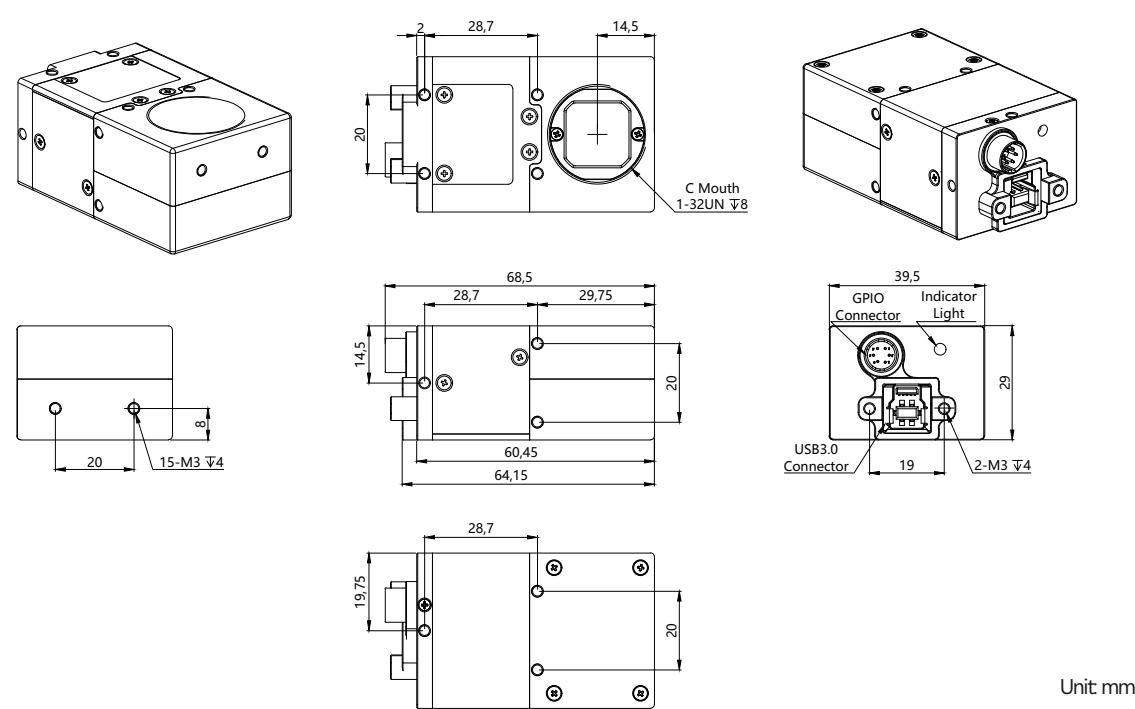
SUA-90 Series Dimension Drawing



GEF-90 Series Dimension Drawing



SUF-90 Series Dimension Drawing



Binocular 3D camera

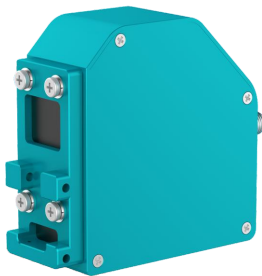
- Field of view and depth of field can be customized
- Camera pixels selectable, 30 to 43million pixels
- Provide development demo based on Halcon
- The baseline can be customized, and the accuracy can be up to 0.1mm
- 3D point cloud data or binocular 2D images can be output
- Provide simple SDK to easily realize secondary development



parameter Model	Pixel	Standard lens focal length	Standard field of view(mm)	Standard accuracy	Transmission interface	Acquisition speed	Point cloud synthesis speed	Object distance (Adjustable)
HT-3D1-130U3MG60-1	1.3MP	6mm	480X360	0.04mm	USB3.0	1.0s	1.0s	Default 600mm
HT-3D1-500U3MG40-1	5MP	16mm	630X530	0.02mm	USB3.0	2.0s	2.0s	Default 600mm

Line laser 3D camera

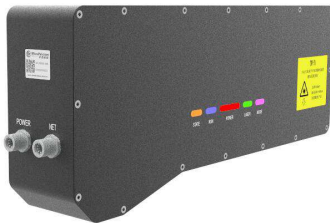
- Accurate calibration before delivery, easy to use
- Support single frame 3D point cloud data output and multi frame stitching contour point cloud output
- High performance 3D camera, integrated with high-precision laser and lens, small size
- Rich SDK software interfaces, supporting c/c++, Halcon, etc



parameter Model	Pixel	Standard lens focal length	Vision (mm)	Precision(mm)	Transmission interface	Scanning frame rate	Object distance (Adjustable)	Laser light source
HT-3DS31GM-L	0.3MP	16mm	40~90	X(0.1) Z(0.01)	GigE	100Hz	150~300mm	Red/ Blue

Line Laser Volumetric 3D Camera

- Precise calibration before leaving the factory, easy to use
- Status light can be controlled by secondary development programming
- Support deep data, 3D point cloud data output
- Built-in Windows system, more functions can be achieved through secondary development while obtaining volumetric data.
- Built-in processor, can directly output the length, width and height of the measured object without an external computer
- The maximum field of view is 2 meters, and the depth of field is 1 meter, applied to the detection of large objects



parameter Model	Pixel	Standard lens focal length	Field of view	Precision	Transmission interface	Scanning frame rate	Working distance	Laser light source
HT-I3D202GM-LN	2MP	8mm	1.2m	1cm	GigE interface	100Hz	60~180 CM	Near infrared

10 gigabit Ethernet card

- GigE Vision standard agreement framing.
- Low latency trasmission and CPU load.
- High-reliability, Data packet auto retransmission.
- PCIE 3.0x8, maximum bandwidth 50Gbps.
- It provides the standard 10gigabit network adapter function.
Not connected with the camera, it can be used as a standard 10 gigabit nic network port..
- High-capacity Onboard RAM, multiple images can be cached.

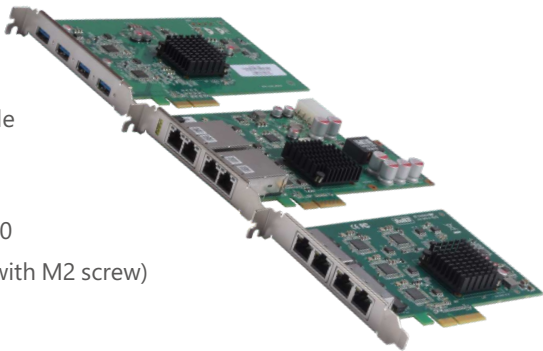


Technical Parameters

Model	HT-CXG40	HT-CXG21
Parameter		
Interface type	10-gigabit Ethernet	
Interface rate	10G/5G/2.5G/1G/100M	
Interface number	4 way	2 way
Transmission media and Interface	RJ45	
Transmission distance	100m	
Bus type	PCIE3.0X8	PCIEgen2.0X4
Frame buffer	4G byte	2G byte
Standard agreement	IEEE 802.3、IEEE 802.3u、IEEE802.3ab、IEEE802.3x、IEEE802.3af、IEEE802.3at	
Driver support	windows7/windows 10/ubuntu	
Features	GigEVision framing	
Power supply	12V PCIE/12V ATX	
power dissipation	<20W	
Heat dissipation	Air-cooling	Radiator heat dissipation
Dimensions	190 × 98.5 × 33.6mm dam-board;121 × 13mm	144.9 × 56.1 × 22.65mm
Weight	<500g	/
Working temperature	0-50degrees	
Storage temperature	20~80%(No condensation)	
Storage humidity	~30~60 degrees	
Dimensions	20~95%(No condensation)	

PCIE10 Gigabit Network Card

- Bus interface: PCI Express X4 Gen
- Interface Quantity: 4 interfaces
- GigE network card: supports 9KB Jumbo frames and supports multiple cards and multiple cameras
- GigE network card: compatible slot PCI-E 16x/8x/4x
- USB interface: 4x USB3.0 interface, compatible with usb2.0/1.1/1.0
- USB connector: 4x USB3.0type-A connector on the front panel (fixed with M2 screw)
- The system supports windows 7/8/10 and Linux



UBS3.0 Technical Parameter

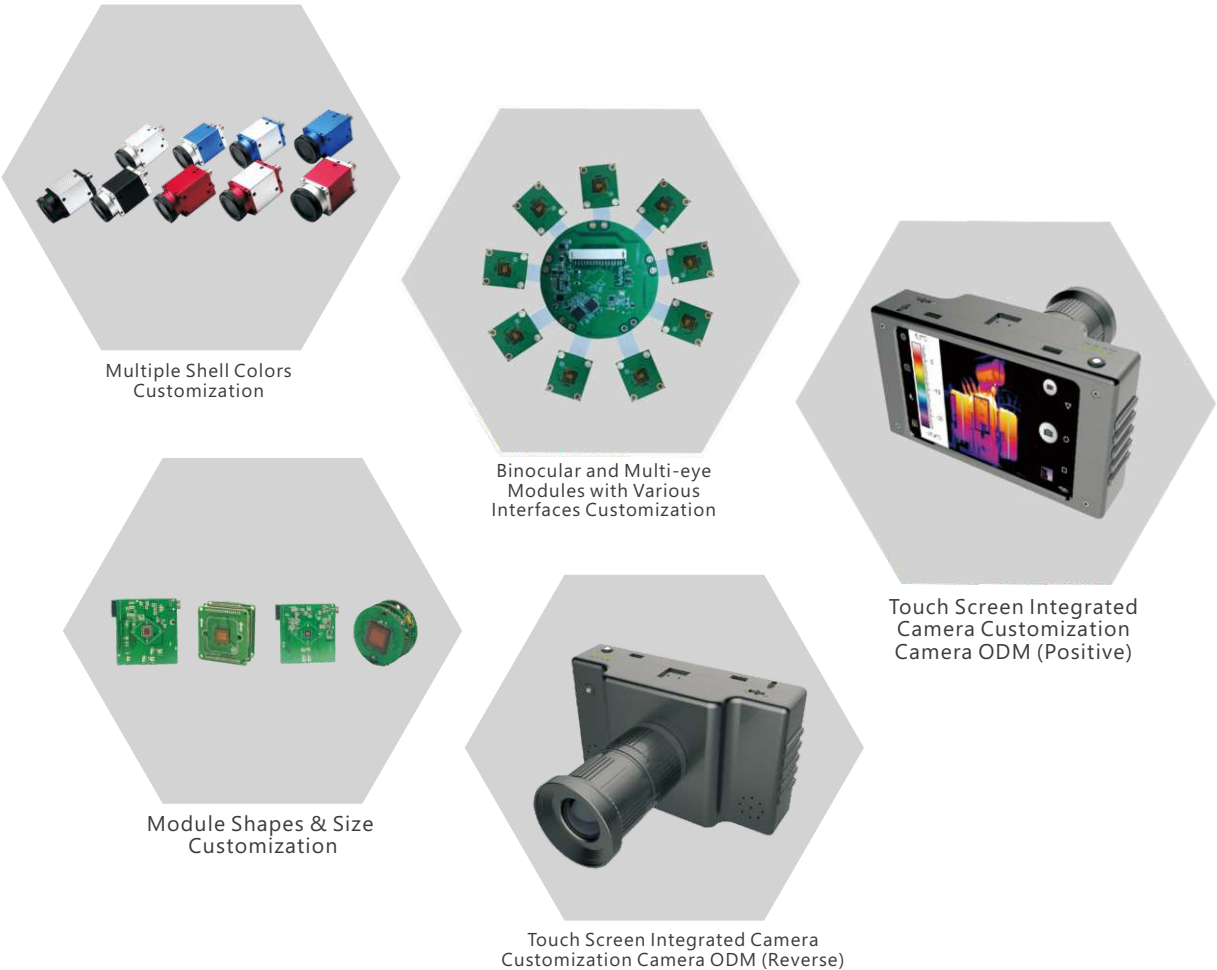
Parameter	Model	PCIE USB3.0 4-port expansion card
Bus interface		PCI Express x4 Gen2 interface; 2.5gb/s total bandwidth, up to 600mb/s for each port
USB interface		4x USB3.0 interface, compatible with usb2.0/ 1.1/ 1.0
USB connector		4X USB 3.0 type-A connector on the front panel (fixed with M2 screw)
USB controller		2x ASM3142 Master controller In compliance with Universal Serial Bus 3.1 specification revision 1.0 standard Accord with Intel xHCI specification revision 1.0 standard
USB per-port current limiting control		The user can select 1500mA current limiting control per port
Power demand		Users can select 1500mA current limiting control per port to obtain power from PCI Express Bus: maximum 2.0A @ 3.3VPower supply from PCI Express Bus: max. 2.8A @ 12V, for external equipment power supply
Working environment		Operating temperature: 0 ℃~60 ℃, with air flow; Storage temperature: -40 ℃~70 ℃; Relative humidity: 5%~90%, non condensing
System support		Windows 7/8/10Linux
Product size		168 mm x 112 mm (Width x length)

Technical Parameters Of Gigabit Network Card

Parameter	Model	PCle 4-port Poe gigabit network card
Bus interface		PCI Express x4 Gen2
Internet interface		4x 10/100/1000m BPS RJ45 network interface full duplex, half duplex intelligent detection and adaptation
Chip		Intel i210AT
Function		Ethernet power supply, compatible with IEEE 802.3af/at; Support 9KB Jumbo frames; Support multi card and multi camera; Poe power management; Poe protection
Compatible slots		PCI-E 16x/8x/4x
Network Environment		Gigabit and below network
Indicator light		ACT: the indicator flashes, and the port has data transceiver link: when the indicator is on for a long time, it indicates that the link is normal POE: when the indicator is on for a long time, it indicates that the external equipment is powered normally
Power requirements		Use 4-pin power connector, maximum 100W Poe power (+ 12V@8A) Using PCIe slots, maximum 20W Poe power (+ 12V@2.1A)
Operating environment		Operating temperature: 0 ℃~70 ℃; Storage temperature: -40 ℃~70 ℃; Relative humidity: 5%~90%, non condensing
System support		Windows XP/7/8/10.Linux
Product size		168 mm x 112 mm (Width x length)

Parameter	Model	PCIE 4-port gigabit network card
Bus interface		PCI Express x4 Gen2
Internet interface		4x 10/100/1000M bps RJ45 network connection; Full duplex, half duplex intelligent detection and adaptation
Chip		Intel i210AT
Function		Compatible with IEEE 802.1p/ 802.1q/802.3 10base-t/802.3u 100base tx/802.3ab 1000BASE-TSupport 9KB Jumbo frames and multi card multi camera
Compatible slots		PCI-E 16x/8x/4x (PCIe 2x/1x Support required connector rear end opening)
Network Environment		Gigabit and below network
Indicator light		ACT: the indicator flashes, and the port has data transceiver link: when the indicator is on for a long time, it indicates that the link is normal
Operating environment		Operating temperature: 0 ℃~70 ℃ storage temperature: -40 ℃~70 ℃ relative humidity: 5%~90%, non condensing
System support		Windows XP/7/8/10.Linux
Product size		140 mm x 100 mm (Width x length)

Camera Customization ODM



Contact information:

Sales email: sales@huatengvision.com

Technical Support: support@huatengvision.com

Office TEL: 0755-28918004

website: www.huatengvision.com

- ◀ Camera Waveband Customization:Ultraviolet, visible light, near infrared, short wave infrared, far infrared,etc.
 - ◀ Customization of cameras interfaces: USB2.0, USB3.0, Gigabit network, optical fiber, 10 Gigabit network, CXP, wireless, etc.
 - ◀ Customization of various additional functions: programmable IO, autofocus, waterproof case, special structural matching, battery power, integrated light source and other control circuits, etc
 - ◀ arious brands and specifications of sensor Customization: SONY ONSENME OV,Centron,etc.
- We welcome the customers to phone or email to consult, let's work together for a win-win cooperation .